

UNIVERSIDAD DE CHILE
FACULTAD DE CIENCIAS FÍSICAS Y MATEMÁTICAS
DEPARTAMENTO DE INGENIERÍA ELÉCTRICA

MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

TESIS PARA OPTAR AL GRADO DE
MAGÍSTER EN CIENCIAS DE LA INGENIERÍA MENCIÓN ELÉCTRICA

MEMORIA PARA OPTAR AL TÍTULO DE
INGENIERO CIVIL ELÉCTRICO

IGNACIO ROMERO ARAVENA

PROFESOR GUÍA:
JAVIER RUIZ DEL SOLAR

PROFESOR CO-GUÍA:
PATRICIO LONCOMILLA

MIEMBROS DE LA COMISIÓN:

SANTIAGO DE CHILE
2026

RESUMEN DE LA TESIS PARA OPTAR AL GRADO DE MAGÍSTER
EN CIENCIAS DE LA INGENIERÍA, MENCIÓN ELÉCTRICA
RESUMEN DE LA MEMORIA PARA OPTAR AL TÍTULO DE
INGENIERO CIVIL ELÉCTRICO
POR: IGNACIO ROMERO ARAVENA
FECHA: 2026
PROFESOR GUÍA: JAVIER RUIZ DEL SOLAR

MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

SPANISH SUMMARY

MITIGATING CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION VIA GAUSSIAN CLASS REPRESENTATIONS

This work proposes a modular Continual Learning framework for object classification that decouples feature extraction from statistical classification. A frozen DINOv2 backbone provides stable high-dimensional embeddings, and each class is independently modelled by a Gaussian distribution estimated in closed form. New categories are incorporated by updating only the sufficient statistics of the new class, leaving all previously learned prototypes unmodified and requiring no past-sample storage. Evaluated on four class-incremental benchmarks, the framework matches or surpasses the prompt-based state of the art, reaching 94.4 % accuracy on Caltech-256. A rigorous statistical characterisation of the embedding space validates the Gaussian modelling assumptions and links the per-class sample regime to the optimal choice of covariance estimator.

Abstract

SPANISH ABSTRACT

Abstract

Traditional deep learning models struggle under non-stationary data: incorporating new classes demands exhaustive retraining or incurs catastrophic forgetting. This thesis introduces a gradient-free Class-Incremental Learning framework that combines a frozen DINOv2 backbone with independent per-class Gaussian models estimated in closed form. New categories are added by computing the mean and covariance of their embeddings in the fixed feature space; previously learned class prototypes are never modified and no past samples need to be retained. The framework therefore achieves zero catastrophic forgetting by construction. Evaluated on Split CIFAR-100, Split ImageNet-R, Split Caltech-256, and a demanding 30-task Joint benchmark, it surpasses every prompt-based baseline on Caltech-256 (94.4 % vs. 92.5 %) and matches state-of-the-art accuracy on the remaining benchmarks. A comprehensive statistical characterisation of the DINOv2 embedding space—encompassing univariate and multivariate Gaussianity tests, spectral analysis, inter-class Bhattacharyya distances, and covariance conditioning analysis—validates the modelling assumptions and demonstrates that the optimal covariance estimator is governed by the ratio of per-class samples to feature dimensionality: diagonal models excel in the severely under-sampled regime, while the hybrid estimator, which decouples the shape and volume terms of the log-likelihood, provides the best accuracy–calibration balance as the sample budget grows.

La aventura nos aguarda

Acknowledgments

Contents

1 Introduction **1**

1.1 Motivation 1

1.2 Hypothesis 2

1.3 General Objective 3

1.4 Specific Objectives 3

1.5 Contributions 3

1.6 Thesis Structure 4

2 Theoretical Framework **6**

2.1 Artificial Intelligence and Machine Learning 6

2.1.1 Deep Learning 7

2.1.2 Computer Vision and Vision Transformers 8

2.1.3 Image Datasets and Benchmarks 9

2.2 Foundations of Probabilistic Modeling 10

2.2.1 Bayes Classification 10

2.2.2 Likelihood-Based Classification 10

2.2.3 Gaussianity Assumptions and Analysis 11

2.2.4 Advanced Covariance Estimation in High Dimensions 15

2.3 Continual Learning 17

2.3.1 Elasticity and Plasticity 19

2.3.2 Main Paradigms 19

2.3.3 Mitigation Strategies 19

3 Methodology **23**

3.1 Proposed Architecture 23

3.1.1 Likelihood-Based Classification 24

3.1.2 Datasets and Experimental Protocol 24

3.1.3 Model Metrics 25

3.2 Covariance Estimation 26

3.2.1 Diagonal Covariances 26

3.2.2 Full Covariances 27

3.2.3 Covariance Conditioning Analysis 27

3.2.4 Cross-Class Frobenius Norm Analysis 28

3.2.5 Hybrid Covariance 29

3.2.6 OAS Covariance 29

3.2.7 Ledoit-Wolff 30

3.2.8 Shared Covariance 31

3.2.9 Mixture of Gaussians 32

3.3 Final-State Confusion Matrix Evaluation 33

3.3.1 Confusion Matrix Definition 33

3.3.2 Mahalanobis-Only Inference 34

3.4 Feature-Space Characterization and Gaussianity Analysis 35

3.4.1 Probabilistic Classification in High Dimension 35

3.4.2 Gaussianity Tests 36

3.4.3 Inter-Class Pairwise Bhattacharyya Distance 38

3.4.4 Spectral Analysis of the Embedding Space 39

3.5 Generative-Discriminative Fusion and Re-ranking Strategies 39

3.5.1 SoftTaskFusion: Soft Class–Task Log-Probability Fusion 40

3.5.2 TaskMeanShrink: Task-Level MAP Mean Shrinkage 40

3.5.3 Top5-CosineReRank: Generative–Angular Cosine Re-ranking 41
3.5.4 OptFusion-MinMax: Optimised Min–Max Score Fusion 42
3.5.5 HardGate: Hierarchical Task Gating 42
3.5.6 Top5-TaskReRank: Top-5 Candidate Re-ranking via Task Likelihood 43
3.5.7 TopK-TaskFilter: Top-K Task Candidate Filtering 43
3.5.8 HardGate-LDA: Hierarchical LDA Subspace Inference 43
3.5.9 SoftGate-LDA: Soft Hierarchical LDA Fusion 44
3.6 Sample Efficiency and Incremental Class Inclusion 44
3.7 Backbones Comparison 45
3.8 Probabilistic Calibration Analysis 45
3.8.1 Confidence and Calibration Metrics 45

4 Results **49**

4.1 Covariance Estimation and Mixtures of Gaussians 49
4.1.1 Final-State Confusion Matrices 50
4.1.2 Effect of the Log-Normalisation Constant 50
4.2 Gaussianity and Feature Space Characterization 53
4.2.1 Score Components Distributions 53
4.2.2 Gaussianity Characterizarion 54
4.2.3 Inter-Class Pairwise Bhattacharyya Distance 57
4.2.4 Spectral Analysis of the Embedding Space 57
4.3 Covariance Matrix Structure 59
4.3.1 Conditioning Analysis 59
4.3.2 Cross-Class Frobenius Norm Analysis 61
4.4 Generative–Discriminative Fusion and Re-ranking Strategies 61
4.5 Sample Efficiency and Incremental Class Inclusion 62
4.6 Backbones Comparison 64
4.7 Confidence and Calibration Metrics 64

Bibliography **70**

A Score Component Distributions **71**

List of Tables

3.1 Summary of the probabilistic calibration and uncertainty metrics used in this work. "Better" indicates the direction of the preferred value; "context" means both high and low values are informative depending on whether predictions are correct. 48

4.1 Continual-learning performance on the three 10-task benchmarks. A_{10} and F_{10} are reported in percentage points. 50

4.2 Performance on the long-horizon **Split Joint Datasets** benchmark (30 tasks aggregated from CIFAR-100, ImageNet-R and Caltech-256). A_{30} and F_{30} are reported in percentage points. 51

4.3 $T_{\max}$ classification accuracy (%) for the Gaussian and Mahalanobis-only inference rules, under diagonal (Diag-Cov) and full (Full-Cov) covariance estimation. 51

4.4 Per-dimension Gaussianity statistics computed over all (*class*, *dimension*) pairs of the frozen DINOv2 embeddings (raw, i.e. taken directly from the backbone). Each dataset yields $C \times d$ pairs ($d = 768$), reported in column **n**. The columns $|\text{Skew}|_\mu$ and $|\text{Kurt}|_\mu$ are the mean absolute values across all pairs; $\mathbb{P}[\cdot]$ denotes the fraction of *(class, dimension) pairs* satisfying each cut-off. 54

4.5 Per-feature normality tests on the training split. All percentages report the fraction of *(class, dimension) pairs* satisfying the criterion, where each dataset contributes $C \times d$ pairs ($d = 768$). *normaltest* is the D'Agostino–Pearson composite test at $\alpha = 0.05$ (H_0: the dimension is normally distributed); *relaxed* is the joint criterion $|\text{Skew}| < 1$ **and** $|\text{Kurt}| < 1$; *both* denotes pairs passing both criteria simultaneously. 55

4.6 Univariate Henze–Zirkler test applied to every (class, dimension) pair on the training split. $\mathbb{E}_{H_0}[T_{\text{HZ}}]$ is the theoretical expectation under exact normality, determined by the per-class sample size via the bandwidth $b^2 = \frac{1}{2}(3/4)^{0.4} n_c^{-0.4}$. 56

4.7 Per-class multivariate Gaussianity via the diagonal Henze–Zirkler test [1] at native $d = 768$, using 100 % of the available samples. $T_{\text{HZ}} \approx 1$ indicates near-multivariate-Gaussianity under the diagonal-covariance model; C is the number of classes and $\tilde{n}_c$ is the median per-class sample count. 56

4.8 Final average accuracy (%) of the Gaussian continual-learning classifier after projecting DINOv2 embeddings to $d \in \{64, 256\}$ dimensions via PCA. *Diag*: diagonal covariance; *Full*: full covariance with adaptive ridge regularisation. Higher is better. 59

4.9 Per-class covariance conditioning metrics across the three benchmarks ($d = 768$, $\varepsilon = 10^{-6}$, $\varepsilon_{\text{thr}} = 10^{-4}$). Degenerate dimensions are those whose eigenvalue falls below ε_{thr}; effective rank is the entropy-based measure of Eq. 3.14. All values are mean $\pm$ std over classes. 61

4.10 Pairwise Frobenius distances $D_{ij}^F = \|\hat{\Sigma}_i^{\text{reg}} - \hat{\Sigma}_j^{\text{reg}}\|_F$ across all class pairs. . . . 61

4.11 Generative–Discriminative Re-ranking Strategies on **Split CIFAR-100**. . . 62

4.12 Generative–Discriminative Re-ranking Strategies on **Split ImageNet-R**. . . 63

4.13 Generative–Discriminative Re-ranking Strategies on **Split Caltech-256**. . . 63

4.14 Generative–Discriminative Re-ranking Strategies on **Split Joint Datasets** (30 tasks). 63

4.15 Backbone ablation. The two best closed-form estimators (Diag-Cov and Full-Cov) are evaluated with three vision backbones across all four benchmarks. . 64

4.16 Probabilistic calibration metrics for the Bayesian classifier at the last after-task checkpoint. Accuracy and ECE are in percentage points; CoE, TC, PM and Entropy are in $[0, 1]$. PM (Probability Margin) is the average difference $p_{(1)} - p_{(2)}$ between the top-1 and top-2 softmax probabilities; TC is the average of $p_{(1)}$ alone. Lower is better for ECE and CoE; higher is better for Accuracy. CIFAR-100 Full was regenerated from the cached DINOv2 features because the original source file was a byte-identical copy of the Diag one. 65

List of Figures

2.1 Machine Learning and Deep Learning Strategies. Image taken from [2]. . . . 7

2.2 Vision Transformer (ViT) architecture overview. The model processes an image as a sequence of patches through a standard Transformer encoder. Image taken from [3]. 8

2.3 Visual representation of kurtosis levels: leptokurtic (> 0), mesokurtic ($= 0$) and platykurtic (< 0) distributions [4]. 13

2.4 Comparison of negatively-skewed, symmetric and positively-skewed distributions [5]. 14

2.5 A sequential task arrival model for Continual Learning, depicting the transition from Task A to Task C over time, where each task represents a unique set of features or data distributions. 18

2.6 Standard Object Classifier comparison. The upper path shows standard inference, while the lower path illustrates the dilemma when a new class is introduced: the trade-off between catastrophic forgetting via fine-tuning and the infeasibility of retraining from scratch. 19

2.7 Class and Task Incremental Learning. Image taken from [6]. 20

3.1 Block diagram of the proposed approach. 23

4.1 Normalised confusion matrices at $T_{\max}$ ($N = 100$ classes) for CIFAR-100. . . 51

4.2 Normalised confusion matrices at $T_{\max}$ ($N = 250$ classes) for Caltech-256. . . 52

4.3 Normalised confusion matrices at $T_{\max}$ ($N = 200$ classes) for ImageNet-R. . . 52

4.4 **CIFAR-100 — Validation set** (Diag-Cov). Empirical distributions of (a) the volume term $\ln A_c$ ($C = 100$ values), (b) the Mahalanobis shape term d_{mah} ($N_{\text{val}} \times C$ sample–class pairs), and (c) the log-likelihood score; Gaussian parameters fitted on the training partition. 53

4.5 **CIFAR-100 — Validation set** (Full-Cov). Score component distributions on held-out samples. 54

4.6 Empirical skewness distribution over the $d \times C$ (class, coordinate) pairs of the DINOv2 embeddings. 54

4.7 Empirical excess-kurtosis distribution over the $d \times C$ (class, coordinate) pairs of the DINOv2 embeddings. 55

4.8 Univariate T_{HZ} distributions over all (class, dimension) pairs for the 80 % training split. The near-zero values confirm that the DINOv2 marginal feature distributions are highly consistent with Gaussianity at the individual coordinate level. 56

4.9 Per-class T_{HZ} distributions from the diagonal HZ test at native $d = 768$. All three benchmarks concentrate tightly near the Gaussian reference value $T_{\mathrm{HZ}} = 1$, with near-zero inter-class variance. 57

4.10 Inter-class separability under the Diag-Cov Gaussian model measured by the Bhattacharyya distance D_B (Eq. 4.4) as a function of the number of classes k. **(a)** Mean pairwise D_B as the class set grows. **(b)** Minimum pairwise D_B; horizontal dashed lines mark $D_B = 20$ and $D_B = 50$. **(c)** Count of pairs below each threshold as k increases. Over 99.8 % of all pairs exceed $D_B = 50$ on every benchmark. 58

4.11 PCA spectral decay of the ℓ_2-normalised DINOv2 features. **(a)** Individual % variance explained per component: steep decay over the first ~ 50 components, near-zero contribution thereafter. **(b)** Cumulative % variance explained: dashed reference lines mark 50, 80, 90 and 95 %; annotated n values show how many components suffice to reach 90 % on each dataset. Despite the $d = 768$ embedding space, 90 % of the variance is captured by fewer than 130 components on every benchmark. 58

4.12 Per-class conditioning metrics across all three benchmarks. Each histogram shows one bar per class; the darkred line marks the median. (a) log-condition number. (b) count of eigenvalues below the degeneracy threshold $\varepsilon_{\mathrm{thr}} = 10^{-4}$. 60

4.13 Effective rank (Eq. 3.14) of the regularised covariance matrices per class. Despite $d = 768$ ambient dimensions, the effective rank is below 110 for all CIFAR-100 classes, and below 80 for most Caltech-256 and ImageNet-R classes. 60

4.14 Pairwise Frobenius distance heatmaps $D_{ij}^F = \|\hat{\Sigma}_i^{\mathrm{reg}} - \hat{\Sigma}_j^{\mathrm{reg}}\|_F$. Colour encodes distance magnitude (darker = smaller = more similar covariance geometry). CIFAR-100 displays rich class-to-class variation; Caltech-256 and ImageNet-R are nearly uniform, indicating that regularisation has collapsed the covariance matrices toward a common εI floor. 61

4.15 Distribution of all pairwise Frobenius distances. The scale difference between CIFAR-100 ($\bar{D}_F \approx 329$) and the other two benchmarks ($\bar{D}_F \approx 0.16$–0.17) spans three orders of magnitude, reflecting the degree to which regularisation dominates the estimated covariances. 62

4.16 Accuracy of a new class as samples arrive incrementally: (**a**) Split CIFAR-100, (**b**) Split ImageNet-R, (**c**) Split Caltech-256. Two target classes per benchmark are shown. 64

A.1 **CIFAR-100 — Training set** (Diag-Cov). Empirical distributions of the volume term $\ln A_c$, the shape term d_{mah}, and the final log-likelihood score, computed on the data used to fit the Gaussian parameters. 71

A.2 **Caltech-256 — Training set** (Diag-Cov). Score component distributions on the training data. 72

A.3 **ImageNet-R — Training set** (Diag-Cov). Score component distributions on the training data. 72

A.4 **Caltech-256 — Validation set** (Diag-Cov). Score component distributions on held-out samples. 72

A.5 **ImageNet-R — Validation set** (Diag-Cov). Score component distributions on held-out samples. 73

A.6 **CIFAR-100 — Training set** (Full-Cov). Score component distributions on the training data. 73

A.7 **Caltech-256 — Training set** (Full-Cov). Score component distributions on the training data. 74

A.8 **ImageNet-R — Training set** (Full-Cov). Score component distributions on the training data. 74

A.9 **Caltech-256 — Validation set** (Full-Cov). Score component distributions on held-out samples. 74

A.10 **ImageNet-R — Validation set** (Full-Cov). Score component distributions on held-out samples. 75

Chapter 1

Introduction

1.1 Motivation

State-of-the-art (SOTA) object classification methods rely on deep learning models that are either trained end-to-end on the target application or initially trained on one task and subsequently fine-tuned for a different one. In both cases the resulting models are *monolithic*: a single network is trained on a large dataset and performs feature extraction and classification jointly. Such models are highly effective when the set of object classes is fixed and known at training or fine-tuning time. When new object classes must be incorporated, however, the entire model must be retrained or fine-tuned. This procedure requires significant computational resources, demands access to the original training dataset and, in many cases, leads to *catastrophic forgetting* of previously learned classes [7].

A natural way to integrate new classes while avoiding catastrophic forgetting is to formulate the problem as continual or class-incremental classification, in which the algorithm must learn to distinguish an ever-growing number of categories [8]. In this paradigm, a *task* corresponds to a distinct portion of the training experience, presented sequentially to the model. Instead of having access to the entire dataset at once, as in conventional supervised learning, the model receives data in sequential chunks [9]. Each task typically corresponds to a set of new classes that must be recognised. The central challenge is that when the model is updated task by task, it tends to overwrite previously acquired knowledge: monolithic models are structurally ill-suited for this scenario, and an alternative paradigm is therefore required.

This limitation has motivated the SOTA in continual classification to focus on balancing *stability* and *plasticity*. Stability refers to a model's ability to preserve previously learned knowledge; plasticity describes its capacity to adapt and incorporate new tasks [10]. Traditional approaches rarely optimise both properties simultaneously: when stability dominates, the model struggles to integrate new knowledge; when plasticity dominates, it loses access to past information. This trade-off is compounded by the limited capacity of neural networks to integrate new representations without interfering with old ones and by the need to continually adjust the network's structure to match the growing number of object classes.

A natural way to overcome this dilemma is to decouple the monolithic model into two distinct components that are trained separately, possibly at different times: one responsible for feature extraction and another for object classification. The feature-extraction model must be trained at system-initialisation time and must produce embeddings that are general enough to characterise every class of interest, including classes unknown at training time. The classification model, by contrast, should be easy to update or extend as new object classes arrive, and ideally should be able to incorporate a new class without modifying the components that already recognise the existing classes. This structural separation mirrors the organisation of the human visual system, in which the structures responsible for low-level feature extraction (simple and complex cells in cortical areas V1 and V2) develop during the early post-natal weeks and remain largely stable thereafter [11]. High-level object recognition, in contrast, occurs in cognitive areas where the ability to learn new categories persists throughout life, occasionally producing slight degradation in the recognition of visually similar classes.

The emergence of large-scale, self-supervised visual foundation models (such as DINOv2 [12]) provides the empirical conditions to put this idea into practice. Foundation models offer general-purpose feature spaces with high transferability, and once frozen they yield a stationary representation in which incremental learning can be reduced to estimating class-conditional statistics rather than adapting network parameters. Building on this observation, we propose a continual learning object recognition system organised as two complementary components. The first component is a feature extractor built from a frozen pre-trained foundation model. The second component is a statistical classifier built from a collection of class-conditional Gaussian models, where each model characterises a specific object class independently of the others. A new Gaussian model can be created and inserted whenever a new class arrives, and classification is performed by assigning the input to the class whose Gaussian yields the highest log-likelihood, in direct analogy with a Bayes classifier.

Gaussian class representations offer two key advantages. First, each class is modelled independently from the rest, so incorporating a new class amounts to estimating a small number of sufficient statistics (mean and covariance) and does not require any modification of the previously stored class models. Second, Gaussian models are compact: they encode each class in a few parameters that can be estimated from a modest sample budget and refined as new samples become available, without any storage of the underlying images.

1.2 Hypothesis

This thesis hypothesises that the combination of an independent class-conditional Gaussian model and a frozen pre-trained feature extractor provides a robust framework for continual object classification. By enabling the addition of new classes without full-model retraining, this design eliminates the conventional source of catastrophic forgetting, while retaining a simple, compact and computationally inexpensive architecture whose accuracy and forgetting are competitive with the state of the art.

1.3 General Objective

The main objective of this thesis is to develop and validate a continual learning framework for object classification that uses pre-trained foundation models as feature extractors and independent class-conditional Gaussian models as classifiers, with the purpose of enabling the incremental incorporation of new classes while preventing catastrophic forgetting of previously learned ones.

1.4 Specific Objectives

For the purpose of fulfilling the general objective and verifying the hypothesis of this work, the following specific objectives are defined:

- To assess the Gaussianity of the class-conditional feature distributions in the DINOv2 latent space, validating the underlying distributional assumptions of the proposed model.
- To characterise the efficacy of independent Gaussian distributions as class representations within a continual-learning framework, and to compare the result against prompt-based state-of-the-art baselines.
- To quantify the impact of various covariance-matrix estimators (diagonal, full, hybrid, shared, Ledoit–Wolf, OAS) and Mixture-of-Gaussian variants on the classification accuracy, the average forgetting and the probabilistic calibration of the resulting classifier.
- To design and evaluate a family of generative–discriminative fusion and re-ranking strategies that combine the Mahalanobis log-likelihood with auxiliary discriminative cues, and to identify which of them improve the baseline reliably and which are brittle.
- To analyse the probabilistic calibration of the proposed model through a comprehensive set of metrics (Expected Calibration Error, Confidence on Errors, Total Confidence, Log-Gap, predictive entropy) that go beyond raw accuracy.
- To evaluate the dependence of the framework on the choice of upstream feature extractor by comparing DINOv2 against supervised Vision Transformers of varying capacity (ViT-base/16, ViT-medium/16) across multiple benchmarks and incremental-learning scenarios.

1.5 Contributions

The principal contributions of this thesis are the following:

1. A modular, gradient-free Bayesian continual-learning framework that decouples feature extraction from classification by combining a frozen DINOv2 backbone with in-

dependent class-conditional Gaussian models, achieving competitive accuracy and low forgetting without any rehearsal mechanism.

2. A systematic comparison of covariance estimators in the high-dimensional, small-sample regime, covering diagonal, full, hybrid, shared, Ledoit–Wolf and OAS variants, with empirical evaluation across four class-incremental benchmarks of varying difficulty.
3. A family of nine re-ranking and fusion strategies that integrate generative log-likelihood evidence with discriminative cosine similarity and log-domain task priors. An empirical taxonomy is derived from the results, separating these strategies into *recommended*, *brittle* and *marginal*.
4. A complete probabilistic-calibration analysis of the framework, including ECE, Total Confidence, Confidence on Errors, Log-Gap and predictive entropy across covariance estimators and benchmarks, with explicit identification of the volume–shape trade-off as the structural cause of the over-confidence observed under the Full-Covariance branch.
5. A backbone-sensitivity study (DINOv2, ViT-base, ViT-medium) and a 30-task heterogeneous Joint benchmark that aggregates CIFAR-100, ImageNet-R and the first 250 categories of Caltech-256 into a single stream, providing the most demanding evaluation regime currently reported for closed-form Bayesian continual classifiers.

1.6 Thesis Structure

The remainder of the document is organised as follows.

Chapter 2 – Theoretical Framework. Establishes the mathematical and conceptual basis of the work. It reviews the foundational paradigms of artificial intelligence, deep learning and computer vision, presents Bayesian classification and the Mixture-of-Gaussian extension, establishes the statistical tools for Gaussianity analysis (PCA spectral decay, skewness, kurtosis, and the Henze–Zirkler test), introduces the high-dimensional covariance estimation methods (regularisation, Ledoit–Wolf and OAS shrinkage), and surveys the continual-learning paradigm together with its dominant mitigation strategies.

Chapter 3 – Methodology. Describes the proposed architecture and its components in detail. It defines the experimental protocol for the four benchmarks—Split CIFAR-100, Split ImageNet-R, Split Caltech-256 (10 tasks of 25 classes each over the first 250 categories) and the 30-task Joint stream—and is structured to mirror the experimental progression. After describing the proposed architecture, it covers the feature-space characterisation methodology (spectral analysis, Bhattacharyya separability, and Gaussianity tests), formalises the six covariance estimators and mixture-of-Gaussian variants, derives the nine generative–discriminative fusion and re-ranking strategies, and closes with sample-efficiency protocols, backbone ablation, and probabilistic calibration metrics.

Chapter 4 – Results. Reports the quantitative results of every experiment carried out in the thesis. It is structured by the same top-down logic used to validate the modelling pipeline: per-dimension Gaussianity moments, feature-space diagnostics (PCA spectral decay and Mahalanobis histograms), continual-learning metrics across covariance estimators, generative–discriminative fusion tables, backbone ablations, calibration metrics, and incremental class-inclusion curves. All interpretation is deferred to the next chapter.

Chapter ?? – Discussion and Analysis. Interprets the experimental evidence. It examines the statistical characterisation of the DINOv2 embedding space, compares the covariance estimators and mixtures under the volume–shape decomposition principle, evaluates the reranking family and identifies which strategies improve the baseline reliably, and discusses backbone sensitivity, calibration and the framework's computational footprint. The chapter closes with a synthesis of the principal findings and a practical recipe.

Chapter ?? – Conclusion. Summarises the main conclusions of the work and outlines two natural extensions of the framework: open-world discovery (using the model's confidence as an out-of-distribution score) and real-time object detection (leveraging the spatial tokens of the DINOv2 backbone to avoid an external region-proposal network).

Chapter 2

Theoretical Framework

This chapter establishes the conceptual and mathematical foundations required to contextualize this research, transitioning from broad paradigms in Artificial Intelligence to the specific statistical mechanics and adaptive learning strategies underpinning our methodology. We begin by reviewing the fundamental principles of Machine Learning, specifically within the realms of image processing and object classification, before providing a rigorous mathematical treatment of Bayesian Inference, Mixture of Gaussians (MoG) and Gaussian Analysis. Furthermore, we detail advanced covariance estimation methods such as OAS and Ledoit-Wolf, which are critical for stability in high-dimensional feature spaces. Finally, we address the challenge of "catastrophic forgetting" by introducing the Continual Learning (CL) paradigm, describing the datasets utilized and evaluating the state-of-the-art mitigation strategies that provide a performance baseline for the proposed work.

2.1 Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is a broad field of computer science dedicated to creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, perception, and language understanding [13]. Within this domain, Machine Learning (ML) focuses on developing algorithms that allow computers to learn patterns from data without explicit programming. ML systems are generally categorized into three paradigms: Supervised Learning, Unsupervised Learning, and Reinforcement Learning [14].

First, in **Supervised Learning**, the model learns a mapping from input variables $\mathbf{x}$ to output variables y based on a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$. The objective is to minimize a loss function that penalizes the discrepancy between the predicted output and the ground truth. This paradigm is typically divided into Classification, where the output is a discrete category, and Regression, where the output is a continuous value.

Second, **Unsupervised Learning** deals with unlabeled data $\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^{N}$. Instead of predicting a target, the goal is to discover the underlying structure or distribution of the data. Common tasks include Clustering, which groups similar data points (e.g., K-Means or

Gaussian Mixture Models), and Dimensionality Reduction, which projects high-dimensional data into a lower-dimensional space while preserving essential information (e.g., PCA or t-SNE).

Finally, **Reinforcement Learning** is based on an agent interacting with an environment to maximize a cumulative reward. Unlike supervised learning, the agent is not told which actions to take but must discover them through trial and error. This process is often modeled as a Markov Decision Process (MDP) [15], where the agent transitions between states based on actions, receiving feedback in the form of numerical rewards. The focus is on finding an optimal policy that balances exploration (trying new actions) and exploitation (using known high-reward actions).

2.1.1 Deep Learning

In traditional Machine Learning, the feature extraction step is a manual process where a human expert must identify and define the most relevant characteristics (such as the shape or edges of the car) before passing them to a classification algorithm. In contrast, as shown in figure 2.1, Deep Learning utilizes neural networks to automate this process; the model performs both feature extraction and classification simultaneously, learning to identify complex patterns directly from the input data without human intervention.

Structurally, this shift allows Deep Learning to handle much more complex, high-dimensional data, though it typically requires significantly larger datasets and more computational power to be effective. While traditional Machine Learning depends on the quality of hand-crafted features, Deep Learning architectures—like multi-layered neural networks—are designed to build hierarchical representations of data, progressively moving from low-level pixels to high-level object concepts.

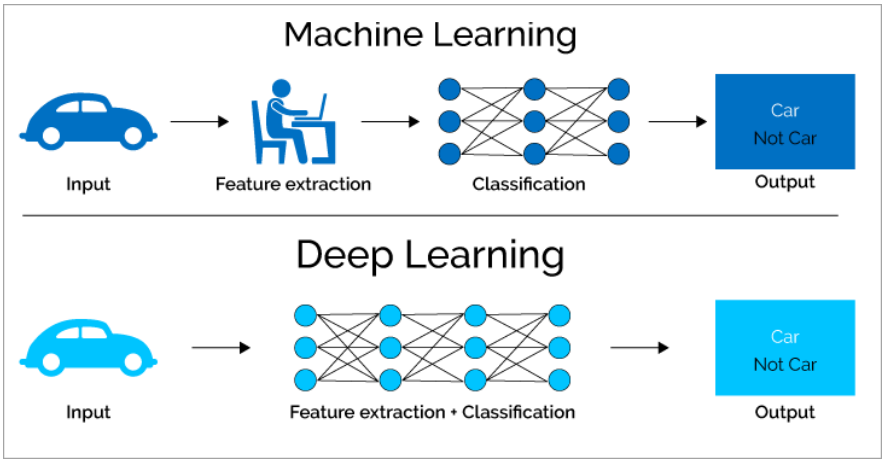


Figure 2.1: Machine Learning and Deep Learning Strategies. Image taken from [2].

2.1.2 Computer Vision and Vision Transformers

Computer Vision is the interdisciplinary subfield of AI that enables computers to gain high-level understanding from digital images or videos, effectively automating tasks performed by the human visual system [16]. While Convolutional Neural Networks (CNNs) [17] historically dominated this field, the Transformer architecture—originally developed for Natural Language Processing—has revolutionized visual recognition through the self-attention mechanism [18].

Vision Transformers (ViT) adapt this architecture by treating an image as a sequence of discrete tokens. As illustrated in Figure 2.2, the process follows a specific pipeline:

- **Patch Partitioning:** The input image is decomposed into fixed-size patches.
- **Linear Projection:** These patches are flattened and mapped to an embedding space.
- **Positional Encoding:** Since Transformers lack inherent knowledge of spatial structure, learnable position embeddings are added to the patch embeddings.
- **Transformer Encoder:** The resulting sequence is processed through multiple layers of multi-head self-attention and MLP blocks to capture global dependencies.

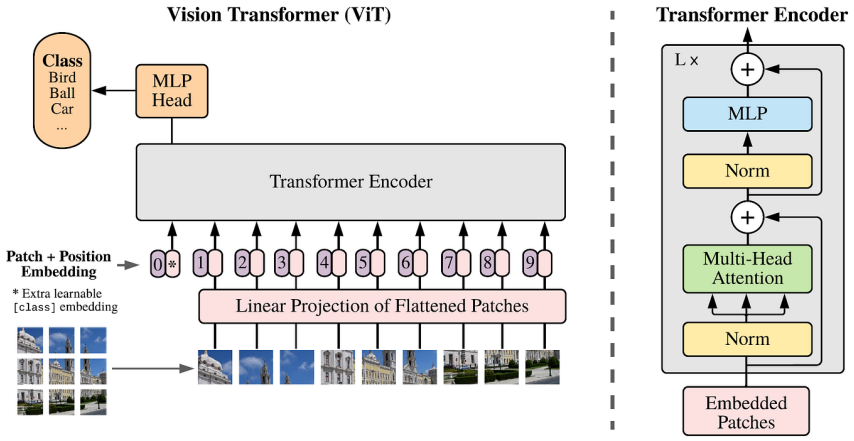


Figure 2.2: Vision Transformer (ViT) architecture overview. The model processes an image as a sequence of patches through a standard Transformer encoder. Image taken from [3].

Object Classification and Representative Power

The primary objective of the original ViT was *Object Classification*: the fundamental task of assigning a single categorical label to an entire input image. This task serves as the primary benchmark for evaluating the representative power of visual models [19]. By processing the global sequence of patches, ViTs have demonstrated an ability to meet or exceed CNN performance, particularly when pre-trained on the large-scale datasets described below.

2.1.3 Image Datasets and Benchmarks

To rigorously test the robustness and generalization of architectures like ViT, this work utilizes several standardized datasets that vary in scale, stylistic diversity, and granularity.

The ImageNet Family

ImageNet [20] is a cornerstone of computer vision, providing over 1.2 million images across 1,000 classes. Its hierarchical structure, based on WordNet, makes it the standard for large-scale classification. To further push the boundaries of model performance, two variants are often employed:

- **ImageNet-21K (and ImageNet-21k-P):** An expansive version containing 14 million images across 21,000 classes [21]. The "P" version is a cleaned subset designed to make pre-training for models like ViT more accessible.
- **ImageNet-R (Renditions):** A dataset of 30,000 images covering 200 classes in non-photographic styles (e.g., art, sketches). It tests a model's robustness to out-of-distribution data [22].

Fine-Grained and Diverse Benchmarks

Beyond large-scale pre-training, fine-grained classification is evaluated using:

- **CIFAR-100:** Consists of 60,000 low-resolution (32×32) images [23]. It is unique for its two-level hierarchy, where 100 "fine" classes are nested within 20 "coarse" superclasses, allowing for the evaluation of both specific and broad categorization.
- **Caltech-256:** Provides 30,607 images across 256 categories [24]. It offers greater category diversity and a higher minimum number of images per class than its predecessors, serving as a robust benchmark for generalisation. In this thesis we use the first 250 categories (approximately 28,100 images) so that the resulting class-incremental protocol can be divided into 10 balanced tasks of 25 classes each (see Chapter 3).

Self-Supervised Advancements: DINOv2

While the aforementioned datasets provide explicit labels for supervision, recent advancements have led to foundation models such as **DINOv2** [12]. Unlike earlier iterations, DINOv2 utilizes a discriminative self-supervised approach that combines contrastive learning with image-level and patch-level objectives. By training on massive, curated datasets without human labels, DINOv2 produces multipurpose visual features that exhibit strong "out-of-the-box" properties, often surpassing models trained with explicit supervision on tasks like semantic segmentation [12].

2.2 Foundations of Probabilistic Modeling

The convergence of statistical learning theory and high-dimensional data analysis has necessitated the development of robust mathematical frameworks to handle the dual challenges of uncertainty quantification and dimensionality reduction. As modern neural architectures, such as the DINOv2 Vision Transformer, extract increasingly complex representations from datasets like ImageNet-R, the subsequent classification and density estimation layers must be grounded in rigorous probabilistic foundations.

2.2.1 Bayes Classification

At its core, Bayes classification is rooted in **Bayes' Theorem**, which provides a mathematical framework for updating the probability of a hypothesis based on observed evidence. For a classification task, we seek the posterior probability $p(y = c \mid \mathbf{x})$—the probability that an observation belongs to class c given its observed feature vector $\mathbf{x}$.

Bayes' Theorem is defined as:

$$p(y = c \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid y = c)\, p(y = c)}{p(\mathbf{x})} \tag{2.1}$$

where:

- $p(y = c \mid \mathbf{x})$ is the **posterior** probability.
- $p(\mathbf{x} \mid y = c)$ is the **likelihood** of the data given the class.
- $p(y = c)$ is the **prior** probability of the class.
- $p(\mathbf{x})$ is the **evidence** or marginal likelihood, acting as a normalization constant.

The objective of a Bayes classifier is to minimize the Bayes risk; in the simplest case, this corresponds to assigning the observation to the class that maximizes the posterior probability.

2.2.2 Likelihood-Based Classification

Likelihood inference provides a coherent framework for updating the probability of a hypothesis as more evidence or information becomes available. In contrast to frequentist approaches that treat parameters as fixed but unknown constants, the Bayesian paradigm treats parameters as random variables characterized by probability distributions.

The connection between Bayes' Theorem and likelihood-based classification lies in the **Maximum A Posteriori (MAP)** decision rule. Since the evidence $p(\mathbf{x})$ is constant across all classes for a given observation, the classification decision depends solely on the product of the likelihood and the prior:

$$p(y = c \mid \mathbf{x}) \propto p(\mathbf{x} \mid y = c)\, p(y = c). \tag{2.2}$$

A generative modeling framework can be adopted and class-conditional Gaussian feature distributions can be assumed:

$$p(\mathbf{x} \mid y = c) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c), \tag{2.3}$$

where $\mathbf{x} \in \mathbb{R}^d$ represents the feature vector, $\boldsymbol{\mu}_c$ is the class mean vector, and $\boldsymbol{\Sigma}_c \succ 0$ is the covariance matrix.

Classification can be performed via MAP inference by maximizing the log-posterior:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\mathbf{x} \mid y = c) + \log \pi_c \right], \tag{2.4}$$

where $\pi_c = p(y = c)$ denotes the class prior.

Substituting the multivariate Gaussian density into the MAP rule, the decision rule becomes equivalent to:

$$\hat{y} = \arg\min_{c} \left[(\mathbf{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) + \log |\boldsymbol{\Sigma}_c| - 2 \log \pi_c \right]. \tag{2.5}$$

Mixture of Gaussians (MoG)

A Mixture of Gaussians models a complex probability density as a convex combination of K distinct Gaussian components. This approach is fundamental for clustering and density estimation in multi-modal distributions. Therefore, to capture multimodal feature distributions, the class-conditional model can be extended to a mixture:

$$p(\mathbf{x} \mid y = c) = \sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}), \tag{2.6}$$

where $\alpha_{c,k} \geq 0$ and $\sum_k \alpha_{c,k} = 1$.

The MAP decision rule becomes

$$\hat{y} = \arg\max_{c} \left[\log \left(\sum_{k=1}^{K_c} \alpha_{c,k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k}) \right) + \log \pi_c \right]. \tag{2.7}$$

2.2.3 Gaussianity Assumptions and Analysis

In the context of generative modeling for foundation feature spaces, the validity of the Bayes classifier depends on how well the class-conditional Gaussian distributions approximate the true data distribution. The following analytical framework establishes the statistical basis for these assumptions.

Principal Component Analysis

Principal Component Analysis (PCA) is a linear dimensionality-reduction technique that finds an orthonormal basis for the data by diagonalising the empirical covariance matrix [25]. Given a centred data matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$ (zero-mean columns), the empirical covariance is

$$\mathbf{S} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}. \tag{2.8}$$

Its eigendecomposition $\mathbf{S} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T$ yields orthonormal eigenvectors $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_d]$ sorted so that the corresponding eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$. Each eigenvector $\mathbf{p}_i$ is a *principal component* (PC), and λ_i equals the variance of the data projected onto $\mathbf{p}_i$.

Dimensionality reduction. Retaining only the leading k eigenvectors defines the projection matrix $\mathbf{P}_k = [\mathbf{p}_1, \ldots, \mathbf{p}_k] \in \mathbb{R}^{d \times k}$. A new observation $\mathbf{x} \in \mathbb{R}^d$ is mapped to a k-dimensional representation via

$$\tilde{\mathbf{x}} = \mathbf{P}_k^T(\mathbf{x} - \boldsymbol{\mu}), \tag{2.9}$$

where $\boldsymbol{\mu}$ is the global mean estimated from the training corpus. The resulting coordinates $\tilde{\mathbf{x}}$ are decorrelated: $\mathrm{Cov}(\tilde{\mathbf{x}}) = \mathrm{diag}(\lambda_1, \ldots, \lambda_k)$. The fraction of total variance preserved by the projection is

$$R(k) = \frac{\sum_{i=1}^{k} \lambda_i}{\sum_{i=1}^{d} \lambda_i}. \tag{2.10}$$

Optimality and geometric interpretation. Among all rank-k linear projections, PCA minimises the mean-squared reconstruction error and maximises the variance of the projected data (Eckart–Young theorem [26]). Geometrically, the projected subspace aligns with the axes of the data ellipsoid along which the observations spread most.

Application to high-dimensional embeddings. When the original dimension d is large (e.g. $d = 768$ for DINOv2 ViT-B/14) but the data occupy a low-dimensional subspace—as evidenced by rapid eigenvalue decay—reducing to $k \ll d$ dimensions before downstream classification offers two complementary benefits: (i) it concentrates the discriminative signal in fewer coordinates, potentially improving the accuracy of subsequent covariance estimators; and (ii) it reduces computational cost by a factor of $\mathcal{O}((d/k)^2)$ for quadratic (Mahalanobis) classifiers. The trade-off is that components beyond k are discarded, which may remove residual discriminative information if the spectrum does not decay quickly enough.

Spectral Decay and Effective Dimensionality

The practical validity of high-dimensional Gaussian modeling is further supported by examining the spectral structure of the feature space through Principal Component Analysis (PCA). The covariance of the class-mean-centred embeddings can be decomposed as $\mathbf{\Sigma} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T$, where the diagonal of $\mathbf{\Lambda}$ contains the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$ and $\mathbf{P}$ holds the corresponding orthonormal eigenvectors.

Variance concentration. If the eigenvalue spectrum decays steeply—most variance is explained by the first $k \ll d$ components—then the effective dimensionality of the feature distribution is much lower than the nominal d. Concretely, the fraction of variance explained by the leading k components is $\sum_{i=1}^{k} \lambda_i \,/\, \sum_{i=1}^{d} \lambda_i$.

Univariate Gaussianity

A second layer of theoretical justification involves examining the univariate properties of individual feature components.

Kurtosis This statistic measures the "tailedness" and peakedness of a probability distribution relative to a normal distribution [4]. As illustrated in Figure 2.3, a perfectly Gaussian distribution has an excess kurtosis of zero (mesokurtic). Distributions with high kurtosis (leptokurtic, $Kurtosis > 0$) exhibit sharper peaks and heavier tails, indicating a higher concentration of data around the mean and a greater likelihood of extreme outliers. Conversely, low kurtosis (platykurtic, $Kurtosis < 0$) indicates a flatter peak and thinner tails. Empirical evidence suggests that for many foundation model embeddings, kurtosis values typically fall within the range of $[-1, 1]$, justifying the Gaussian approximation.

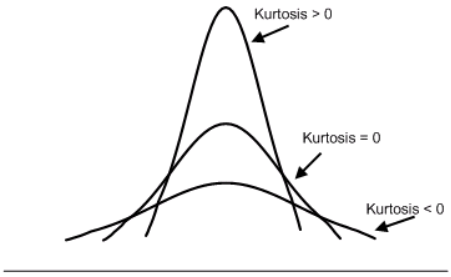


Figure 2.3: Visual representation of kurtosis levels: leptokurtic (> 0), mesokurtic ($= 0$) and platykurtic (< 0) distributions [4].

Skewness This measure quantifies the asymmetry of a distribution around its mean. As shown in Figure 2.4, a symmetric distribution (center) is a fundamental assumption of the Gaussian model, implying that the mean $\boldsymbol{\mu}_c$, median, and mode coincide, providing an adequate representation of the class prototype within the embedding space. A positive skew

(right) indicates a distribution with a long tail extending toward higher values, while a negative skew (left) indicates a long tail extending toward lower values. Significant skewness would suggest that the class-conditional mean is no longer a central representation of the feature distribution.

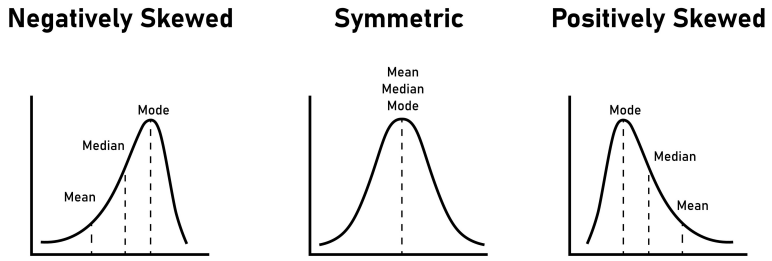


Figure 2.4: Comparison of negatively-skewed, symmetric and positively-skewed distributions [5].

Multivariate Gaussianity

Per-dimension normality is necessary but not sufficient for the joint d-dimensional distribution to be Gaussian. The Henze–Zirkler (HZ) test [1] is an omnibus multivariate diagnostic that addresses this gap by comparing the empirical characteristic function of the data against the theoretical characteristic function of a multivariate normal distribution.

Given n centred samples $\mathbf{z}_i = \mathbf{x}_i - \hat{\boldsymbol{\mu}}_c$ and a smoothing bandwidth $b > 0$, the test statistic is built from three terms evaluated using the squared Mahalanobis distances $d_M^2(\cdot, \cdot)$ induced by the class covariance $\boldsymbol{\Sigma}_c$:

$$A = \frac{1}{n^2} \sum_{j=1}^{n} \sum_{k=1}^{n} e^{-\frac{b^2}{2} d_M^2(\mathbf{z}_j, \mathbf{z}_k)}, \tag{2.11}$$

$$B = \frac{2}{n} (1 + b^2)^{-d/2} \sum_{j=1}^{n} e^{-\frac{b^2}{2(1+b^2)} d_M^2(\mathbf{z}_j, \mathbf{0})}, \tag{2.12}$$

$$C = (1 + 2b^2)^{-d/2}. \tag{2.13}$$

Term A measures the average pairwise kernel similarity of the observed samples; term B quantifies how consistently the empirical density decays away from the class centroid compared to a Gaussian; and term C is the asymptotic self-similarity constant of a true d-dimensional Gaussian. The final statistic is:

$$T_{n,b} = n \cdot [A - B + C]. \tag{2.14}$$

Under the null hypothesis of multivariate normality, $T_{n,b}$ converges to a fixed value close to 1; departures above 1 indicate that the data is more structured (tighter clusters or heavier tails) than a Gaussian. The bandwidth b controls the resolution of the characteristic function comparison and is set optimally as a function of n and d (see Section 3.4.2).

2.2.4 Advanced Covariance Estimation in High Dimensions

In many modern applications, such as image feature analysis using DINOv2, the dimensionality d (often in the hundreds or thousands) often exceeds the sample size n. In this "large d, small n" regime, the empirical sample covariance matrix (SCM) S is a poor estimator: its eigenvalues are biased, and the matrix is singular if $n < d$. This singularity prevents the inversion of the matrix required for the quadratic form in classification and density estimation.

Regularization and Numerical Stability

To stabilize the inversion of the covariance matrix $\mathbf{\Sigma}_c$ for each class, regularization techniques are employed. A primary method is diagonal loading, which adds a small constant ε (e.g., 10^{-6}) to the diagonal elements:

$$\mathbf{\Sigma}_{reg} = \mathbf{\Sigma}_c + \varepsilon I \tag{2.15}$$

This adjustment ensures that the matrix remains strictly positive definite. However, for higher numerical precision in calculating the inverse (precision matrix) and the log-determinant, we utilize Spectral Filtering via eigendecomposition.

Spectral Filtering and Precision Matrix Inversion

For each class, the system computes the eigenvectors P and eigenvalues λ such that $\mathbf{\Sigma}_{reg} = P\Lambda P^T$. To ensure stability against near-zero eigenvalues that survive initial regularization, we apply a floor: $\hat{\lambda} = \max(\lambda, \varepsilon)$. The precision matrix (inverse covariance) is then reconstructed as:

$$\mathbf{\Sigma}_c^{-1} = P\text{diag}\left(\frac{1}{\hat{\lambda}_1}, \ldots, \frac{1}{\hat{\lambda}_d}\right) P^T \tag{2.16}$$

The log-determinant, required for the Gaussian log-likelihood, is computed directly from these filtered eigenvalues to avoid underflow:

$$\ln|\mathbf{\Sigma}_c| = \sum_{j=1}^{d} \ln(\hat{\lambda}_j) \tag{2.17}$$

Finally, the normalization constant $\ln A_c$ (representing the constant term of the multivariate Gaussian log-density) is stored:

$$\ln A_c = -\frac{1}{2}\left(d\ln(2\pi) + \ln|\mathbf{\Sigma}_c|\right) \tag{2.18}$$

This approach eliminates "noise floor" issues that would otherwise cause the matrix to become singular or produce infinite values during log-likelihood estimation.

Linear Shrinkage Estimators

Linear shrinkage methods propose a compromise between the SCM S and a highly structured target matrix F, such as the identity matrix or a diagonal matrix of sample variances.

Ledoit-Wolf Estimator The Ledoit-Wolf estimator [27] determines an optimal shrinkage intensity ρ to minimize the expected Frobenius norm between the estimate and the true covariance matrix $\mathbf{\Sigma}$. It is defined as:

$$\hat{\mathbf{\Sigma}}_{LW} = (1-\rho)S + \rho F \tag{2.19}$$

Ledoit-Wolf is computationally efficient and provides an asymptotically optimal estimate without assuming a specific data distribution.

Oracle Approximating Shrinkage (OAS) The OAS estimator [28] refines the Ledoit-Wolf approach by assuming the data follows a Gaussian distribution. This assumption allows for a more precise estimation of the shrinkage intensity, particularly for small sample sizes. The OAS shrinkage parameter ρ_{OAS} is given by:

$$\rho_{OAS} = \min\left(1, \frac{(1-\frac{2}{d})\mathrm{tr}(S^2) + \mathrm{tr}^2(S)}{(n+1-\frac{2}{d})(\mathrm{tr}(S^2) - \frac{\mathrm{tr}^2(S)}{d})}\right) \tag{2.20}$$

OAS is specifically designed to minimize Mean Squared Error (MSE) under Gaussianity and often outperforms standard shrinkage when the data distribution is approximately normal.

Frobenius Norm and Matrix Distance

The **Frobenius norm** of a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ is defined as the square root of the sum of its squared entries:

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2} = \sqrt{\mathrm{tr}(\mathbf{A}^T\mathbf{A})}. \tag{2.21}$$

It is the natural extension of the Euclidean vector norm to matrices and satisfies all norm axioms (non-negativity, homogeneity, triangle inequality). For symmetric positive semi-definite

matrices such as covariance matrices, it has two useful equivalent forms. Writing the eigendecomposition $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^T$,

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{d} \lambda_i^2}, \tag{2.22}$$

so the Frobenius norm measures the ℓ_2 norm of the eigenvalue vector.

Frobenius distance between covariance matrices. Given two regularised covariance matrices $\hat{\mathbf{\Sigma}}_i, \hat{\mathbf{\Sigma}}_j \in \mathbb{R}^{d\times d}$, their *Frobenius distance* is:

$$D_{ij}^F = \|\hat{\mathbf{\Sigma}}_i - \hat{\mathbf{\Sigma}}_j\|_F = \sqrt{\mathrm{tr}\Big[(\hat{\mathbf{\Sigma}}_i - \hat{\mathbf{\Sigma}}_j)^2\Big]}. \tag{2.23}$$

This quantity measures how geometrically similar the two class-conditional distributions are in their second-order structure, independently of mean displacement. A small D_{ij}^F indicates that the two classes share nearly the same spread and correlation structure; a large value indicates that one class occupies a substantially different region of the feature covariance landscape.

Relationship to Ledoit-Wolf optimality. The Ledoit-Wolf shrinkage estimator [27] is derived precisely by minimising the *expected* Frobenius loss between the estimated and the true covariance:

$$\hat{\rho}^* = \arg\min_{\rho} \ \mathbb{E}\Big[\|\hat{\mathbf{\Sigma}}_{LW}(\rho) - \mathbf{\Sigma}\|_F^2\Big], \tag{2.24}$$

which makes D^F the natural loss criterion for evaluating covariance estimator quality. In the HDLSS regime ($n_c \ll d$), the sample covariance deviates substantially from the truth in Frobenius sense, and shrinkage reduces this deviation by pulling outlying eigenvalues toward a structured target.

Sensitivity to regularisation. When diagonal loading $\hat{\mathbf{\Sigma}}_c^{\text{reg}} = \hat{\mathbf{\Sigma}}_c + \varepsilon I$ is applied uniformly across classes, any class-to-class difference in $\hat{\mathbf{\Sigma}}_c$ is preserved exactly in D_{ij}^F, since the εI terms cancel:

$$D_{ij}^F = \|\hat{\mathbf{\Sigma}}_i^{\text{reg}} - \hat{\mathbf{\Sigma}}_j^{\text{reg}}\|_F = \|\hat{\mathbf{\Sigma}}_i - \hat{\mathbf{\Sigma}}_j\|_F. \tag{2.25}$$

However, when $n_c \ll d$ and the majority of eigenvalues of $\hat{\mathbf{\Sigma}}_c$ are near zero, $\hat{\mathbf{\Sigma}}_c \approx \mathbf{0}$ for most directions, so $\hat{\mathbf{\Sigma}}_i - \hat{\mathbf{\Sigma}}_j \approx \mathbf{0}$ and $D_{ij}^F \to 0$ regardless of class identity. This regularisation-collapse effect—analysed empirically in Section 4.3.2—provides a diagnostic for whether full covariance estimation can provide meaningful discriminative structure beyond the diagonal approximation.

2.3 Continual Learning

Continual Learning (CL), also known as lifelong learning, constitutes the core focus of this work. It represents a machine learning paradigm designed to handle the progressive acquisi-

tion of knowledge from sequences of tasks or data streams over time. As illustrated in Figure 2.5, a **task** in this context is defined as a specific learning experience or dataset –such as Task A, B, or C. Each task is potentially characterized by unique data distributions or objectives, as symbolized by the varying shapes (e.g., circles vs. hearts) above the task labels.

Unlike conventional training approaches that rely on a fixed dataset and a stationary environment, CL emphasizes the arrival of these tasks at distinct intervals along the **time** axis. This sequential arrival requires the model to be inherently non-stationary, adapting its internal representations as environments evolve. The primary challenge in this paradigm lies in the ongoing integration of new information from a current task (e.g., Task B) while simultaneously preserving the knowledge previously acquired from Task A.

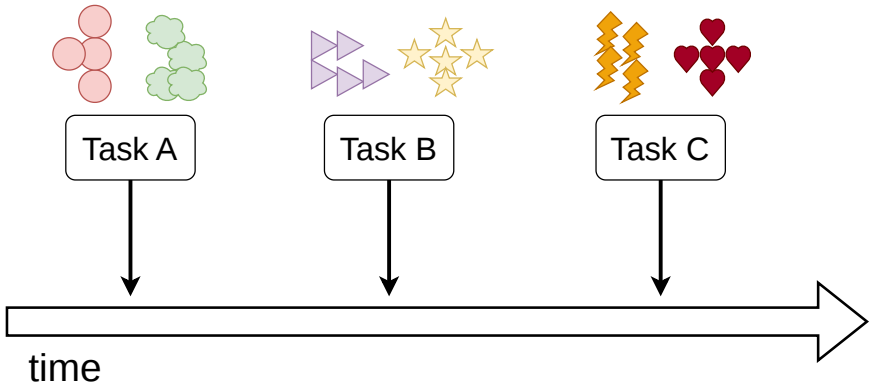


Figure 2.5: A sequential task arrival model for Continual Learning, depicting the transition from Task A to Task C over time, where each task represents a unique set of features or data distributions.

Without specific architectural or algorithmic safeguards, this sequential learning process inevitably triggers **catastrophic forgetting** [29]. This phenomenon occurs when the assimilation of new knowledge leads to the degradation or total loss of prior learning. As illustrated in Figure 2.6, the dilemma of forgetting becomes apparent when a *New Object Class* is introduced to an existing classifier, forcing a choice between two traditional but flawed strategies:

- **Fine-Tuning:** Updating the model using only the current task data is computationally efficient but leads directly to catastrophic forgetting [29]. Because the optimization process focuses solely on the new objective, the internal representations associated with previously learned classes are overwritten, causing the model to lose its original predictive capabilities.
- **Retraining from Scratch:** While joint training on the full historical dataset ($\mathcal{D}_{old} \cup \mathcal{D}_{new}$) technically prevents forgetting, it faces two significant hurdles. First is the issue of **Data Privacy and Availability**: in many real-world scenarios, the original data $\mathcal{D}_{old}$ is often deleted or inaccessible due to privacy regulations. Second is the **Computational Bottleneck**: as the system's lifespan increases, the cumulative dataset grows, making the energy and time required to retrain the model unsustainable in continuously evolving environments.

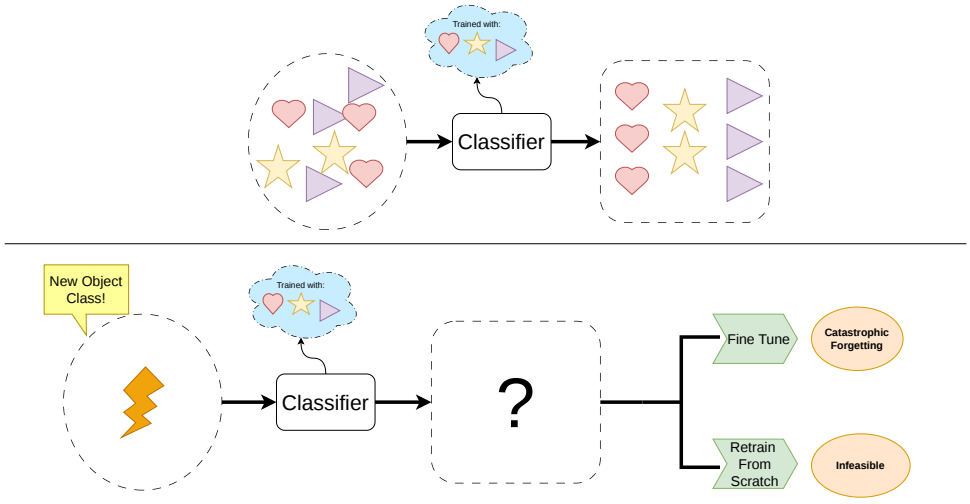


Figure 2.6: Standard Object Classifier comparison. The upper path shows standard inference, while the lower path illustrates the dilemma when a new class is introduced: the trade-off between catastrophic forgetting via fine-tuning and the infeasibility of retraining from scratch.

2.3.1 Elasticity and Plasticity

A central concept in continual learning is the balance between elasticity and plasticity. **Plasticity** refers to the ability of a model to adapt rapidly to new information, ensuring that it can effectively learn novel tasks. **Elasticity** (or stability), in contrast, emphasizes preserving the parameters and representations that are crucial for retaining old knowledge [10]. Achieving an appropriate equilibrium between these two properties is known as the **stability–plasticity dilemma** [29].

2.3.2 Main Paradigms

As shown in Figure 2.7, two principal paradigms define how continual learning problems are formulated. **Task-incremental learning (Task-IL)** assumes that the task identity is available during inference, allowing for task-specific components to be isolated [30]. **Class-incremental learning (Class-IL)**, on the other hand, does not provide the task identity at test time, requiring the model to distinguish between all classes seen across all tasks simultaneously [6].

2.3.3 Mitigation Strategies

To address the stability-plasticity dilemma, CL strategies are generally categorized into four main families: rehearsal-based, regularization-based, architecture-based, and representation-based (specifically prompt-based) methods.

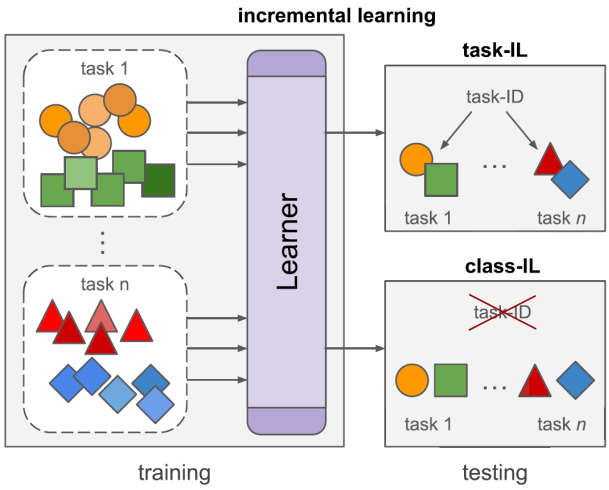


Figure 2.7: Class and Task Incremental Learning. Image taken from [6].

Rehearsal Methods

To preserve past performance, these methods reuse samples from previously learned tasks while training on the current task.

- **Experience Replay (ER)** [31]: Maintains a small episodic memory buffer containing a sample of old, representative examples to be interleaved with new data.
- **Average Gradient Episodic Memory (A-GEM)** [32]: Improves efficiency by using the buffer to constrain the direction of parameter updates, ensuring they do not interfere with past tasks.
- **Dark Experience Replay (DER)** [33]: Enhances ER by incorporating knowledge distillation on the logits of replayed examples to preserve the "dark knowledge" of the model.
- **Generative Replay** [34]: A non-data-storing method where a generator is trained to produce synthetic features or samples of previously learned classes, avoiding the privacy concerns of raw data storage.

Regularization Methods

These methods limit updates to parameters identified as important for prior learning by introducing penalty terms in the loss function.

- **Elastic Weight Consolidation (EWC)** [35]: Identifies critical parameters using the Fisher Information Matrix and restricts their movement during new task training.
- **Synaptic Intelligence (SI)** [36]: Calculates importance weights online by accumulating their contribution to the reduction of the loss.

- **Learning Without Forgetting (LwF)** [37]: A functional regularization method that uses knowledge distillation to ensure the model's outputs for new data remain consistent with the previous model's predictions.
- **Adaptive Plasticity Improvement (API)** [10]: Enforces an update direction orthogonal to the important subspaces of previous tasks using Dual Gradient Projection Memory (Dual-GPM).

Architecture-based Methods

These methods mitigate forgetting by modifying the network architecture to increase or reorganize its capacity.

- **Parameter Isolation**: Identifies and freezes parameters important for previous tasks –often through pruning or gradient masking– so only a subset of weights is updated for the current task [38].
- **Dynamic Expansion**: The architecture grows as new tasks arrive. For instance, [39] uses an expandable super-feature extractor followed by pruning, while [40] expands the network only when the current capacity is insufficient.

Representation and Prompt-based Methods

The emergence of foundation models (e.g., ViTs) has enabled parameter-efficient fine-tuning (PEFT) where the backbone remains frozen and learning is contained within small "prompts".

- **Learning to Prompt (L2P)** [41]: Uses a prompt pool where prompts are selected based on the similarity between input embeddings and learned keys.
- **Dual Prompt** [42]: Refines L2P by separating representations into fixed G-prompts (General) and task-specific E-prompts (Expert).
- **CODA-Prompt** [43]: Introduces a fully differentiable method where the prompt is a weighted sum of components dynamically calculated via attention mechanisms.
- **Advanced Prompting**: Recent methods like **InFLoRA** [44] and **DOLFIN** [45] apply low-rank adaptation and distributed learning, while **APT** [46] injects key-value tokens directly into transformer layers to reduce inference overhead.

Classifier Adaptation on Frozen Representations

A complementary line of work investigates classifier adaptation on top of frozen feature extractors, remaining fundamentally discriminative or metric-based. **SLCA** [47] proposes a slow learner with classifier alignment for continual learning on pre-trained models, enabling efficient incremental learning without backpropagation through the backbone. **ACIL** [48]

introduces analytic class-incremental learning with closed-form updates for linear classifiers, providing absolute memorisation and privacy protection without exemplar storage. **FeCAM** [49] exploits the heterogeneity of class distributions in exemplar-free continual learning by performing classification using a Mahalanobis distance metric, highlighting the importance of modelling heterogeneous feature distributions. In contrast to these approaches, which remain fundamentally discriminative or metric-based, the framework proposed in this thesis adopts a fully probabilistic perspective and formulates class-incremental learning as recursive estimation of class-conditional densities in a stationary embedding space.

Chapter 3

Methodology

3.1 Proposed Architecture

As shown in 3.1, our CIL framework is built upon a simple yet structurally motivated decomposition: (i) a frozen feature extractor based on a visual foundation model, and (ii) a probabilistic classification layer that models each object category through an independent Gaussian distribution in the embedding space. Given an input image, the feature extractor maps it to a fixed high-dimensional representation. For each class, we maintain a class-conditional multivariate Gaussian characterized by its mean vector and covariance matrix, estimated incrementally from observed samples. Classification is performed according to the maximum likelihood principle, assigning the input to the class whose Gaussian model yields the highest likelihood under the stationary embedding representation. This modular separation between representation learning and statistical modeling enables incremental updates without backpropagation or parameter adaptation: adding a new class requires only estimating its distribution in the fixed feature space, while previously learned classes remain unaffected at the representation level.

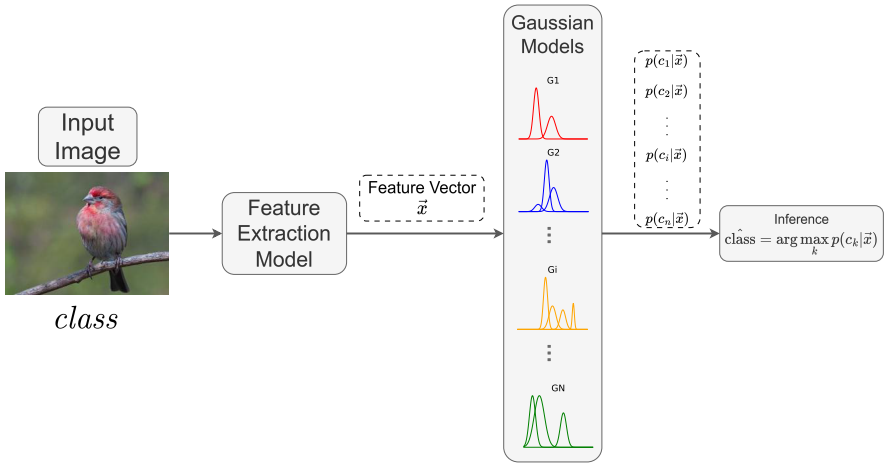


Figure 3.1: Block diagram of the proposed approach.

3.1.1 Likelihood-Based Classification

Given an input image $x \in \mathcal{X}$, the frozen foundation model $\mathcal{F}$ produces a fixed-dimensional embedding

$$\mathbf{z} = \mathcal{F}(x) \in \mathbb{R}^d, \quad d = 768, \tag{3.1}$$

and all learning is performed directly in the stationary feature space $\mathbb{R}^d$. We adopt a generative modelling framework and assume class-conditional Gaussian feature distributions:

$$p(\mathbf{z} \mid y = c) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_c, \Sigma_c), \tag{3.2}$$

where $\boldsymbol{\mu}_c \in \mathbb{R}^d$ and $\Sigma_c \succ 0$ are the class-specific mean and covariance matrix, estimated incrementally from observed samples. Classification is performed via Maximum A Posteriori (MAP) inference:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} \left[\log p(\mathbf{z} \mid y = c) + \log \pi_c \right], \tag{3.3}$$

where $\pi_c = p(y = c)$ is the class prior. Under a uniform prior and the Gaussian density model, the decision rule is equivalent to the quadratic discriminant:

$$\hat{y} = \arg\min_{c \in \mathcal{Y}} \left[(\mathbf{z} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{z} - \boldsymbol{\mu}_c) + \log |\Sigma_c| \right]. \tag{3.4}$$

This quadratic decision function is the core of the proposed framework. The Mahalanobis term $(\mathbf{z} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{z} - \boldsymbol{\mu}_c)$ encodes the *shape* of the class distribution while $\log |\Sigma_c|$ encodes its *volume*. The covariance parameterisations introduced in Section 3.2 are specialisations of Equation (3.4) that differ in the structure imposed on Σ_c; the interplay between the shape and volume components is the central design axis of the covariance estimation study and is examined in detail in the analysis of results.

Adding a new class to the system requires only estimating the sufficient statistics $(\boldsymbol{\mu}_c, \Sigma_c)$ in the fixed feature space; previously learned class statistics are never modified. This formulation therefore enables genuinely incremental updates without backpropagation, replay buffers, or exemplar storage.

3.1.2 Datasets and Experimental Protocol

We follow the standard Continual Learning (CL) protocol where the model learns a sequence of disjoint tasks and calculates the average accuracy and the forgetting. The specific configurations for the datasets are as follows:

- **Split CIFAR-100:** This dataset is a widely utilized benchmark in the continual learning literature. It partitions the original CIFAR-100 [23] dataset into 10 disjoint tasks, with each task containing 10 distinct classes.
- **Split ImageNet-R:** This benchmark is constructed from ImageNet-R [22], a dataset containing renditions of ImageNet classes such as art, cartoons, and sketches. We randomly divide the 200 classes into 10 sequential tasks, with 20 classes per task, introducing a significant domain shift challenge.

- **Split Caltech-256:** Designed to evaluate the model on higher-resolution images with greater object variability, this benchmark utilises the Caltech-256 [24] dataset. The first 250 of the original 256 categories are used and sequentially divided into 10 disjoint tasks of 25 classes each; the remaining six categories are discarded so that the benchmark has a fixed, balanced number of classes per task. The resulting protocol contains approximately 28 100 images with an average of $\overline{n_c} \approx 112$ training samples per class, which makes the per-class budget the smallest of the three constituent datasets and stresses the high-dimensional covariance estimators introduced in Section 3.2.
- **Split Joint Dataset:** This benchmark serves as a comprehensive stress test by aggregating the 30 tasks from the three aforementioned datasets into a single, heterogeneous stream. The tasks are presented to the system in a randomized order, forcing the model to navigate extreme distribution shifts between natural, artistic, and high-resolution images. This long-horizon sequence is specifically designed to evaluate the model's resilience against catastrophic forgetting over an extended period of diverse learning objectives.

3.1.3 Model Metrics

As mentioned, to quantify performance, SOTA works [41] utilize two standard metrics derived from the test accuracy scores: the average accuracy and the average forgetting.

On the one hand, **Average Accuracy (A_t)**, presented on equation (3.5), measures overall performance after task t. It is the mean test accuracy across all tasks learned up to that point.

$$A_t = \frac{1}{t} \sum_{\tau=1}^{t} S_{t,\tau} \tag{3.5}$$

On the other hand, **Forgetting (F_t)**, presented on equation (3.6), quantifies how much the model has forgotten earlier tasks. It computes, for each past task, how much accuracy has dropped compared to its best historical score, and averages those drops.

$$F_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} \left(\max_{k \leq t-1} S_{k,\tau} - S_{t,\tau} \right) \tag{3.6}$$

Regarding forgetting, the proposed framework does not exhibit forgetting in the conventional sense, as previously learned class statistics are never modified when new classes are introduced. The measured forgetting values therefore primarily reflect the increased classification difficulty as the number of competing classes grows, rather than degradation of previously learned representations.

3.2 Covariance Estimation

As mentioned, due to the high the dimensionality of the extracted feature, the empirical sample covariance matrix (SCM) is a poor estimator: its eigenvalues are biased, and the matrix is singular if $n < d$. Our benchmarks are tested using different convariance estimation methods:

3.2.1 Diagonal Covariances

In this configuration, we assume that the features within each class c are independent, leading to a diagonal covariance matrix $\Sigma_c = \text{diag}(\sigma_{c,1}^2, \sigma_{c,2}^2, \ldots, \sigma_{c,d}^2)$. This simplification significantly reduces the number of parameters to estimate from $O(d^2)$ to $O(d)$, which is particularly advantageous when the dimensionality $d = 768$ is large relative to the number of samples per class.For each class $c \in \{1, \ldots, C\}$, the mean vector μ_c and the diagonal elements of Σ_c are computed as:

$$\mu_c = \frac{1}{n_c} \sum_{i=1}^{n_c} x_i, \quad \sigma_{c,j}^2 = \frac{1}{n_c} \sum_{i=1}^{n_c} (x_{i,j} - \mu_{c,j})^2 + \varepsilon \tag{3.7}$$

where $\varepsilon = 10^{-6}$ is a small constant added to ensure numerical stability and strictly positive variances. The log-likelihood of a test sample $\mathbf{x}$ belonging to class c is determined by the discriminant function:

$$g_c(\mathbf{x}) = \ln P(\mathbf{x}|c) + \ln P(c) \tag{3.8}$$

Under the assumption of a uniform prior $P(c)$ and Gaussian distribution, this simplifies to:

$$g_c(\mathbf{x}) = -\frac{1}{2} \sum_{j=1}^{d} \left(\ln(2\pi\sigma_{c,j}^2) + \frac{(x_j - \mu_{c,j})^2}{\sigma_{c,j}^2} \right) \tag{3.9}$$

In the implementation, this is computed using a pre-calculated constant $\ln A_c = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \sum \ln(\sigma_{c,j}^2)$ and the squared Mahalanobis distance weighted by the inverse of the diagonal variances.

Score component analysis. To characterise how each term contributes to classification, the two components of the score are evaluated separately on the held-out validation partition. Gaussian parameters are estimated exclusively from the training partition and remain fixed. At evaluation time, every validation sample is scored against *all* C classes simultaneously, not only its ground-truth class. This produces a C-dimensional vector of volume terms $\{\ln A_c\}$ (one value per class, independent of any test sample) and a full $N_{\text{val}} \times C$ matrix for both $d_{\text{mah}}(\mathbf{x}, c)$ and $\text{score}(\mathbf{x}, c)$. Inspecting the marginal distributions of these matrices reveals

whether the classifier's decisions are dominated by the volume penalty (covariance geometry) or by the shape term (distance to the class mean), and whether the two components interact as expected under the Gaussian model.

For the **diagonal-covariance** estimator the full $N_{\text{val}} \times C$ matrix is computed in closed form via vectorised operations, since the precision matrix is diagonal. For the **full-covariance** estimator, the analysis is carried out on a random subsample of the validation partition to keep the computation tractable.

3.2.2 Full Covariances

To account for the dependencies between different dimensions of the feature vector, we estimate a full covariance matrix $\Sigma_c \in \mathbb{R}^{d\times d}$ for each class c. While this provides a richer representation of the class distribution, the estimation of d^2 parameters (where $d = 768$) is prone to ill-conditioning when the number of samples n_c is limited.The mean vector μ_c and the empirical covariance Σ_c are calculated as:

$$\mu_c = \frac{1}{n_c}\sum_{i=1}^{n_c} x_i, \quad \Sigma_c = \frac{1}{n_c - 1}\sum_{i=1}^{n_c}(x_i - \mu_c)(x_i - \mu_c)^T \tag{3.10}$$

To ensure the matrix is invertible and to improve numerical stability, a regularizatio is applied by adding a small identity term to the diagonal: $\hat{\Sigma}_c = \Sigma_c + \varepsilon I$, with $\varepsilon = 10^{-6}$.The implementation performs the inversion and determinant calculation using the eigendecomposition of the symmetric covariance matrix, $\hat{\Sigma}_c = P\Lambda P^T$, where Λ contains the eigenvalues λ_j. To further ensure stability, eigenvalues are clipped to a minimum value of 10^{-6}. The inverse and the log-determinant are then computed as:

$$\hat{\Sigma}_c^{-1} = P\Lambda^{-1}P^T, \quad \ln|\hat{\Sigma}_c| = \sum_{j=1}^{d}\ln(\lambda_j) \tag{3.11}$$

The discriminant function for classification is the full quadratic form:

$$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T\hat{\Sigma}_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.12}$$

where $\ln A_c = -\frac{1}{2}(d\ln(2\pi) + \ln|\hat{\Sigma}_c|)$.

3.2.3 Covariance Conditioning Analysis

The sample covariance matrix $\hat{\Sigma}_c \in \mathbb{R}^{d\times d}$ estimated from n_c training samples is rank-deficient whenever $n_c < d$. In our setting $d = 768$ while n_c ranges from approximately 40 to 400 depending on the benchmark, so the matrix always has at most $n_c - 1$ non-zero eigenvalues out of $d = 768$, leaving $d - (n_c - 1)$ eigenvalues that are structurally zero before any regularisation is applied. To quantify this ill-conditioning we analyse the eigenvalue spectrum of the regularised matrix $\hat{\Sigma}_c^{\text{reg}} = \hat{\Sigma}_c + \varepsilon I$, $\varepsilon = 10^{-6}$, and compute the following metrics per class.

Condition number. The condition number is defined as the ratio of the largest to the smallest eigenvalue:

$$\kappa(\hat{\Sigma}_c^{\text{reg}}) = \frac{\lambda_{\max}}{\lambda_{\min}}, \tag{3.13}$$

where $\lambda_{\max}$ and $\lambda_{\min}$ are the extremal eigenvalues of $\hat{\Sigma}_c^{\text{reg}}$. A large κ indicates that the matrix is nearly singular in the sense that directions associated with small eigenvalues contribute negligible variance and inflate the Mahalanobis distance. We report $\log_{10}(\kappa)$ to compress the dynamic range.

Effective dimensions. A dimension j is classified as *effective* if $\lambda_j \geq \varepsilon_{\text{thr}} = 10^{-4}$, and as *degenerate* otherwise. The number of degenerate dimensions per class measures directly how many of the $d = 768$ directions are driven to the regularisation floor and thus carry no meaningful covariance information.

Effective rank. The entropy-based effective rank [**?**] captures the spread of the eigenvalue distribution:

$$r_{\text{eff}}(\hat{\Sigma}_c^{\text{reg}}) = \exp\left(-\sum_{j=1}^{d} p_j \ln p_j\right), \qquad p_j = \frac{\lambda_j}{\sum_k \lambda_k}. \tag{3.14}$$

This equals d for a perfectly isotropic matrix ($\lambda_j = \text{const}$) and equals 1 when a single eigenvalue dominates. In practice it provides a soft count of the directions that carry the bulk of the variance.

3.2.4 Cross-Class Frobenius Norm Analysis

To characterise the geometric diversity of the class-conditional covariance matrices after regularisation, we compute all pairwise Frobenius distances:

$$D_{ij}^F = \|\hat{\Sigma}_i^{\text{reg}} - \hat{\Sigma}_j^{\text{reg}}\|_F = \sqrt{\sum_{k,l} (\hat{\sigma}_{i,kl} - \hat{\sigma}_{j,kl})^2}, \tag{3.15}$$

where $\hat{\Sigma}_c^{\text{reg}} = \hat{\Sigma}_c + \gamma I$, $\gamma = 10^{-6}$. The resulting $C \times C$ symmetric matrix $\mathbf{D}^F$ with zero diagonal captures how similar or dissimilar any two class covariances are in the ambient Frobenius geometry.

A small D_{ij}^F indicates that classes i and j share nearly identical second-order statistics, implying that the Mahalanobis-based classifier must rely primarily on the mean displacement $\boldsymbol{\mu}_i - \boldsymbol{\mu}_j$ rather than covariance contrast. Conversely, a large D_{ij}^F indicates that the covariance geometry itself provides additional discriminative information beyond the first-order term.

We summarise the resulting distribution via its mean, standard deviation, and percentile bands, and visualise the full $C \times C$ heatmap for each benchmark. Results are reported in Section 4.3.

3.2.5 Hybrid Covariance

The hybrid covariance estimator decouples the estimation of the Mahalanobis distance from the normalisation constant.For each class c, the full covariance matrix $\hat{\Sigma}_c$ is estimated and regularized as described in the previous section:

$$\hat{\Sigma}_c = \Sigma_c + \varepsilon I \tag{3.16}$$

The quadratic form of the log-likelihood (the Mahalanobis distance) uses the full inverse $\hat{\Sigma}_c^{-1}$, calculated via eigendecomposition to capture the feature interactions. However, the log-determinant—which represents the "volume" of the distribution—is computed using only the diagonal elements of the covariance matrix (the variances):

$$\ln |\Sigma_c|_{\text{diag}} = \sum_{j=1}^{d} \ln(\sigma_{c,j}^2) \tag{3.17}$$

where $\sigma_{c,j}^2$ are the diagonal entries of $\hat{\Sigma}_c$. The resulting discriminant function is defined as:

$$g_c(\mathbf{x}) = -\frac{1}{2}\left(d\ln(2\pi) + \sum_{j=1}^{d} \ln(\sigma_{c,j}^2) + (\mathbf{x}-\boldsymbol{\mu}_c)^T \hat{\Sigma}_c^{-1} (\mathbf{x}-\boldsymbol{\mu}_c) \right) \tag{3.18}$$

This formulation avoids the numerical issues that arise when very small eigenvalues inflate the log-determinant, at the cost of using an approximation for the volume term.

3.2.6 OAS Covariance

Oracle Approximating Shrinkage (OAS) is a shrinkage technique that computes an structurally optimized covariance matrix by finding an optimal compromise between the sample covariance matrix and a structured target (the identity matrix scaled by the average trace).The OAS estimator provides a more accurate estimation of the covariance Σ_c by minimizing the Mean Squared Error (MSE) under the assumption of Gaussianity. The estimated covariance is defined as:

$$\hat{\Sigma}_{\text{OAS},c} = (1-\rho_c)\hat{\Sigma}_{\text{SCM},c} + \rho_c \hat{F}_c \tag{3.19}$$

where ρ_c is the shrinkage coefficient calculated specifically for class c, $\hat{\Sigma}_{\text{SCM},c}$ is the sample covariance matrix, and $\hat{F}_c = \frac{\text{tr}(\hat{\Sigma}_{\text{SCM},c})}{d} I$ is the shrinkage target.In our implementation, we compute the precision matrix $\Psi_c = \hat{\Sigma}_{\text{OAS},c}^{-1}$ directly via the OAS algorithm. This precision matrix is then used to calculate the Mahalanobis distance in the log-likelihood discriminant function:

$$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_c)^T \Psi_c (\mathbf{x}-\boldsymbol{\mu}_c) \tag{3.20}$$

where $\ln A_c$ is the normalization constant derived from the log-determinant of the OAS-estimated covariance matrix, evaluated numerically via the stable log-determinant of the eigendecomposition. This approach ensures that the model captures class-specific correlations while remaining robust to the "curse of dimensionality."

3.2.7 Ledoit-Wolff

The Ledoit-Wolf Diagonal Anchor estimator addresses the instability of the sample covariance matrix S by shrinking it toward a target matrix T.

The selection of a target matrix T represents a prior assumption regarding the feature distribution. Both methods stabilize the estimator by pulling the noisy off-diagonal elements of the sample covariance matrix S toward zero, but they differ in how they treat the diagonal elements (the variances).

Diagonal Target

The diagonal anchor is defined as $T_{\text{diag}} = \text{diag}(s_{11}, s_{22}, \ldots, s_{dd})$. This approach assumes that the features are independent but allows for heteroscedasticity, meaning each dimension can have a different variance. Shrinkage toward this target preserves the scale of each individual feature as observed in the sample data while suppressing only the correlations. This is particularly useful when certain dimensions of the DINOv2 embeddings are naturally more informative or volatile than others.

Identity Target

The identity anchor is more restrictive, defined as $T_{\text{ident}} = \mu I$, where $\mu = \frac{1}{d}\text{tr}(S)$ is the average variance across all dimensions. This target assumes that the features are not only independent but also share a common variance (homoscedasticity). The resulting distribution is spherical. Because the embeddings used in this work are L2-normalized, the variances across dimensions tend to be similar, making the identity anchor a highly robust regularizer that prevents any single dimension from disproportionately influencing the distance metric.

Structural Summary

The fundamental difference between these two estimators lies in the flexibility of the variance components:

- **Diagonal Anchor:** $\hat{\Sigma}_{ij} = (1-\gamma)s_{ij}$ for $i \neq j$, and $\hat{\Sigma}_{ii} = s_{ii}$. It yields an axis-aligned ellipsoidal distribution.
- **Identity Anchor:** $\hat{\Sigma}_{ij} = (1-\gamma)s_{ij}$ for $i \neq j$, and $\hat{\Sigma}_{ii} = (1-\gamma)s_{ii} + \gamma\mu$. It yields a distribution that is pulled toward a perfectly spherical geometry.

3.2.8 Shared Covariance

The Shared Covariance framework assumes that the internal noise or feature variability is homoscedastic across all classes. In this paradigm, all classes share a single covariance matrix Σ, which captures the global structure of the feature space. This significantly reduces the total number of parameters to be estimated compared to class-specific models. Depending on the constraints placed on Σ, we evaluate three distinct variations:

Shared Diagonal Covariance

By enforcing a global diagonal covariance matrix $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_d^2)$, we simplify the model to assume that feature dimensions are independent. The shared variance for each feature j is estimated by pooling the squared residuals from all C classes:

$$\sigma_j^2 = \frac{1}{N} \sum_{c=1}^{C} \sum_{i=1}^{n_c} (x_{i,j} - \mu_{c,j})^2 + \varepsilon \tag{3.21}$$

where N is the total number of training samples and $\mu_{c,j}$ is the mean of class c for feature j. The classification of a test sample $\mathbf{x}$ reduces to finding the class c that minimizes the weighted Euclidean distance:

$$c^* = \arg\min_c \sum_{j=1}^{d} \frac{(x_j - \mu_{c,j})^2}{\sigma_j^2} \tag{3.22}$$

Shared Full Covariance

In the shared full covariance model (equivalent to the generative assumption of Linear Discriminant Analysis), we estimate a dense matrix $\Sigma \in \mathbb{R}^{d \times d}$ that captures the correlations between all feature dimensions. To ensure numerical stability in high-dimensional spaces, we utilize the Oracle Approximating Shrinkage (OAS) estimator to compute the pooled covariance. The classification rule utilizes the shared precision matrix $\Psi = \Sigma^{-1}$ to minimize the full squared Mahalanobis distance:

$$c^* = \arg\min_c (\mathbf{x} - \boldsymbol{\mu}_c)^T \Psi (\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.23}$$

Shared Hybrid Covariance

The shared hybrid approach serves as a regularized middle ground. It utilizes the full shared precision matrix Ψ for the quadratic distance calculation to capture feature interactions, but simplifies the normalization constant (the log-determinant) by considering only the diagonal

elements of the shared matrix. This prevents the classification scores from being dominated by extreme eigenvalues while still benefiting from the descriptive power of the full correlation structure.

3.2.9 Mixture of Gaussians

In order to capture potential multi-modality within the feature space of a single class, we employ Gaussian Mixture Models (GMM). We implemented the following models using the previously defined diagonal, full and hybrid covariances approaches.

Mixture of 2 Gaussians

Forcing each class to be defined as a GMM with $K = 2$ components, each class c can be modeled by a probability density function:

$$p(\mathbf{x}|c) = \sum_{k=1}^{K} \pi_{c,k} \mathcal{N}(x|\mu_{c,k}, \Sigma_{c,k}) \tag{3.24}$$

where $\pi_{c,k}$ represents the mixing weight of the k-th component for class c, subject to $\sum \pi_{c,k} = 1$. In this specific variant, the covariance matrices $\Sigma_{c,k}$ are restricted to be diagonal to maintain computational efficiency and prevent over-fitting in the high-dimensional DINOv2 space.The parameters $\{\pi_{c,k}, \mu_{c,k}, \Sigma_{c,k}\}$ are estimated using the Expectation-Maximization (EM) algorithm. To ensure numerical stability, a small constant $\varepsilon = 10^{-5}$ is added to the diagonal of each covariance matrix. The log-likelihood for a test sample $\mathbf{x}$ is calculated using the *LogSumExp* operator to avoid underflow:

$$\ln p(\mathbf{x}|c) = \text{lnaddexp}\left(\ln(\pi_{c,1}\mathcal{N}(x|\mu_{c,1}, \Sigma_{c,1})), \ln(\pi_{c,2}\mathcal{N}(x|\mu_{c,2}, \Sigma_{c,2}))\right) \tag{3.25}$$

In the implementation, this is expanded as:

$$\ln p(\mathbf{x}|c) = \ln\left(\exp(\ln \pi_{c,1} + \ln A_{c,1} - \frac{1}{2} d_{M,1}) + \exp(\ln \pi_{c,2} + \ln A_{c,2} - \frac{1}{2} d_{M,2})\right) \tag{3.26}$$

where $d_{M,k}$ is the diagonal Mahalanobis distance to the k-th component and $\ln A_{c,k}$ is the normalization constant. This approach allows the model to better represent classes that may contain significantly different visual sub-types or clusters within the latent embedding space.

Mixture of up-to-4 Gaussians

To adapt to the varying complexity of different classes within the different datasets, we implement a flexible Gaussian Mixture Model where the number of components K_c for each

class c is not fixed. Instead, the model selects the most appropriate number of components $K_c \in \{1, 2, 3, 4\}$ by minimizing the Bayesian Information Criterion (BIC).The BIC is defined as:

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L}) \tag{3.27}$$

where n is the number of samples in the class, k is the number of free parameters to be estimated, and $\hat{L}$ is the maximized value of the likelihood function. By using diagonal covariance matrices, the number of parameters per component is kept low ($2d + 1$), allowing the selection process to favor multi-modal distributions only when the increase in likelihood justifies the additional parameters. For each component k in class c, the diagonal covariance $\Sigma_{c,k} = \text{diag}(\sigma^2_{c,k,1}, \ldots, \sigma^2_{c,k,d})$ is regularized with $\varepsilon = 10^{-6}$. The classification of a test sample $\mathbf{x}$ involves calculating the weighted log-likelihood across all active classes:

$$\ln p(\mathbf{x}|c) = \text{LSE}_{k=1}^{K_c} \left(\ln \pi_{c,k} - \frac{1}{2} \left(\sum_{j=1}^{d} \ln(2\pi\sigma^2_{c,k,j}) + \sum_{j=1}^{d} \frac{(x_j - \mu_{c,k,j})^2}{\sigma^2_{c,k,j}} \right) \right) \tag{3.28}$$

where LSE denotes the LogSumExp operator. This methodology allows for a more granular representation of class distributions—using a single Gaussian for simple, compact classes and up to four Gaussians for classes with high intra-class variance or distinct visual subclusters—while preserving the numerical efficiency of diagonal estimators.

3.3 Final-State Confusion Matrix Evaluation

The confusion matrix at the final incremental state $T_{\max}$ provides a diagnostic complementary to the scalar accuracy and calibration metrics reported in Sections 3.2 and 3.8. While average accuracy collapses all per-class errors into a single scalar, the confusion matrix reveals the full joint distribution of true and predicted labels, exposing structured misclassification patterns—such as visual or semantic similarity between classes—that are otherwise invisible.

3.3.1 Confusion Matrix Definition

Let $\mathcal{D}_{\text{test}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{M}$ be the held-out test partition, where $y_i \in \mathcal{C}_{T_{\max}}$ is the ground-truth label and $\hat{y}_i = \arg\max_c \log p(\mathbf{x}_i \mid c)$ is the label predicted by maximum-likelihood classification. The confusion matrix $\mathbf{M} \in \mathbb{Z}_{\geq 0}^{N \times N}$ is defined entry-wise as

$$M_{c,\hat{c}} = \big|\{i : y_i = c \ \wedge \ \hat{y}_i = \hat{c}\}\big|, \qquad c, \hat{c} \in \{0, \ldots, N-1\}. \tag{3.29}$$

Rows index true classes and columns index predicted classes; a perfect classifier yields a diagonal matrix. For visualisation the row-normalised version

$$\tilde{M}_{c,\hat{c}} = \frac{M_{c,\hat{c}}}{\sum_{\hat{c}'} M_{c,\hat{c}'}} \tag{3.30}$$

is displayed, so that each row sums to one and represents the empirical misclassification distribution for a single true class.

3.3.2 Mahalanobis-Only Inference

The standard Gaussian classifier assigns each class c the log-posterior score

$$s_c(\mathbf{x}) = \log A_c - \tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^\top \Sigma_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c), \tag{3.31}$$

where the log-normalisation constant $\log A_c = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_c|$ depends on the determinant of Σ_c. For the diagonal model, $\log A_c = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\sum_{j=1}^{d}\log\sigma_{c,j}^2$; for the full model, $\log A_c = -\frac{d}{2}\log(2\pi) - \frac{1}{2}\sum_k \log\lambda_{c,k}$, where $\{\lambda_{c,k}\}$ are the eigenvalues of Σ_c.

The *Mahalanobis-only* variant removes the normalisation term entirely, reducing the decision rule to the squared Mahalanobis distance alone:

$$\tilde{s}_c(\mathbf{x}) = -\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^\top \Sigma_c^{-1}(\mathbf{x} - \boldsymbol{\mu}_c). \tag{3.32}$$

The predicted class is still $\hat{y} = \arg\max_c \tilde{s}_c(\mathbf{x})$. Geometrically, each class boundary is now determined solely by the inverse covariance shape (the ellipsoidal metric) rather than by the combination of shape and volume encoded in $\log A_c$.

This ablation isolates the contribution of the log-determinant term. When all classes share the same covariance (e.g. a common LDA-style matrix), $\log A_c$ is identical for all c and cancels in the $\arg\max$; removing it has no effect. Differences emerge when per-class covariances have varying determinants: a class whose feature distribution occupies a large volume has a small $\log A_c$ (large $|\Sigma_c|$), which acts as a penalty in the full Gaussian rule. Removing $\log A_c$ therefore biases decisions toward high-volume classes and can degrade accuracy when class volumes differ meaningfully—as is the case for the diagonal model, where per-dimension variances are learned independently. Conversely, for the full-covariance model on balanced datasets, the eigenvalues tend to be similar across classes (all classes see the same number of training samples and similar covariance structure), so $\log A_c$ is approximately constant and the Mahalanobis-only and full-Gaussian rules are nearly equivalent.

The ablation is implemented by substituting $\log A_c \leftarrow 0$ for all c in the scoring function (Eq. 3.31) while keeping the estimated parameters $\{\boldsymbol{\mu}_c, \Sigma_c^{-1}\}$ unchanged.

3.4 Feature-Space Characterization and Gaussianity Analysis

3.4.1 Probabilistic Classification in High Dimension

In high-dimensional feature spaces such as those produced by foundational vision models, probabilistic classification faces a systematic calibration challenge. While the Maximum A Posteriori (MAP) decision rule correctly identifies the most likely class, the resulting posterior probabilities $p(y = c \mid \mathbf{x})$ often exhibit extreme overconfidence: the model assigns near-unit probability to its top prediction even when that prediction is wrong, effectively masking the true uncertainty of the classification.

Uncertainty Collapse in High Dimensions Under diagonal covariance Gaussian models, the log-likelihood for class c is:

$$\log p(\mathbf{x} \mid y = c) = \ln \mathcal{A}_c - \frac{1}{2} \underbrace{\sum_{i=1}^{d} \frac{(x_i - \mu_{c,i})^2}{\sigma_{c,i}^2}}_{m_c^2(\mathbf{x})} \tag{3.33}$$

where $\ln \mathcal{A}_c = -\frac{d}{2} \ln(2\pi) - \frac{1}{2} \sum_i \ln(\sigma_{c,i}^2)$ is the log-normalisation constant and $m_c^2(\mathbf{x})$ is the squared Diag-Cov Mahalanobis distance. For truly Gaussian data this distance concentrates tightly around its expected value d (see Section 3.4.2), but the *difference* in log-likelihoods between the winning class and any other class grows with d. Intuitively, each of the $d = 768$ standardised coordinates contributes a small increment to the distance gap, and these increments accumulate. When the raw log-likelihoods are converted into a posterior, even a modest advantage translates into a near-one-hot posterior: the probability mass collapses almost entirely onto the single best-fitting class. This *uncertainty collapse* makes the raw posterior probabilities unreliable as confidence estimates, which is particularly harmful in continual learning, where the number of competing classes grows with each new task.

Bayesian Temperature Scaling To recover reliable uncertainty estimates without altering the MAP decision boundary, we introduce a Temperature Scaling parameter $T > 0$ directly into the Bayesian inference process. We treat T as a global smoothing factor applied to the log-likelihoods before softmax normalisation. The calibrated posterior is:

$$p(y = c \mid \mathbf{x}; T) = \exp\left(\frac{1}{T} \log p(\mathbf{x} \mid y = c) - \mathcal{L}_E \right) \tag{3.34}$$

where $\mathcal{L}_E$ is the log-evidence computed via the numerically stable logsumexp operator:

$$\mathcal{L}_E = \text{logsumexp}_{j \in \mathcal{Y}} \left(\frac{1}{T} \log p(\mathbf{x} \mid y = j) \right) \tag{3.35}$$

Choosing $T > 1$ divides all log-likelihoods by T, shrinking the differences between classes proportionally and *softening* the posterior distribution. Choosing $T < 1$ sharpens it. Crucially,

T does not change the rank order of the log-likelihoods, so the MAP prediction is preserved. The effect on uncertainty is captured by the calibration metrics defined in Section 3.8: a well-chosen T reduces the Expected Calibration Error (ECE), lowers the Confidence on Errors (CoE), and brings the predictive entropy closer to the difficulty of the task.

3.4.2 Gaussianity Tests

Before committing to a class-conditional Gaussian classifier we must verify that the assumption is empirically defensible on the DINOv2 embedding space. We employ three complementary tests, arranged from weakest (moment-based) to strongest (formal hypothesis test) for the univariate case, plus a separate multivariate test.

Per-Dimension Gaussianity: Moments and Formal Test

To verify Gaussianity at the level of individual feature coordinates, we use three complementary indicators applied to every (class c, dimension j) pair across the 80 % training split.

Shape moments. Following the normality diagnostics of Pearson [5] and D'Agostino and Pearson [50, 51], we compute the sample skewness and excess kurtosis for each class c along each of the $d = 768$ coordinates:

$$\widehat{\mathrm{Skew}}_{c,j} \;=\; \frac{\frac{1}{n_c}\sum_i (x_{i,j}-\mu_{c,j})^3}{\hat{\sigma}^3_{c,j}}, \qquad \widehat{\mathrm{Kurt}}_{c,j} \;=\; \frac{\frac{1}{n_c}\sum_i (x_{i,j}-\mu_{c,j})^4}{\hat{\sigma}^4_{c,j}} - 3, \tag{3.36}$$

so that both quantities vanish exactly for a Gaussian [5, 4]. We report their mean, standard deviation, mean absolute value, and the fraction of coordinates satisfying the informal cut-offs $|\mathrm{Skew}| < 0.5$ and $|\mathrm{Kurt}| < 1.0$.

D'Agostino–Pearson omnibus test. The moments alone do not constitute a formal statistical test. We therefore apply the D'Agostino–Pearson omnibus test to every (class, dimension) pair [50, 51]. The test combines the two shape statistics into a single composite statistic:

$$K^2 \;=\; Z_1\Big(\widehat{\mathrm{Skew}}\Big)^2 + Z_2\Big(\widehat{\mathrm{Kurt}}\Big)^2 \;\sim\; \chi^2_2 \quad \text{under } H_0, \tag{3.37}$$

where Z_1 and Z_2 are normalised transformations of the skewness and kurtosis estimators respectively, and H_0 is the hypothesis that the data are normally distributed. We reject H_0 at $\alpha = 0.05$ and report the fraction of pairs that *pass* the test (i.e. fail to reject normality).

Relaxed criterion. Because the omnibus test is sensitive to the relatively large class sizes available ($n_c \approx 80$–400)—even minor departures from perfect normality accumulate into significant K^2 values—we supplement it with a size-independent indicator: a dimension is classified as near-Gaussian if $|\mathrm{Skew}| < 1$ *and* $|\mathrm{Kurt}| < 1$ hold simultaneously. The strict test and the relaxed criterion together provide a lower and an upper bound on the fraction of approximately Gaussian feature dimensions; both are reported in Section 4.2.2.

Multivariate Gaussianity: Henze–Zirkler Test

Per-dimension normality is necessary but not sufficient for the joint d-dimensional distribution to be Gaussian. A set of coordinates can each be marginally normal while the joint distribution departs from multivariate normality due to non-linear dependence or higher-order cross-moment structure. We therefore complement the univariate analysis with the Henze–Zirkler (HZ) test [1], whose theoretical basis is described in Section 2.2.3.

Diagonal-covariance HZ at native resolution ($d = 768$). As a first pass, the test is applied in the native $d = 768$ feature space using a manual diagonal-covariance formulation. The Mahalanobis distances required by the HZ statistic are computed analytically by inverting only the d per-coordinate sample variances (circumventing the rank-deficiency that arises from full matrix inversion when $n_c < d$), with a numerical floor of $\varepsilon = 10^{-10}$ applied to guard against inactive dimensions. All available samples for each class are used (100 % of the data, no train/test split). Under exact multivariate normality, $\mathbb{E}[T_{\mathrm{HZ}}] = 1 - (1 + 2b^2)^{-d/2} \to 1$ for large d (Eq. 3.40), so $T_{\mathrm{HZ}} \approx 1$ is the Gaussian reference at native dimensionality. The resulting T_{HZ} values, together with their per-dataset histograms, are reported in Section 4.2.2.

PCA-reduced HZ (valid multivariate analysis). To obtain a statistically informative multivariate HZ test, we project each class to a K-dimensional PCA subspace where the sample covariance is full rank ($K = \min(\lfloor n_c/3 \rfloor, 50)$ ensures $n_c \geq 3K$). The bandwidth b follows the standard HZ recommendation [1], and the same training split as the univariate tests is used. Because $K \leq 50 \ll d$, the kernel terms no longer underflow and the resulting T_{HZ} values—close to 1 for Gaussian data and growing monotonically with distributional complexity—are reported together with per-dataset histograms in Section 4.2.2.

Reported quantities (multivariate). For each benchmark and each analysis variant (diagonal $d = 768$ and PCA-reduced), we report the mean and standard deviation of T_{HZ} across all classes, together with the $[\mathrm{P}_{10}, \mathrm{P}_{90}]$ percentile band.

Univariate Henze–Zirkler Test per Dimension

As a complement to both the moment-based diagnostics of Section 3.4.2 and the multivariate HZ test above, we introduce a univariate HZ analysis applied independently to every (class c, dimension j) pair. This bridges the two levels: it uses the same formal HZ statistic as the multivariate test but at $d = 1$, providing a stricter hypothesis-based criterion than the D'Agostino–Pearson omnibus test while also being more interpretable than the full-dimensional multivariate result.

Bandwidth. For $d = 1$ the standard HZ smoothing parameter reduces to:

$$b^2 \;=\; \frac{1}{2}\left(\frac{3}{4}\right)^{2/5} n_c^{-2/5}, \tag{3.38}$$

which decays as $n_c^{-2/5}$ and is small ($b^2 \approx 0.04$) for the class sizes encountered here ($n_c \in [60, 400]$).

Efficient vectorised computation. For $d = 1$ the pairwise kernel term in T_{HZ} can be expanded via its Taylor series around $b^2 x_j x_k = 0$:

$$A_m \;=\; \frac{1}{n_c^2} \sum_{j,k} e^{-\frac{b^2}{2}(z_{j,m} - z_{k,m})^2} \;=\; \frac{1}{n_c^2} \sum_{\ell=0}^{L} \frac{b^{2\ell}}{\ell!}\, S_\ell^2 \;+\; \mathcal{O}\big((b^2)^{L+1}\big)\,, \tag{3.39}$$

where $S_\ell = \sum_j v_{j,m}\, z_{j,m}^\ell$ and $v_{j,m} = \exp(-b^2 z_{j,m}^2/2)$. All $d = 768$ dimensions are processed simultaneously; the memory footprint is $\mathcal{O}(n_c \cdot d)$ (no $n_c \times n_c$ matrix is formed). With $L = 8$ terms the approximation error is below 10^{-6} for all encountered bandwidths, verified against the exact per-dimension computation.

Expected value under H_0 and p-value. Under exact normality the expected value of T_{HZ} for a d-dimensional problem is:

$$\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \;=\; 1 - (1 + 2b^2)^{-d/2}, \tag{3.40}$$

which unifies both the univariate and multivariate cases. For $d = 1$ and the small bandwidths encountered here ($b^2 \approx 0.04$), the term $(1+2b^2)^{-1/2} \approx 0.96$ remains close to 1, so $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \approx 0.04$. For large d (e.g. the $d = 768$ multivariate case), even a modest bandwidth causes $(1 + 2b^2)^{-d/2} \to 0$, driving $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}] \to 1$. In short, the Gaussian reference value is near 0 for $d = 1$ and near 1 for large d; both cases follow from the same formula with different exponents. For the univariate analysis, with $d = 1$, the expected value is approximately 0.04 for $n_c = 400$ and increases toward 1 as $n_c \to 0$ (fewer samples, wider kernel). Empirically, the ratio $T_{\mathrm{HZ}} \,/\, \mathbb{E}_{H_0}[T_{\mathrm{HZ}}]$ follows a $\chi^2(1)$ distribution under H_0 (verified by simulation with coefficient of variation $\approx \sqrt{2}$), yielding the one-tailed p-value:

$$p \;=\; 1 - F_{\chi^2(1)}\!\left(\frac{T_{\mathrm{HZ}}}{\mathbb{E}_{H_0}[T_{\mathrm{HZ}}]}\right). \tag{3.41}$$

A dimension j of class c passes the test if $p > \alpha = 0.05$, i.e. $T_{\mathrm{HZ}} < 3.841\, \mathbb{E}_{H_0}[T_{\mathrm{HZ}}]$.

Reported quantities (univariate HZ). For each benchmark we report the distribution of T_{HZ} values over all $(C \times d)$ pairs (histograms), the mean per-class T_{HZ}, and the fraction of pairs passing the test at $\alpha = 0.05$. Results are in Section 4.2.2.

3.4.3 Inter-Class Pairwise Bhattacharyya Distance

While the Henze–Zirkler test of Section 2.2.3 measures the goodness-of-fit of individual classes to the diagonal-Gaussian model, the *inter-class* pairwise Bhattacharyya distance measures how distinguishable any two class-conditional Gaussians are from one another. The Bhattacharyya distance is symmetric and provides a direct connection to the Bayes error rate via

the Bhattacharyya coefficient $\rho_{ij} = e^{-D_B(i,j)}$. Under the diagonal-Gaussian parameterisation, the Bhattacharyya distance between classes i and j is:

$$D_B(i,j) = \frac{1}{8}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)^\top \bar{\Sigma}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j) + \frac{1}{2}\ln\left(\frac{\det \bar{\Sigma}}{\sqrt{\det \Sigma_i \, \det \Sigma_j}}\right), \tag{3.42}$$

where $\bar{\Sigma} = (\Sigma_i + \Sigma_j)/2$. Because both Σ_i and $\bar{\Sigma}$ are diagonal, all matrix operations reduce to element-wise arithmetic: determinants become sums of logs, and the matrix inverse reduces to element-wise reciprocals. $D_B = 0$ indicates complete distributional overlap; larger values correspond to greater separability and a lower Bhattacharyya coefficient $\rho = e^{-D_B}$. We track three statistics as the class count grows: (a) the mean pairwise D_B with a $[\mathrm{P}_{10}, \mathrm{P}_{90}]$ percentile band, (b) the minimum pairwise D_B (the hardest class pair), and (c) the count of pairs below selected threshold values to characterise the tail of the distribution. Together, these three curves characterise the class-confusion landscape that the Bayesian classifier must navigate as the continual-learning stream progresses.

3.4.4 Spectral Analysis of the Embedding Space

To characterise the global structure of the DINOv2 feature space, independently of the class-conditional model, we compute the PCA spectrum of the ℓ_2-normalised embeddings pooled across all training classes. The $d = 768$ eigenvalues of the empirical data covariance are sorted in decreasing order; we report both the individual percentage-of-variance contribution of each principal component and the cumulative explained variance. This diagnostic measures the *effective intrinsic dimensionality* of the shared representation: a rapid eigenvalue decay implies that the discriminative signal is concentrated in a low-dimensional subspace, whereas a slow decay indicates that the signal is distributed across the full ambient space. The spectral profile also constrains the effective rank available to the per-class covariance estimators: a per-class estimator can only populate directions already covered by the pooled class spectrum. Results are reported in Section 4.2.4.

3.5 Generative-Discriminative Fusion and Re-ranking Strategies

In high-dimensional classification tasks, relying on a single metric often fails to capture the full complexity of the feature space. While generative approaches (Mahalanobis-based) model the underlying class-conditional distributions, discriminative approaches (Cosine-based) excel at capturing angular separability on the unit hypersphere. To leverage the strengths of both, we evaluate several fusion and re-ranking strategies.

Importantly, these strategies are agnostic to the specific structure of the class-conditional distributions; they can be applied whether the underlying generative model utilizes a **diagonal covariance** matrix for computational efficiency or a **full covariance** matrix to capture complex feature correlations.

The nine strategies are presented in the order used throughout the rest of the thesis: the *recommended* group (SoftTaskFusion, TaskMeanShrink, Top5-CosineReRank and OptFusion-MinMax) comes first; the *brittle* group (HardGate, Top5-TaskReRank, TopK-TaskFilter) follows; and the two *marginal* LDA variants (HardGate-LDA, SoftGate-LDA) close the section. The corresponding empirical taxonomy is derived from the analysis of the experimental results.

3.5.1 SoftTaskFusion: Soft Class–Task Log-Probability Fusion

The Soft Task-Aware Fusion strategy mitigates the rigidity of hierarchical gating by treating the task assignment as a latent variable in a joint probability framework. In this approach, the final decision score for a class c is determined by combining its class-conditional log-likelihood with the log-likelihood of its parent task $t(c)$. Mathematically, given a feature vector $\mathbf{x}$, the final score S_c for each active class is defined as:

$$S_c(\mathbf{x}) = \ln P(\mathbf{x}|c) + \ln P(\mathbf{x}|t(c)) \tag{3.43}$$

where $\ln P(\mathbf{x}|c)$ and $\ln P(\mathbf{x}|t(c))$ represent the Gaussian log-likelihoods at the class and task levels, respectively. These are computed as:

$$\ln P(\mathbf{x}|i) = \ln A_i - \frac{1}{2}(x-\mu_i)^T\Sigma_i^{-1}(x-\mu_i), \quad i \in c, t \tag{3.44}$$

During inference, the task-level log-likelihoods $\ln P(\mathbf{x}|t)$ are calculated for all tasks currently seen in the stream. These values are then "expanded" or broadcasted to all constituent classes within those tasks. The classification rule is given by:

$$c^* = \arg\max_{c \in \mathcal{C}_{active}} [S_c(\mathbf{x})] \tag{3.45}$$

By performing an additive fusion in the log-domain (equivalent to a product of probabilities), this method allows the model to "boost" the scores of classes whose parent task distribution highly matches the input, while still allowing a class to be selected even if its task-level likelihood is not the absolute maximum, provided the class-level evidence is sufficiently strong. This approach effectively balances global class competition with coarse task-level guidance.

3.5.2 TaskMeanShrink: Task-Level MAP Mean Shrinkage

To improve the robustness of the class-level generative models, we implement a Maximum A Posteriori (MAP) estimation for the mean vectors. This strategy, often referred to as Mean Shrinkage, addresses the high variance in class-specific estimates—especially when data is scarce—by anchoring them to a task-level prior.

In this framework, each class c belonging to a task t is modeled using a task-aware mean $\mu_{c,\text{MAP}}$. This mean is a convex combination of the empirical class mean $\hat{\mu}_c$ and the pooled task mean μ_t:

$$\mu_{c,\text{MAP}} = \frac{n_c\hat{\mu}_c + \lambda\mu_t}{n_c + \lambda} \tag{3.46}$$

where n_c is the number of samples in class c, and λ is a shrinkage hyperparameter (set to 10.0 in our experiments) that controls the influence of the task-level prior. The task mean μ_t is computed by pooling the features of all classes within task t:

$$\mu_t = \frac{1}{\sum_{i \in \text{Task}_t} n_i} \sum_{i \in \text{Task}_t} \sum_{j=1}^{n_i} x_{i,j} \tag{3.47}$$

By shrinking the class means toward the task centroid, the model reduces over-fitting to class-specific noise while emphasizing features that characterize the broader task domain. Once the MAP means are calculated, they are used to compute the Mahalanobis distance in the log-likelihood score:

$$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{c,\text{MAP}})^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_{c,\text{MAP}}) \tag{3.48}$$

This approach provides a "Bayesian" regularization effect, ensuring that classes with very few samples or high intra-class variance are pull towards a more stable task-level representation, thereby improving generalizability in the global classification space.

3.5.3 Top5-CosineReRank: Generative–Angular Cosine Re-ranking

The Generative-Angular Re-ranking strategy leverages both the statistical modeling of Mahalanobis-based distances and the angular separability of Cosine similarity. In high-dimensional latent spaces, particularly those generated by models trained with contrastive or foundation objectives like DINOv2, the direction of the feature vector often carries significant discriminative information that the Euclidean-based Mahalanobis distance might under-prioritize.

The inference procedure is structured as a two-stage pipeline:

1. **Generative Candidate Filtering:** For an input feature $\mathbf{x}$, we first compute the log-likelihood scores across all active classes $\mathcal{C}_{active}$ using the Gaussian generative model (utilizing either diagonal or full covariance matrices):

   $$g_c(\mathbf{x}) = \ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) \tag{3.49}$$

   From these scores, a candidate set $\mathcal{K}$ containing the Top-5 classes with the highest likelihood is extracted.

2. **Angular Re-ranking:** Within the subset $\mathcal{K}$, the final prediction c^* is determined by maximizing the Cosine similarity between the input feature $\mathbf{x}$ and the class mean vectors μ_c:

   $$c^* = \arg\max_{c \in \mathcal{K}} \frac{\mathbf{x} \cdot \boldsymbol{\mu}_c}{\|\mathbf{x}\| \, \|\boldsymbol{\mu}_c\|} \tag{3.50}$$

By using the Gaussian model as a broad filter and Cosine similarity as a fine-grained refiner, this approach ensures that the chosen class is not only statistically probable given the distribution volume but also geometrically aligned with the class centroid on the unit hypersphere. This hybrid approach is particularly effective in mitigating the "over-confidence" of Gaussian models in regions where class distributions overlap significantly.

3.5.4 OptFusion-MinMax: Optimised Min–Max Score Fusion

The Optimized Score Fusion strategy seeks to integrate the probabilistic rigor of generative Gaussian modeling with the angular robustness of Cosine similarity. Because log-likelihoods from a high-dimensional Gaussian distribution and Cosine similarity scores operate on vastly different numerical scales and ranges, a direct combination is mathematically unsound. To address this, we implement a dynamic fusion pipeline utilizing Min-Max Normalization and automated hyperparameter optimization.

For a given input $\mathbf{x}$, we first compute two independent score vectors across all active classes:

1. **Generative Score (S_{gen}):** The Mahalanobis-based log-likelihood $g_c(\mathbf{x})$.
2. **Discriminative Score (S_{dis}):** The Cosine similarity between x and the class mean μ_c:

$$\cos(\theta_c) = \frac{\mathbf{x} \cdot \boldsymbol{\mu}_c}{\|\mathbf{x}\| \, \|\boldsymbol{\mu}_c\|} \tag{3.51}$$

To perform the fusion, both score vectors are independently mapped to the range $[0, 1]$ using a Min-Max normalization function:

$$\text{Norm}(S) = \frac{S - \min(S)}{\max(S) - \min(S) + \varepsilon} \tag{3.52}$$

The final decision score S_{final} is then computed as a convex combination of the normalized vectors:

$$S_{final} = \alpha \cdot \text{Norm}(S_{gen}) + (1 - \alpha) \cdot \text{Norm}(S_{dis}) \tag{3.53}$$

The fusion weight $\alpha \in [0, 1]$ is determined through an automated Bayesian hyperparameter search over a validation subset (20% of the training data). By maximising the Average Accuracy on this validation set, the optimal balance between the generative and discriminative evidence is identified prior to the final evaluation on the held-out test set. This approach ensures that the fusion weight reflects the geometric properties of the DINOv2 feature space rather than a fixed heuristic.

3.5.5 HardGate: Hierarchical Task Gating

This strategy implements a strict two-step decision process. First, a global gating mechanism identifies the most probable task t^* for a given input $\mathbf{x}$. Once the task is fixed, the final classification is restricted solely to the classes belonging to that task.The task-level distribution is modeled as a Gaussian $\mathcal{N}(\mu_t, \Sigma_t)$, where the parameters are estimated by pooling all samples from the classes belonging to task t. The inference logic is:

1. **Task Selection:** $t^* = \arg\max_t \left(\ln A_t - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^T \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t) \right)$
2. **Local Classification:** $c^* = \arg\max_{c \in \text{Task}_{t^*}} \left(\ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) \right)$

This approach significantly reduces the search space during inference but is highly sensitive to errors in the initial gating step (Task-IL vs. Class-IL interference).

3.5.6 Top5-TaskReRank: Top-5 Candidate Re-ranking via Task Likelihood

Unlike the strict gating, this strategy performs an initial global search across all active classes but uses task-level information to refine the final choice among the most likely candidates.The process follows these steps:

1. **Candidate Selection:** The model identifies the Top-5 classes with the highest log-likelihoods across the entire set of active classes.
2. **Task-Guided Verification:** For each candidate in the Top-5, the model retrieves the likelihood of the parent task $p(\mathbf{x}|t)$.
3. **Selection:** The final class c^* is chosen as the one among the Top-5 candidates whose parent task yields the highest task-level likelihood.

This strategy provides a "safety net" where the fine-grained class similarity is cross-referenced with the coarser task-level distribution to resolve ambiguities.

3.5.7 TopK-TaskFilter: Top-K Task Candidate Filtering

This strategy is a hybrid between strict gating and global search. It expands the search space of Gating by considering the N most probable tasks (where $N = 3$ in the implementation) and evaluating all classes within those selected tasks.The decision rule is formulated as:

$$c^* = \arg \max_{c \in \bigcup \text{Top-}N(\text{Tasks})} \left(\ln A_c - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_c)^T \Sigma_c^{-1} (\mathbf{x} - \boldsymbol{\mu}_c) \right) \tag{3.54}$$

By evaluating a "neighborhood" of likely tasks, this method mitigates the risk of a single incorrect gating decision while maintaining computational efficiency by ignoring tasks that are highly unlikely according to the pooled task-level Gaussian models.

3.5.8 HardGate-LDA: Hierarchical LDA Subspace Inference

This strategy combines global task gating with local expert models optimized via Linear Discriminant Analysis (LDA). By projecting high-dimensional DINOv2 features ($d = 768$) into a task-specific 9-dimensional subspace, the model maximizes the separability of classes within that task while reducing noise.

The process is defined as follows:

1. **Subspace Projection:** For each task t, we compute the within-class scatter matrix S_w and the between-class scatter matrix S_b using the classes belonging to that task. We solve the generalized eigenvalue problem $S_b v = \lambda S_w v$ to obtain the projection matrix $W_t \in \mathbb{R}^{768\times 9}$, which contains the top 9 eigenvectors.
2. **Stage 1 (Hard Gating):** A test sample $\mathbf{x}$ is first assigned to a task t^* by maximizing the log-likelihood of task-pooled Gaussian distributions in the original 768D space:
$$t^* = \arg\max_t \left(\ln A_t - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_t)^T \Sigma_t^{-1} (\mathbf{x} - \boldsymbol{\mu}_t) \right) \tag{3.55}$$
3. **Stage 2 (Local Subspace Classification):** The feature is projected into the specific subspace of task t^*: $\mathbf{x}' = W_{t^*}^T \mathbf{x}$. The final class is predicted using non-separable (full) covariance Gaussians modeled within this 9D space:
$$c^* = \arg\max_{c \in \text{Task}_{t^*}} \left(\ln A_{c,\text{proj}} - \frac{1}{2}(z' - \mu_{c,\text{proj}})^T \Sigma_{c,\text{proj}}^{-1} (z' - \mu_{c,\text{proj}}) \right) \tag{3.56}$$

3.5.9 SoftGate-LDA: Soft Hierarchical LDA Fusion

This other strategy improves upon the previous one by replacing the hard gating decision with a joint probabilistic fusion. This prevents a single error in the task selection stage from irrecoverably leading to a misclassification.

Instead of selecting t^* and then calculating the class likelihood, SoftGate-LDA evaluates a total score for every class across all tasks. The decision rule combines the global gating evidence with the local subspace likelihood:

$$c^* = \arg\max_{c \in \mathcal{C}_{\text{active}}} \left(\text{Score}_{\text{gating}}(t(c)) + \text{Score}_{\text{LDA}}(c | W_{t(c)}) \right) \tag{3.57}$$

where $t(c)$ is the task to which class c belongs. For every class, the feature $\mathbf{x}$ is projected into the subspace associated with its task. The scores are computed as:

- $\textbf{Score}_{\textbf{gating}}$**:** The log-likelihood of the parent task in the original 768D space.
- $\textbf{Score}_{\textbf{LDA}}$**:** The log-likelihood of the class in its task-specific 9D LDA subspace.

This additive fusion in the log-domain (equivalent to a product of probabilities) allows the local expert models to influence the task selection. If a class in Task A has an extremely high likelihood in its 9D subspace, it can override a slightly higher task-level likelihood for Task B, leading to more robust global classification.

3.6 Sample Efficiency and Incremental Class Inclusion

To quantify how quickly each closed-form estimator converges as training data accumulates, we design an incremental class inclusion experiment. Starting from a base classifier trained

on nine classes, a tenth target class is introduced and its training samples are presented in successive steps of 10 % of the total per-class budget. After each step, the updated classifier is evaluated on the held-out test set of all ten active classes and the accuracy on the new class is recorded. Two representative target classes per benchmark are selected to span the range of class indices. This protocol directly measures the *sample efficiency* of each covariance estimator—how many samples are required before the classifier reaches its asymptotic accuracy—which is directly relevant to the continual-learning setting where new classes may arrive with limited data. The corresponding results are reported in Section 4.5.

3.7 Backbones Comparison

Throughout this work, DINOv2 [12] is used as the default backbone feature extractor for the Gaussian-based classifiers because of its self-supervised training regime and the strong empirical evidence of approximately Gaussian, well-conditioned class-conditional distributions presented in Section 3.4.2. However, the proposed framework is agnostic to the specific feature extractor used: any frozen encoder that produces fixed-dimensional embeddings can plug in unchanged into the downstream Gaussian classifier. To quantify the dependence of the framework on the choice of backbone, we further analyse the impact of the backbone architecture by re-running the Diag-Cov and Full-Cov variants using two supervised Vision Transformers from the `timm` library [52]: **ViT-base/16** (ViT-b, $d = 768$) and **ViT-medium/16** (ViT-m, $d = 512$), both pre-trained on ImageNet-21k [21]. The covariance estimation, hyper-parameters and evaluation protocol are kept identical across backbones to isolate the effect of the upstream representation; the results are reported in Section 4.6.

3.8 Probabilistic Calibration Analysis

3.8.1 Confidence and Calibration Metrics

While accuracy and forgetting measure the correctness of predictions, they do not account for the model's reliability or its tendency to over-confide in incorrect decisions—a common issue in Class-Incremental Learning. To address this, we introduce five complementary metrics that jointly characterise the probabilistic behaviour of the classifier: the Expected Calibration Error (ECE) measures the global alignment between confidence and accuracy; the Confidence on Errors (CoE) quantifies how dangerously overconfident the model is when it fails; Total Confidence (TC) and Probability Margin (PM) both capture the decisiveness of the posterior distribution from two complementary angles; and the Average Predictive Entropy (H) measures the spread of probability mass across all classes. All five are defined below in the order in which they appear in Table 3.1.

Expected Calibration Error (ECE)

This metric quantifies the mismatch between predictive confidence and empirical accuracy. We partition the predictions into M equally spaced bins based on their confidence scores. The ECE is defined as the weighted average of the absolute difference between the accuracy and confidence of each bin:

$$ECE_t = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| acc(B_m) - conf(B_m) \right| \tag{3.58}$$

where N is the total number of samples, B_m is the set of indices of samples whose prediction confidence falls into the m-th interval, and $acc(B_m)$ and $conf(B_m)$ represent the mean accuracy and mean confidence within that bin, respectively.

Confidence on Errors (CoE)

To specifically analyze the model's behavior when it fails, we measure the average confidence assigned to the winning (but incorrect) class. High CoE indicates a model that is "dangerously wrong", which is a critical factor in safety-oriented applications. It is computed for a task t as:

$$CoE_t = \frac{1}{|\mathcal{E}_t|} \sum_{i \in \mathcal{E}_t} \max(\hat{p}_i) \tag{3.59}$$

where $\mathcal{E}_t$ is the set of misclassified samples and $\hat{p}_i$ is the predicted probability distribution for sample i.

Total Confidence (TC)

Total Confidence measures how certain the model is on average about its top-1 prediction, regardless of whether that prediction is correct:

$$TC_t = \frac{1}{N} \sum_{i=1}^{N} \max_c \, p_{i,c} \tag{3.60}$$

A value of $TC = 1$ means the model assigns all probability mass to a single class for every sample—it is always maximally decisive. A value close to $1/C$ would indicate near-uniform posteriors. TC is a measure of *decisiveness*, not correctness: a well-calibrated model should have $TC \approx A_T$ (confident when right, uncertain when wrong), whereas a pathologically overconfident model can reach $TC \approx 1$ even at low accuracy.

Probability Margin (PM)

The Probability Margin is the natural, non-logarithmic measure of how decisively the model separates its top-1 prediction from its second-best alternative:

$$PM_t = \frac{1}{N}\sum_{i=1}^{N}\Big(p_{i,(1)} - p_{i,(2)}\Big) \tag{3.61}$$

where $p_{i,(1)} \geq p_{i,(2)}$ are the two highest posterior probabilities for sample i. A margin near 1 means the model almost entirely ignores its second candidate; a margin near 0 means the model is nearly tied between its top-two guesses and one should treat the prediction with caution. PM is bounded in $[0, 1]$ and directly interpretable: a value of 0.7 means the winning class is, on average, assigned 70% more probability than the runner-up.

Average Predictive Entropy (H)

We utilize Shannon entropy to measure the uncertainty of the model's predictions. A higher entropy suggests that the model is distributing its probability mass across multiple classes, indicating lower certainty. For task t, the average entropy is calculated as:

$$H_t = \frac{1}{N}\sum_{i=1}^{N}\left(-\sum_{c=1}^{C} p_{i,c}\log(p_{i,c})\right) \tag{3.62}$$

where $p_{i,c}$ is the probability assigned to class c for sample i.

Summary of calibration and uncertainty metrics

Table 3.1 collects all six metrics (accuracy and the five calibration/uncertainty indicators), their range, the direction of improvement, and the key question each one answers. Together they provide a multi-dimensional view of the model's reliability throughout continual learning: accuracy tells us *how often* the model is right; ECE tells us *how honest* its probabilities are; CoE tells us *how dangerous* its mistakes are; TC tells us *how decisive* it is overall; PM tells us *how clearly* it separates its top-two candidates; and entropy tells us *how spread* its uncertainty is across all classes.

Table 3.1: Summary of the probabilistic calibration and uncertainty metrics used in this work. "Better" indicates the direction of the preferred value; "context" means both high and low values are informative depending on whether predictions are correct.

Metric	**Symbol**	**Range**	**Better**	**Key question**
Accuracy	A_T	$[0, 1]$	$\uparrow$	What fraction of predictions are correct?
Expected Calibration Error	ECE	$[0, 1]$	$\downarrow$	How well does confidence match actual accuracy?
Confidence on Errors	CoE	$[0, 1]$	$\downarrow$	How confident is the model when it is *wrong*?
Total Confidence	TC	$[0, 1]$	context	How decisive is the model on its top-1 choice?
Probability Margin	PM	$[0, 1]$	context	How much more likely is top-1 than top-2?
Predictive Entropy	H	$[0, \log C]$	$\downarrow$	How spread is probability mass over all C classes?

Chapter 4

Results

This chapter presents the quantitative results of the experimental campaign in the following order: Section 4.1 reports continual-learning metrics for the covariance estimators and mixture-of-Gaussian variants, including final-state confusion matrices and the effect of the log-normalisation constant; Section 4.2 characterises the feature space, consolidating score component distributions, marginal and multivariate Gaussianity tests (Section 4.2.2), inter-class separability via pairwise Bhattacharyya distance (Section 4.2.3), and the PCA spectral decay of the DINOv2 embeddings together with the accuracy under PCA-reduced representations (Section 4.2.4); Section 4.3 reports the covariance matrix conditioning and cross-class Frobenius analysis; Section 4.4 covers the generative–discriminative re-ranking strategies; Section 4.5 presents the sample efficiency and incremental class-inclusion curves; Section 4.6 ablates the choice of feature extractor; and Section 4.7 reports the probabilistic confidence and calibration metrics. All interpretation is deferred to Chapter **??**.

For every benchmark we report the average accuracy after the last task, A_T, and the forgetting score F_T as defined in Chapter 3. Whenever possible we report mean and variance over three independent runs (different task-order seeds). Since the proposed Gaussian classifiers maintain only sufficient statistics $(\boldsymbol{\mu}_c, \Sigma_c)$ and do not involve parameter optimisation during incremental updates, the resulting classifier is independent of the specific class-presentation order; this explains the ± 0.00 standard deviation reported for our methods throughout. State-of-the-art prompt-based competitors are reproduced from the published values reported in their respective papers. Methods proposed in this thesis are highlighted in grey. The best result per column is shown in bold.

4.1 Covariance Estimation and Mixtures of Gaussians

Table 4.1 reports the average accuracy and the forgetting score for every covariance estimator (diagonal, full, hybrid, Ledoit–Wolf identity/diagonal target [27], OAS [28]) and for the mixture-of-Gaussian variants MOG-k with $k \in \{2, 4\}$ components and three sub-variants (D, F, H for diagonal, full and hybrid covariance respectively). The first block compares our methods against prompt-based state-of-the-art baselines reproduced from the literature.

Table 4.2 reports the same family on the 30-task Joint stream.

Table 4.1: Continual-learning performance on the three 10-task benchmarks. A_{10} and F_{10} are reported in percentage points.

Method	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech-256**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
L2P	83.86 ± 1.17	7.35 ± 0.39	61.57 ± 0.40	9.73 ± 0.42	92.11 ± 0.32	2.94 ± 0.26
DualPrompt	86.51 ± 0.49	5.16 ± 0.20	68.13 ± 0.52	4.68 ± 0.18	92.50 ± 0.16	2.50 ± 0.16
DualPrompt-PGP	86.92 ± 0.05	5.35 ± 0.19	69.34 ± 0.05	4.53 ± 0.04	—	—
CODA-Prompt	87.00 ± 0.38	4.78 ± 0.24	74.24 ± 0.56	4.92 ± 0.21	—	—
CPrompt	87.83 ± 0.37	4.88 ± 0.79	76.71 ± 0.61	4.66 ± 0.93	—	—
ConvPrompt	88.87 ± 0.33	4.75 ± 0.15	77.86 ± 0.25	4.33 ± 0.24	—	—
RainbowPrompt	$\mathbf{89.86 \pm 0.11}$	$\mathbf{3.44 \pm 0.26}$	79.09 ± 0.13	3.90 ± 0.23	—	—
APT	88.88 ± 0.65	3.47 ± 0.55	$\mathbf{79.40 \pm 0.47}$	4.38 ± 0.46	—	—
Diag-Cov (ours)	86.07 ± 0.00	4.30 ± 0.12	77.12 ± 0.00	7.44 ± 0.17	90.06 ± 0.00	$\mathbf{1.69 \pm 0.36}$
Full-Cov (ours)	88.48 ± 0.00	4.18 ± 0.00	75.08 ± 0.00	5.07 ± 0.03	90.01 ± 0.01	2.16 ± 0.05
Hybrid-Cov (ours)	88.49 ± 0.00	4.18 ± 0.00	75.12 ± 0.00	5.03 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
Shared-Diag-Cov (ours)	86.54 ± 0.00	4.61 ± 0.10	75.12 ± 0.01	4.80 ± 0.24	93.00 ± 0.00	2.00 ± 0.11
Shared-Full-Cov (ours)	88.56 ± 0.00	3.76 ± 0.11	77.06 ± 0.01	$\mathbf{3.97 \pm 0.30}$	93.54 ± 0.00	2.05 ± 0.29
MOG2-D (ours)	86.96 ± 0.00	4.24 ± 0.02	70.15 ± 0.00	4.93 ± 0.02	91.41 ± 0.00	1.67 ± 0.30
MOG2-F (ours)	85.01 ± 0.00	5.13 ± 0.17	73.42 ± 0.00	5.49 ± 0.26	93.95 ± 0.00	2.14 ± 0.11
MOG2-H (ours)	84.68 ± 0.00	5.26 ± 0.22	73.52 ± 0.00	5.47 ± 0.23	93.93 ± 0.00	2.13 ± 0.10
MOG4-D (ours)	87.80 ± 0.00	4.02 ± 0.06	72.52 ± 0.01	5.06 ± 0.15	91.12 ± 0.00	1.63 ± 0.31
MOG4-F (ours)	88.87 ± 0.00	3.99 ± 0.00	75.07 ± 0.00	5.07 ± 0.05	$\mathbf{94.39 \pm 0.00}$	2.12 ± 0.08
MOG4-H (ours)	88.88 ± 0.00	4.04 ± 0.00	75.01 ± 0.00	5.12 ± 0.04	94.36 ± 0.00	2.09 ± 0.08
OAS (ours)	84.38 ± 0.00	3.90 ± 0.07	71.47 ± 0.00	4.57 ± 0.29	86.70 ± 0.00	1.90 ± 1.77
LW-I (ours)	84.78 ± 0.00	3.93 ± 0.09	71.91 ± 0.00	4.54 ± 0.43	89.89 ± 0.00	1.50 ± 0.08
LW-D (ours)	82.85 ± 0.00	3.94 ± 0.28	70.35 ± 0.00	4.62 ± 0.95	90.04 ± 0.00	1.66 ± 0.30

4.1.1 Final-State Confusion Matrices

Figures 4.1–4.3 show the normalised confusion matrices at $T_{\max}$ for the three benchmarks and Diag and Full covariance estimation methods, produced from the pre-extracted DINOv2 features as described in Section 3.3. The diagonal of each matrix corresponds to correct classifications; off-diagonal mass reveals systematic misclassification between visually or semantically similar classes.

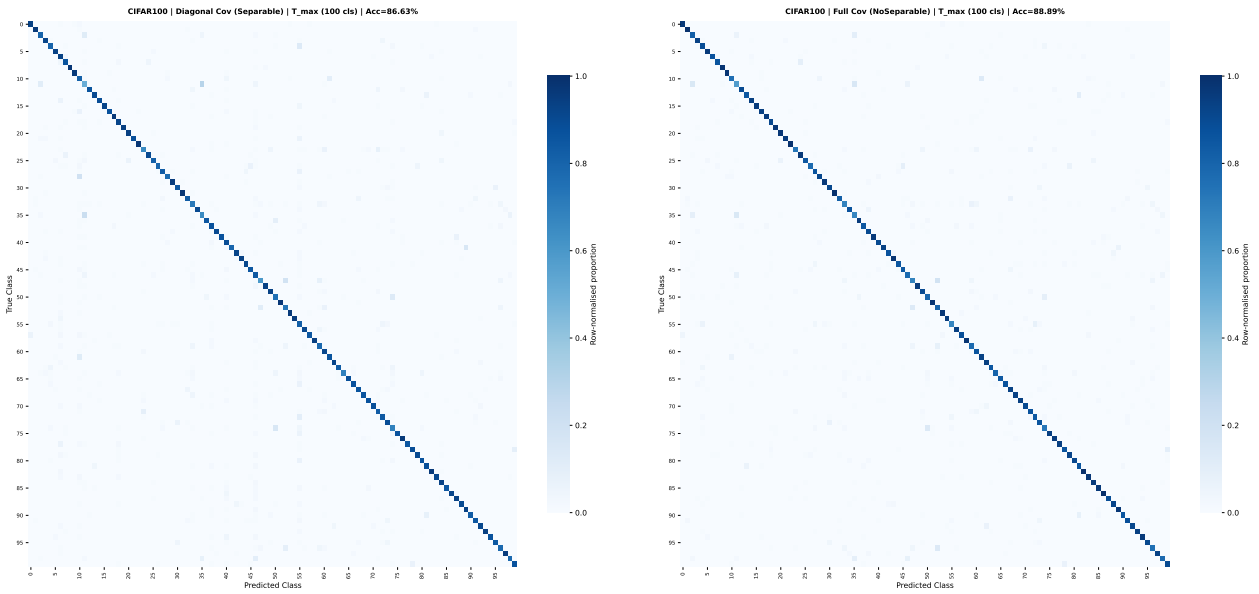
The Diag-Cov classifier achieves $T_{\max}$ accuracies of 86.6%, 90.6%, and 72.2% for CIFAR-100, Caltech-256, and ImageNet-R respectively, with off-diagonal mass that is diffuse across all benchmarks. The NoSeparable (Full-Cov) classifier achieves 88.9% on CIFAR-100, 94.8% on Caltech-256, and 73.6% on ImageNet-R. These $T_{\max}$ single-split accuracies are not directly comparable to the CIL average accuracies in Table 4.1, which are averaged over all T task checkpoints and three random task orderings.

4.1.2 Effect of the Log-Normalisation Constant

Table 4.3 compares the full Gaussian scoring rule (Eq. 3.31) with the Mahalanobis-only variant (Eq. 3.32), which sets $\log A_c = 0$ for all classes. Both rules use the same estimated parameters; the difference is whether the log-determinant penalty is included.

Table 4.2: Performance on the long-horizon **Split Joint Datasets** benchmark (30 tasks aggregated from CIFAR-100, ImageNet-R and Caltech-256). A_{30} and F_{30} are reported in percentage points.

Method	**Split Joint Datasets**	
	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
Diag-Cov	78.32 ± 0.21	6.05 ± 0.12
Full-Cov	78.65 ± 0.00	4.69 ± 0.02
Hybrid-Cov	$\mathbf{83.42 \pm 0.00}$	$\mathbf{3.86 \pm 0.04}$
Shared-Full-Cov	80.77 ± 0.00	4.47 ± 0.08
Shared-Diag-Cov	79.52 ± 0.00	4.95 ± 0.18
MOG2-D	73.91 ± 0.02	5.69 ± 0.02
MOG2-F	78.41 ± 0.02	5.85 ± 0.04
MOG2-H	80.27 ± 0.02	5.04 ± 0.04
MOG4-D	80.28 ± 0.01	4.67 ± 0.00
MOG4-F	59.68 ± 0.25	5.88 ± 0.16
MOG4-H	58.01 ± 0.30	5.97 ± 0.09
OAS	77.46 ± 0.00	4.60 ± 0.10
LW-I	76.41 ± 0.01	5.83 ± 0.31
LW-D	77.42 ± 0.02	5.41 ± 0.38



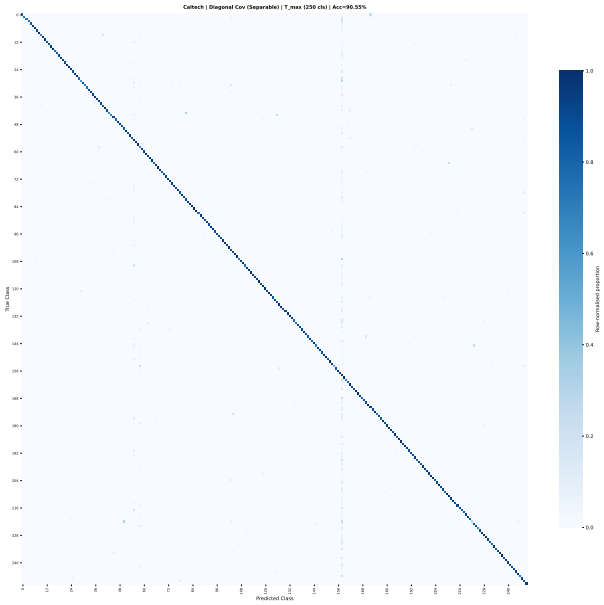
(a) Diag-Cov, Acc = 86.6%

(b) Full-Cov, Acc = 88.9%

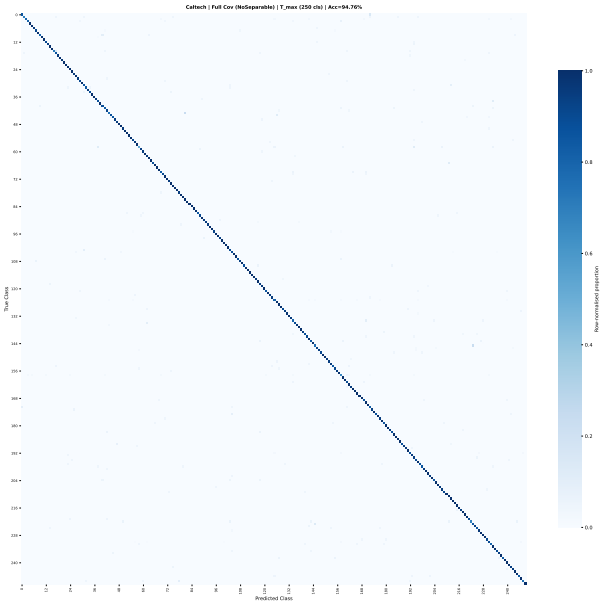
Figure 4.1: Normalised confusion matrices at $T_{\max}$ ($N = 100$ classes) for CIFAR-100.

Table 4.3: $T_{\max}$ classification accuracy (%) for the Gaussian and Mahalanobis-only inference rules, under diagonal (Diag-Cov) and full (Full-Cov) covariance estimation.

Cov. Model	**Inference**	**CIFAR-100**	**Caltech-256**	**ImageNet-R**
Diag-Cov	Gaussian	86.07	90.06	77.12
	Mahal-only	78.75	72.64	65.52
Full-Cov	Gaussian	88.48	90.01	75.08
	Mahal-only	88.48	90.01	75.08

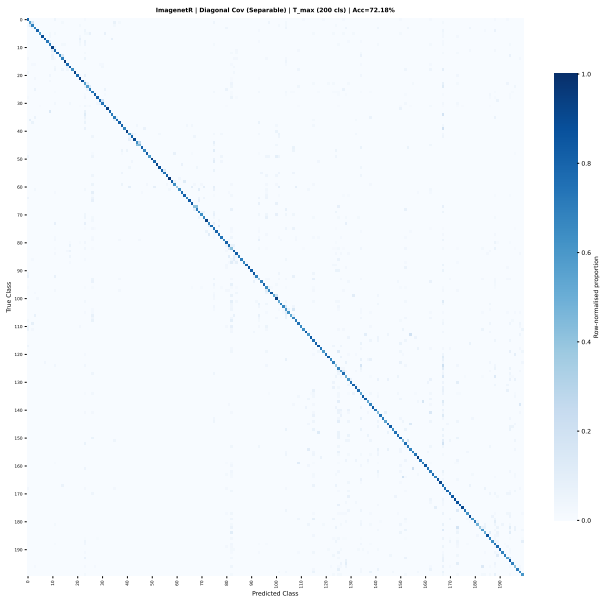


(a) Diag-Cov, Acc = 90.6%

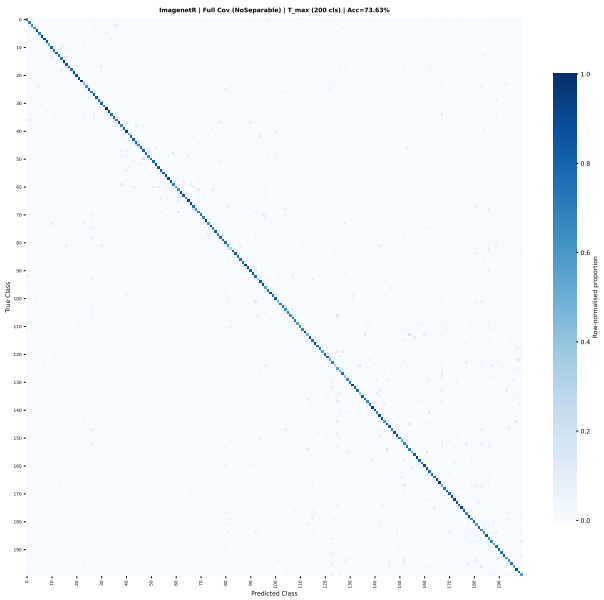


(b) NoSeparable (Full-Cov), Acc = 94.8%

Figure 4.2: Normalised confusion matrices at $T_{\max}$ ($N = 250$ classes) for Caltech-256.



(a) Diag-Cov, Acc = 72.2%



(b) Full-Cov, Acc = 73.6%

Figure 4.3: Normalised confusion matrices at $T_{\max}$ ($N = 200$ classes) for ImageNet-R.

4.2 Gaussianity and Feature Space Characterization

4.2.1 Score Components Distributions

The diagonal-covariance Gaussian classifier assigns each test sample $\mathbf{x}$ to the class that maximizes the log-likelihood score:

$$\text{score}(\mathbf{x}, c) = \underbrace{\ln A_c}_{\text{(a) volume}} - \tfrac{1}{2} \underbrace{d_{\text{mah}}(\mathbf{x}, c)}_{\text{(b) shape}}, \tag{4.1}$$

where the two components are

$$\ln A_c = -\tfrac{D}{2} \ln(2\pi) - \tfrac{1}{2} \sum_{d=1}^{D} \ln \sigma_{c,d}^2, \tag{4.2}$$

$$d_{\text{mah}}(\mathbf{x}, c) = \sum_{d=1}^{D} \frac{(x_d - \mu_{c,d})^2}{\sigma_{c,d}^2}. \tag{4.3}$$

$\ln A_c$ depends only on the class variances (volume term); d_{mah} depends on the distance from the sample to the class mean weighted by the inverse variances (shape term). Each figure shows three panels: (a) the C values of $\ln A_c$, one per class; (b) the $N_{\text{val}} \times C$ pairwise distances $d_{\text{mah}}(\mathbf{x}, c)$, with every validation sample scored against every class; and (c) the corresponding log-likelihood scores.

Figures 4.4 and 4.5 show these distributions on the held-out validation set for CIFAR-100 under the Diag-Cov and Full-Cov estimators respectively.

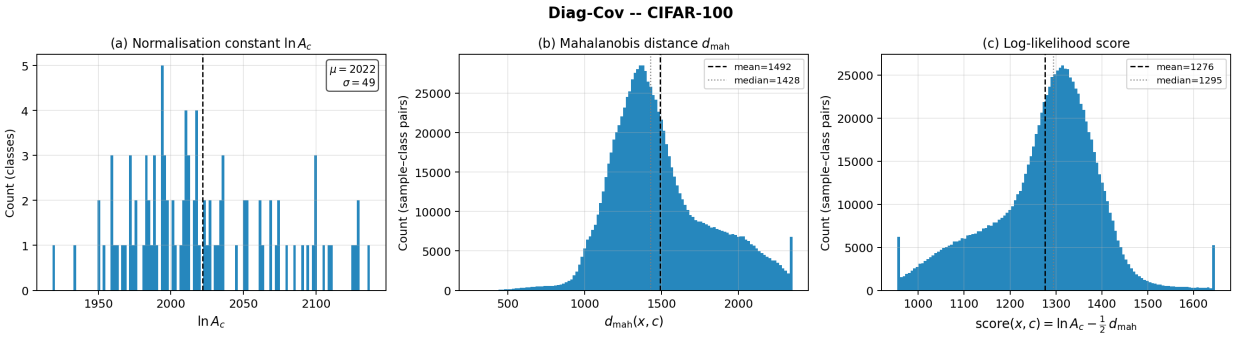


Figure 4.4: **CIFAR-100 — Validation set** (Diag-Cov). Empirical distributions of (a) the volume term $\ln A_c$ ($C = 100$ values), (b) the Mahalanobis shape term d_{mah} ($N_{\text{val}} \times C$ sample–class pairs), and (c) the log-likelihood score; Gaussian parameters fitted on the training partition.

Figure 4.5 repeats the same analysis under the **full covariance** estimator, where the precision matrix Σ_c^{-1} is obtained via eigendecomposition. The score distributions are computed on a subsample of the validation partition; fitting is still performed on the full training set.

The corresponding distributions for the training splits and for Caltech-256 and ImageNet-R are provided in Appendix A for completeness.

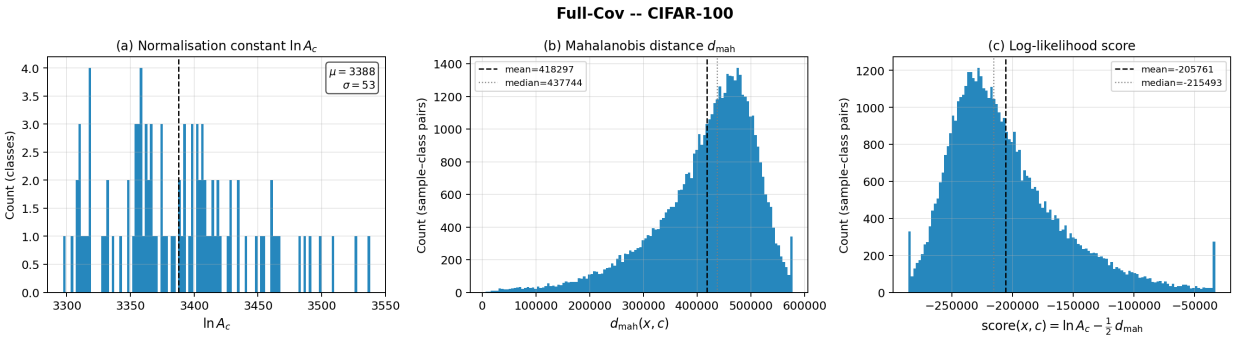


Figure 4.5: **CIFAR-100 — Validation set** (Full-Cov). Score component distributions on held-out samples.

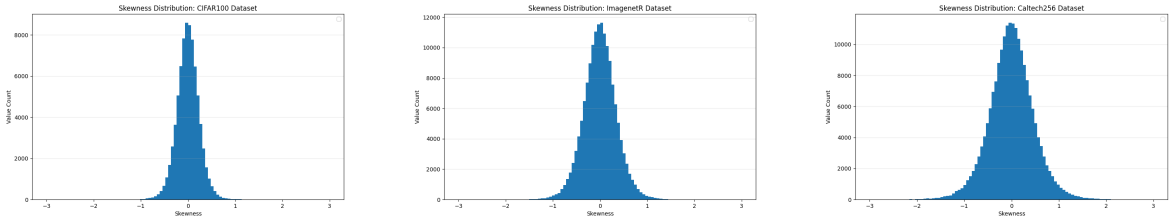
4.2.2 Gaussianity Characterizarion

Univariate Gaussianity

Table 4.4: Per-dimension Gaussianity statistics computed over all $(class, dimension)$ pairs of the frozen DINOv2 embeddings (raw, i.e. taken directly from the backbone). Each dataset yields $C \times d$ pairs ($d = 768$), reported in column **n**. The columns $|\text{Skew}|_\mu$ and $|\text{Kurt}|_\mu$ are the mean absolute values across all pairs; $\mathbb{P}[\cdot]$ denotes the fraction of *(class, dimension) pairs* satisfying each cut-off.

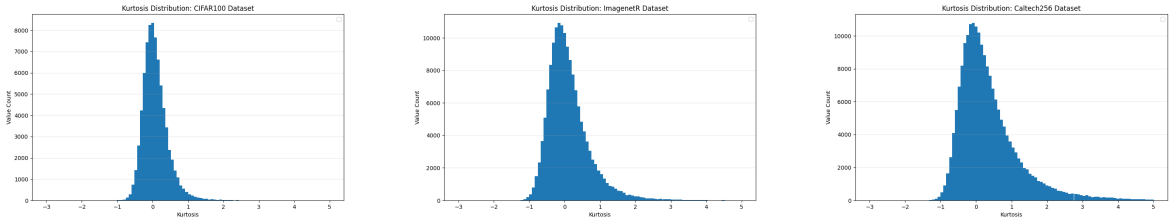
Dataset	**n**	$\textbf{Skew}_\mu$	$\lvert\text{Skew}\rvert_\mu$	$\textbf{Skew}_\sigma$	$\mathbb{P}[\lvert\text{Skew}\rvert<0.5]$	$\textbf{Kurt}_\mu$	$\lvert\text{Kurt}\rvert_\mu$	$\mathbb{P}[\lvert\text{Kurt}\rvert<1]$
CIFAR-100	76 800	−0.001	0.181	0.236	0.961	0.078	0.264	0.982
ImageNet-R	153 600	−0.002	0.269	0.348	0.859	0.115	0.432	0.924
Caltech-256	192 000	−0.001	0.360	0.472	0.747	0.402	0.659	0.822

Table 4.4 reports the aggregate per-dimension Gaussianity statistics on each benchmark. The mean skewness is essentially zero on all three datasets ($|\text{Skew}_\mu| \leq 0.002$). The fraction of (class, dimension) pairs satisfying $|\text{Skew}| < 0.5$ drops from 96.1 % on CIFAR-100 to 85.9 % on ImageNet-R and 74.7 % on Caltech-256, while the mean absolute excess kurtosis rises from 0.26 to 0.43 and 0.66, respectively. Figures 4.6 and 4.7 visualise the full empirical distributions.



(a) CIFAR-100 (b) ImageNet-R (c) Caltech-256

Figure 4.6: Empirical skewness distribution over the $d \times C$ (class, coordinate) pairs of the DINOv2 embeddings.



(a) CIFAR-100 (b) ImageNet-R (c) Caltech-256

Figure 4.7: Empirical excess-kurtosis distribution over the $d \times C$ (class, coordinate) pairs of the DINOv2 embeddings.

Table 4.5: Per-feature normality tests on the training split. All percentages report the fraction of *(class, dimension) pairs* satisfying the criterion, where each dataset contributes $C \times d$ pairs ($d = 768$). *normaltest* is the D'Agostino–Pearson composite test at $\alpha = 0.05$ (H_0: the dimension is normally distributed); *relaxed* is the joint criterion $|\text{Skew}| < 1$ **and** $|\text{Kurt}| < 1$; *both* denotes pairs passing both criteria simultaneously.

Dataset	**normaltest (% pairs)**	**relaxed (% pairs)**	**both (% pairs)**
CIFAR-100	70.0	97.8	70.0
ImageNet-R	80.8	91.6	79.7
Caltech-256	73.4	82.3	71.3

Two complementary per-feature normality assessments are reported in Table 4.5. The D'Agostino–Pearson normaltest at $\alpha = 0.05$ provides the strictest formal criterion; it combines standardised skewness and kurtosis into a χ^2_2 statistic and is therefore more sensitive than inspecting either moment alone. The relaxed criterion—$|\text{Skew}| < 1$ *and* $|\text{Kurt}| < 1$ simultaneously—is a lenient but transparent indicator of near-Gaussianity. With training splits of $\bar{n}_c \approx 400$, 120 and 90 samples per class for CIFAR-100, ImageNet-R and Caltech-256 respectively, even minor deviations from perfect normality are detectable, so the normaltest is conservative. 70–81 % of all (class, dimension) pairs pass the strict test, and the median p-value lies well above 0.05 on every benchmark. The relaxed criterion yields 82–98 % of pairs near-Gaussian in both shape moments simultaneously.

Univariate Henze–Zirkler Test per Dimension

Table 4.6 and Figure 4.8 report the univariate HZ test applied independently to every (class c, dimension j) pair (Section 3.4.2).

The reference point for interpreting the observed T_{HZ} values is the theoretical expectation under exact normality, $\mathbb{E}[T_{\text{HZ}}] = 1-(1+2b^2)^{-1/2}$ (Eq. 3.40 with $d = 1$), where the bandwidth $b^2 = \frac{1}{2}\left(\frac{3}{4}\right)^{0.4} n_c^{-0.4}$ depends only on the per-class sample size n_c. Larger classes have smaller b^2 and thus smaller $\mathbb{E}[T_{\text{HZ}}]$: for CIFAR-100 ($n_c \approx 400$) the expected value is approximately 0.04, while for Caltech-256 ($n_c \approx 80$) it rises to approximately 0.07. A distribution that is perfectly Gaussian should therefore produce a T_{HZ} close to these values; values substantially above them indicate departures from normality.

Table 4.6: Univariate Henze–Zirkler test applied to every (class, dimension) pair on the training split. $\mathbb{E}_{H_0}[T_{\mathrm{HZ}}]$ is the theoretical expectation under exact normality, determined by the per-class sample size via the bandwidth $b^2 = \frac{1}{2}(3/4)^{0.4} n_c^{-0.4}$.

Dataset	**pairs**	$\mathbb{E}_{H_0}[T_{\mathrm{HZ}}]$	T_{HZ} mean $\pm$ std	$[\mathrm{P}_{10}, \mathrm{P}_{90}]$
CIFAR-100	76 800	≈ 0.04	0.00052 ± 0.00095	[0.00002, 0.00132]
ImageNet-R	153 600	≈ 0.06	0.00134 ± 0.00210	[0.00006, 0.00346]
Caltech-256	192 000	≈ 0.07	0.00234 ± 0.00379	[0.00009, 0.00605]

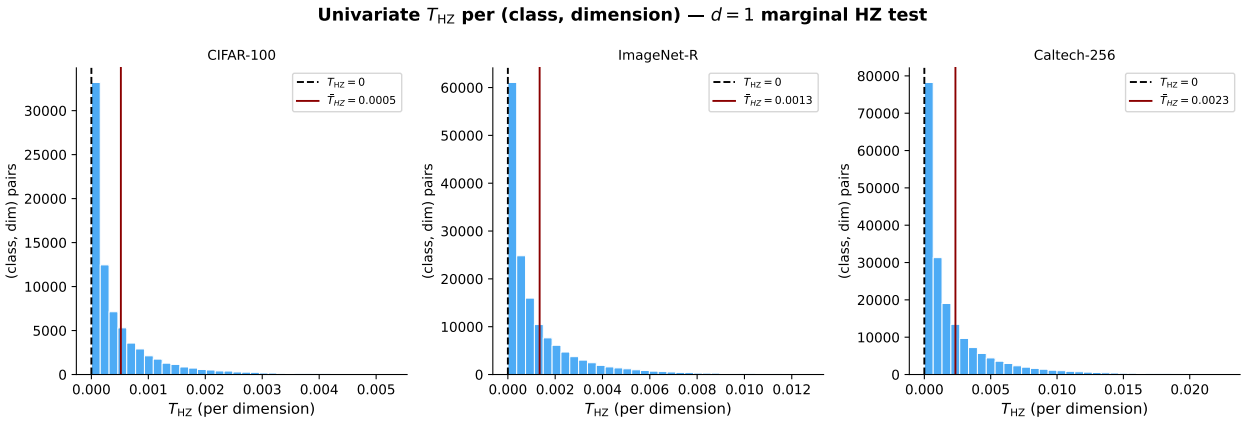


Figure 4.8: Univariate T_{HZ} distributions over all (class, dimension) pairs for the 80 % training split. The near-zero values confirm that the DINOv2 marginal feature distributions are highly consistent with Gaussianity at the individual coordinate level.

Multivariate Gaussianity: Henze–Zirkler Test

Table 4.7 reports the per-class HZ test applied in the native $d = 768$ feature space using a diagonal covariance formulation (Section 3.4.2). Rather than inverting a full 768×768 sample covariance matrix—which is rank-deficient whenever $n_c < d$—only the per-coordinate sample variances are used. All available samples per class are used (100 % of the data, no train/test split). Under exact multivariate normality $T_{\mathrm{HZ}} = 1$; values close to 1 indicate near-Gaussianity. Figure 4.9 shows the per-class T_{HZ} distributions.

Table 4.7: Per-class multivariate Gaussianity via the diagonal Henze–Zirkler test [1] at native $d = 768$, using 100 % of the available samples. $T_{\mathrm{HZ}} \approx 1$ indicates near-multivariate-Gaussianity under the diagonal-covariance model; C is the number of classes and $\tilde{n}_c$ is the median per-class sample count.

Dataset	C	$\tilde{n}_c$	d	T_{HZ} mean $\pm$ std	$[\mathrm{P}_{10}, \mathrm{P}_{90}]$
Split CIFAR-100	100	500	768	1.0002 ± 0.0020	[1.0000, 1.0000]
Split ImageNet-R	200	137	768	1.0033 ± 0.0073	[1.0000, 1.0128]
Split Caltech-256	250	100	768	1.0016 ± 0.0089	[1.0000, 1.0000]

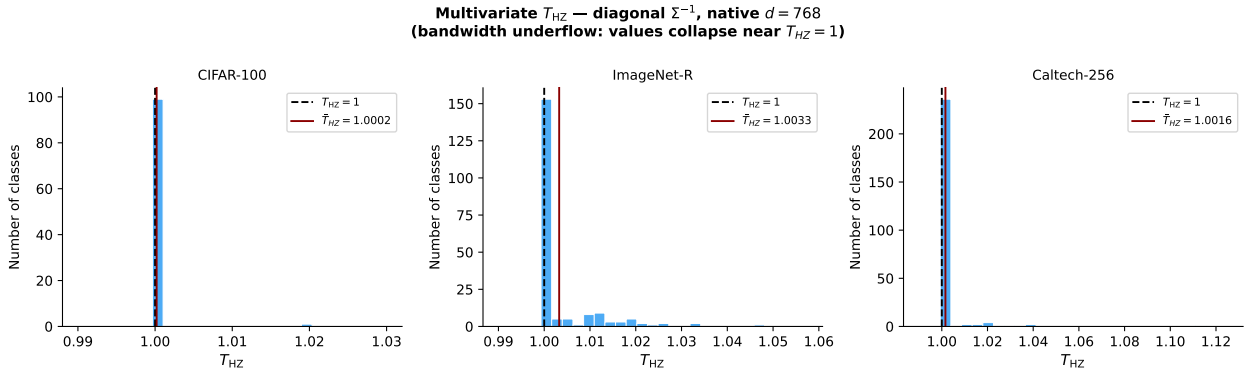


Figure 4.9: Per-class T_{HZ} distributions from the diagonal HZ test at native $d = 768$. All three benchmarks concentrate tightly near the Gaussian reference value $T_{\mathrm{HZ}} = 1$, with near-zero inter-class variance.

4.2.3 Inter-Class Pairwise Bhattacharyya Distance

Under the diagonal-covariance Gaussian model each class c is represented by $\mathcal{N}(\boldsymbol{\mu}_c, \Sigma_c)$ with $\Sigma_c = \mathrm{diag}(\sigma_{c,1}^2, \ldots, \sigma_{c,D}^2)$. The Bhattacharyya distance between two such distributions is

$$D_B(i,j) = \frac{1}{8}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j)^\top \bar{\Sigma}^{-1}(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j) + \frac{1}{2}\ln\left(\frac{\det\bar{\Sigma}}{\sqrt{\det\Sigma_i \,\det\Sigma_j}}\right), \tag{4.4}$$

where $\bar{\Sigma} = (\Sigma_i + \Sigma_j)/2$. Because both Σ_i and $\bar{\Sigma}$ are diagonal, all matrix operations reduce to element-wise arithmetic: $\det\Sigma = \prod_d \sigma_d^2$, $\bar{\Sigma}^{-1} = \mathrm{diag}(2/(\sigma_{i,d}^2 + \sigma_{j,d}^2))$, and log-determinants become sums of logs. This avoids costly dense matrix inversions and keeps computation in $\mathcal{O}(C \cdot D)$ memory per class pair. $D_B = 0$ indicates complete distributional overlap; larger values correspond to greater separability and a lower Bhattacharyya coefficient $\rho = e^{-D_B}$.

Figure 4.10 characterises the inter-class separability as the class set grows from $k = 2$ to the full benchmark size. The mean pairwise D_B at the full class-set size is: CIFAR-100 (89.43, [P_{10}=56.85, P_{90}=126.30]), ImageNet-R (84.81, [P_{10}=49.45, P_{90}=122.85]), and Caltech-256 (220.61, [P_{10}=113.63, P_{90}=350.79]). The minimum pairwise D_B at the full class-set size reaches 3.57 on CIFAR-100, 11.54 on ImageNet-R, and 20.96 on Caltech-256.

Pairs with $D_B < 20$ account for only 0.20 % (20/9 900) of CIFAR-100 pairs, 0.12 % (48/39 800) of ImageNet-R pairs, and 0.00 % of Caltech-256 pairs. Broadening the threshold to $D_B < 50$ captures 4.77 %, 10.40 %, and 0.15 % of pairs respectively.

4.2.4 Spectral Analysis of the Embedding Space

Figure 4.11 reports the PCA spectral decay of the ℓ_2-normalised DINOv2 embeddings in percentage-of-variance units. Panel (a) shows the individual variance contribution of each component: the first component alone explains around 8–10 % of the total variance, and the contribution decays steeply over the first $\sim$50 components before levelling off near zero.

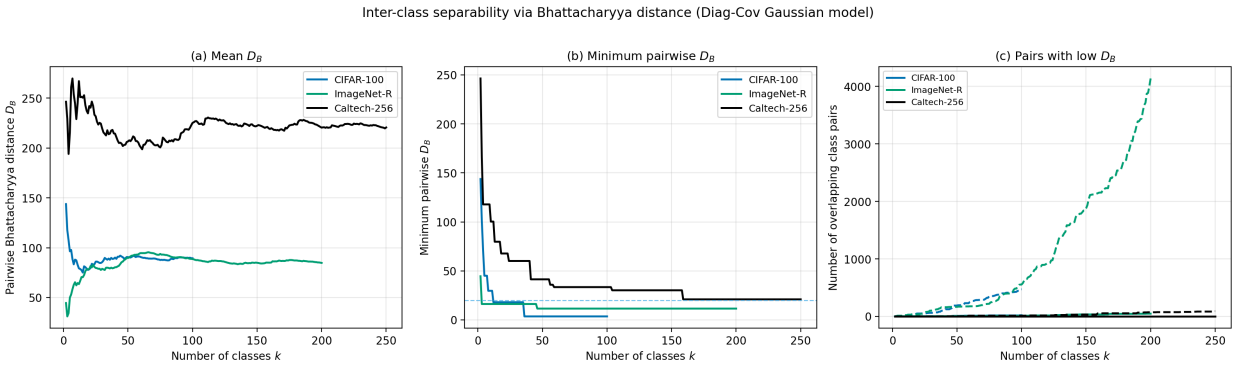


Figure 4.10: Inter-class separability under the Diag-Cov Gaussian model measured by the Bhattacharyya distance D_B (Eq. 4.4) as a function of the number of classes k. **(a)** Mean pairwise D_B as the class set grows. **(b)** Minimum pairwise D_B; horizontal dashed lines mark $D_B = 20$ and $D_B = 50$. **(c)** Count of pairs below each threshold as k increases. Over 99.8 % of all pairs exceed $D_B = 50$ on every benchmark.

Panel (b) shows the cumulative explained variance: roughly 90 % of the total variance is captured by the leading 100–130 components on all three benchmarks, despite the embedding space being $d = 768$-dimensional. The three datasets share remarkably similar spectra across both panels.

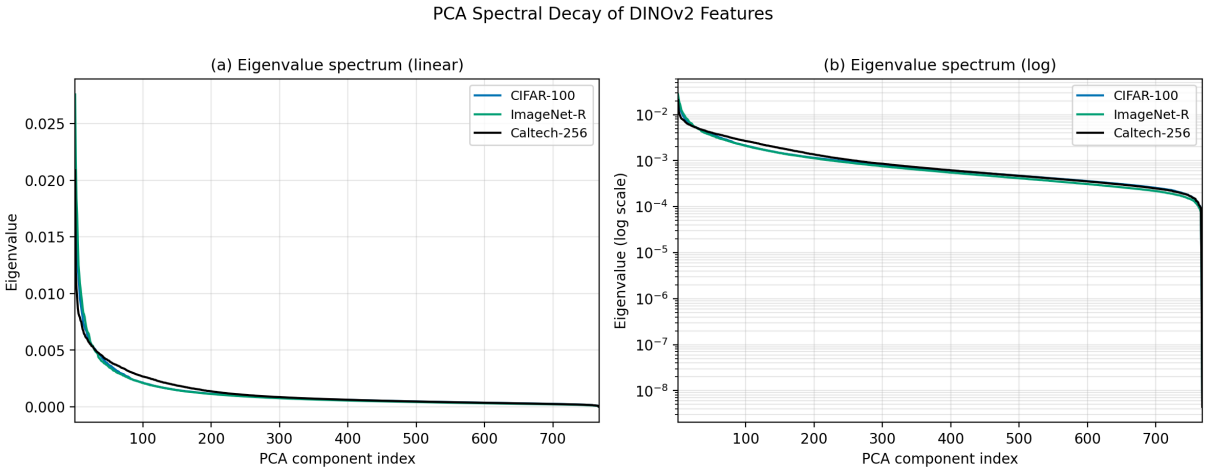


Figure 4.11: PCA spectral decay of the ℓ_2-normalised DINOv2 features. **(a)** Individual % variance explained per component: steep decay over the first ~ 50 components, near-zero contribution thereafter. **(b)** Cumulative % variance explained: dashed reference lines mark 50, 80, 90 and 95 %; annotated n values show how many components suffice to reach 90 % on each dataset. Despite the $d = 768$ embedding space, 90 % of the variance is captured by fewer than 130 components on every benchmark.

PCA-Reduced Continual-Learning Accuracy

Table 4.8 reports the final average accuracy of the Gaussian classifier after completing all tasks, evaluated under two PCA projection dimensionalities ($d = 64$ and $d = 256$) and two

covariance estimators (diagonal and full). Features were ℓ_2-normalised DINOv2 embeddings projected using the shared PCA weights estimated from the pooled training corpus; the train/test partition uses an 80/20 per-class split of the pre-extracted feature cache. Each cell reports the mean final average accuracy across three independent class-ordering runs (standard deviation is zero for all reported entries because the class-conditional statistics are estimated from the fixed training split, making the final-row accuracy deterministic regardless of task permutation; inter-run variance therefore reflects only ordering effects on the intermediate trajectory, not on the converged accuracy).

Table 4.8: Final average accuracy (%) of the Gaussian continual-learning classifier after projecting DINOv2 embeddings to $d \in \{64, 256\}$ dimensions via PCA. *Diag*: diagonal covariance; *Full*: full covariance with adaptive ridge regularisation. Higher is better.

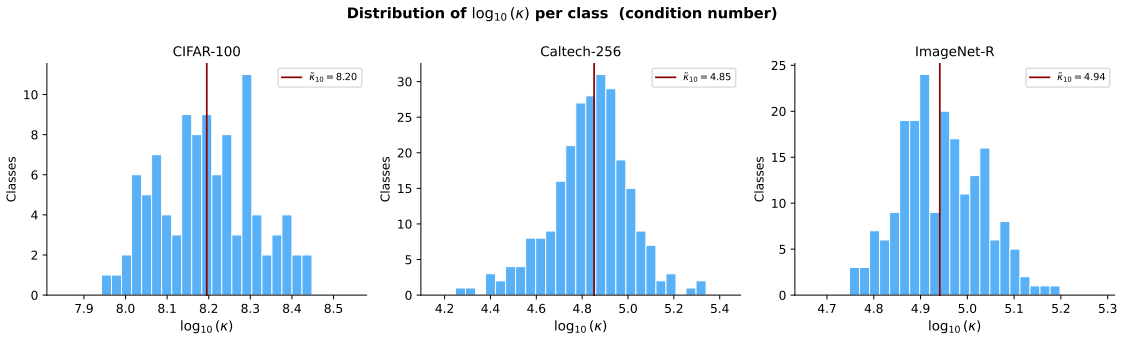
	PCA $d = 64$		**PCA** $d = 256$	
Benchmark	Diag	Full	Diag	Full
CIFAR-100	78.48	**83.05**	**85.22**	82.29
ImageNet-R	**63.86**	49.74	**71.64**	61.27
Caltech-256	**87.18**	44.09	89.98	**91.06**

On CIFAR-100, increasing PCA dimensionality from 64 to 256 improves diagonal-covariance accuracy by +6.7 pp; the corresponding gains are +7.8 pp on ImageNet-R and +2.8 pp on Caltech-256. On CIFAR-100 at $d = 64$, Full-Cov gains +4.6 pp over Diag-Cov, but at $d = 256$ this margin collapses to −2.9 pp. On ImageNet-R and Caltech-256 at $d = 64$, Full-Cov degrades by 14.1 and 43.1 pp respectively relative to Diag-Cov; both gaps narrow at $d = 256$.

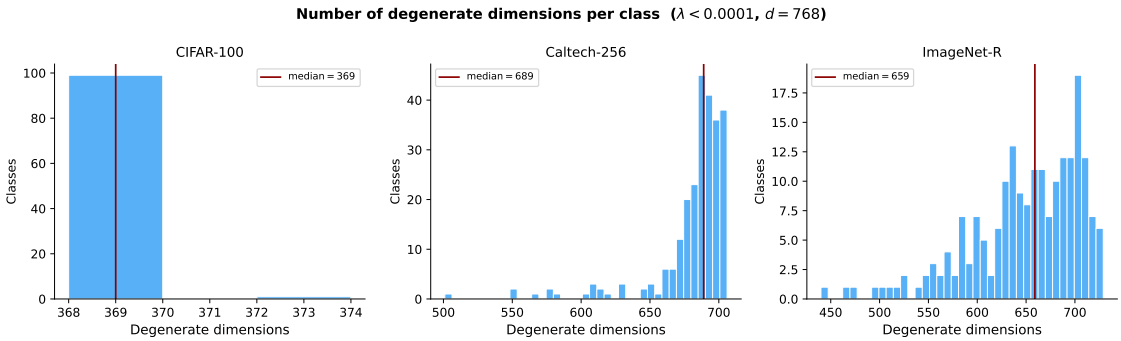
4.3 Covariance Matrix Structure

4.3.1 Conditioning Analysis

Table 4.9 summarises the conditioning metrics of the regularised full covariance matrices $\hat{\Sigma}_c^{\text{reg}} = \hat{\Sigma}_c + 10^{-6} I$ computed on the training split (Section 3.2.3). The three reported metrics characterise different aspects of how well-conditioned the per-class covariance is. The **condition number** $\kappa = \lambda_{\max}/\lambda_{\min}$ measures the spread between the largest and smallest eigenvalue; because it can span many orders of magnitude it is reported on a $\log_{10}$ scale. A high $\log_{10}(\kappa)$ means the matrix is nearly singular, so small perturbations in the data can cause large changes in the estimated inverse. **Degenerate dimensions** counts how many of the $d = 768$ eigenvalues fall below the threshold $\varepsilon_{\text{thr}} = 10^{-4}$; these directions carry negligible variance and are effectively noise. **Effective rank** is the entropy-based soft count of informative directions (Eq. 3.14): unlike the hard rank it penalises unequal eigenvalue distributions, so a matrix whose variance is concentrated in a few directions will have a low effective rank even if many eigenvalues are technically non-zero.



(a) Condition number $\log_{10}(\kappa)$



(b) Degenerate dimensions ($\lambda < 10^{-4}$)

Figure 4.12: Per-class conditioning metrics across all three benchmarks. Each histogram shows one bar per class; the darkred line marks the median. (a) log-condition number. (b) count of eigenvalues below the degeneracy threshold $\varepsilon_{\text{thr}} = 10^{-4}$.

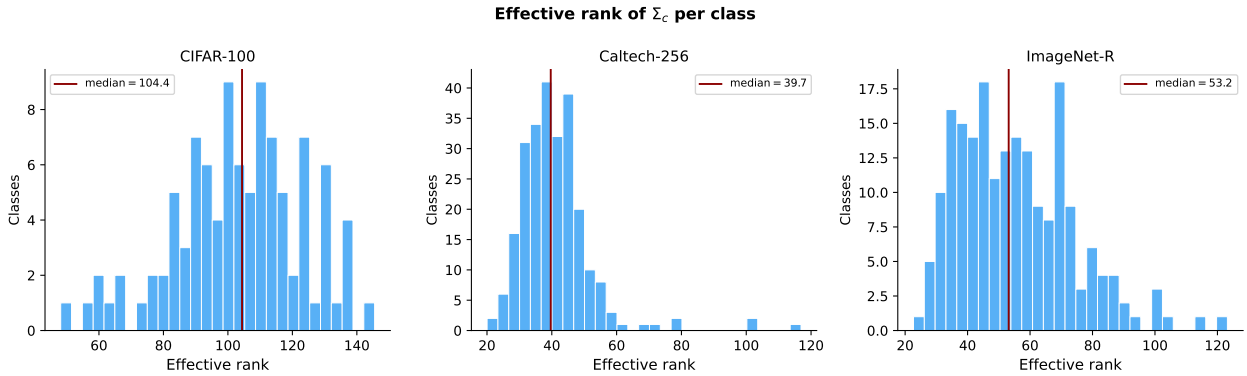


Figure 4.13: Effective rank (Eq. 3.14) of the regularised covariance matrices per class. Despite $d = 768$ ambient dimensions, the effective rank is below 110 for all CIFAR-100 classes, and below 80 for most Caltech-256 and ImageNet-R classes.

Table 4.9: Per-class covariance conditioning metrics across the three benchmarks ($d = 768$, $\varepsilon = 10^{-6}$, $\varepsilon_{\text{thr}} = 10^{-4}$). Degenerate dimensions are those whose eigenvalue falls below ε_{thr}; effective rank is the entropy-based measure of Eq. 3.14. All values are mean $\pm$ std over classes.

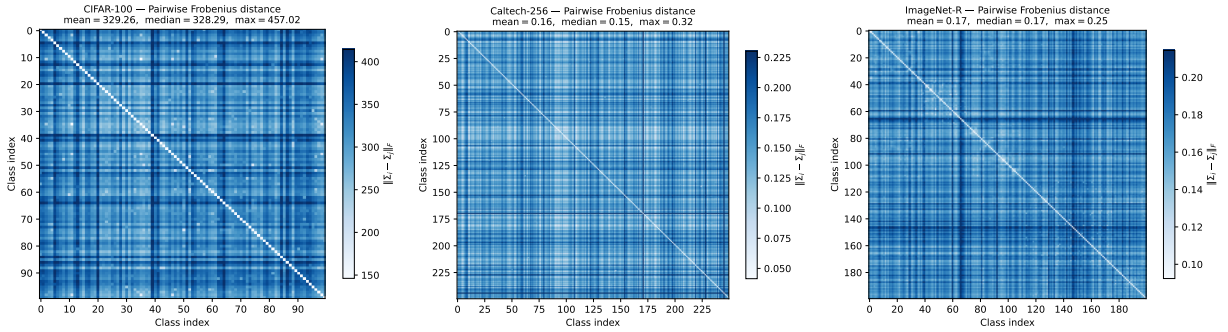
Dataset	$\tilde{n}_c$	$\log_{10}(\kappa)$	Degen. dims	Eff. rank
CIFAR-100	400	8.20 ± 0.11	369 ± 0	103.3 ± 19.8
Caltech-256	80	4.84 ± 0.17	682 ± 28	41.2 ± 11.4
ImageNet-R	110	4.94 ± 0.09	650 ± 57	55.4 ± 18.3

4.3.2 Cross-Class Frobenius Norm Analysis

Figure 4.14 shows the $C \times C$ pairwise Frobenius distance matrices for all three benchmarks; Figure 4.15 shows the marginal distribution of pairwise distances; Table 4.10 summarises the key statistics.

Table 4.10: Pairwise Frobenius distances $D_{ij}^F = \|\hat{\Sigma}_i^{\text{reg}} - \hat{\Sigma}_j^{\text{reg}}\|_F$ across all class pairs.

Dataset	Mean	Std	Min	Median	Max
CIFAR-100	329.26	41.05	146.15	328.29	457.02
Caltech-256	0.16	0.03	0.04	0.15	0.32
ImageNet-R	0.17	0.02	0.09	0.17	0.25



(a) CIFAR-100 (b) Caltech-256 (c) ImageNet-R

Figure 4.14: Pairwise Frobenius distance heatmaps $D_{ij}^F = \|\hat{\Sigma}_i^{\text{reg}} - \hat{\Sigma}_j^{\text{reg}}\|_F$. Colour encodes distance magnitude (darker = smaller = more similar covariance geometry). CIFAR-100 displays rich class-to-class variation; Caltech-256 and ImageNet-R are nearly uniform, indicating that regularisation has collapsed the covariance matrices toward a common εI floor.

4.4 Generative–Discriminative Fusion and Re-ranking Strategies

Tables 4.11–4.14 report the effect of the nine re-ranking and fusion strategies introduced in Chapter 3, paired with both the diagonal and the full covariance branch on every benchmark. The rows of each table are organised into three groups separated by horizontal rules.

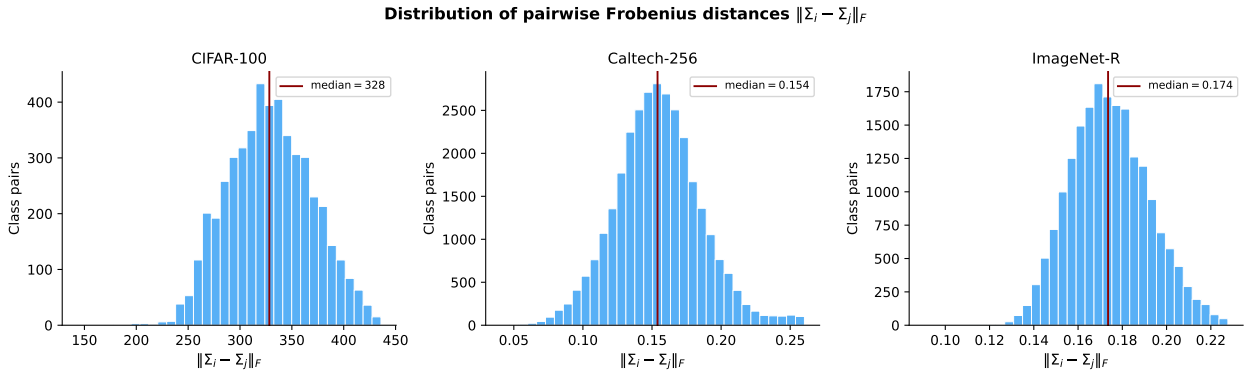


Figure 4.15: Distribution of all pairwise Frobenius distances. The scale difference between CIFAR-100 ($\bar{D}_F \approx 329$) and the other two benchmarks ($\bar{D}_F \approx 0.16$–0.17) spans three orders of magnitude, reflecting the degree to which regularisation dominates the estimated covariances.

The *recommended* group (SoftTaskFus, TaskMeanShr, Top5-CosRR, OptFusion-MM) contains strategies that consistently match or improve the unrestricted baseline across both covariance branches and all benchmarks, without requiring task identity at test time. The *brittle* group (HardGate, Top5-TaskRR, TopK-TaskFilt) contains strategies that rely on hard task-level gating. The *marginal* group (HardGate-LDA, SoftGate-LDA) contains LDA-based subspace methods.

Table 4.11: Generative–Discriminative Re-ranking Strategies on **Split CIFAR-100**.

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
SoftTaskFus	86.16 ± 0.00	4.41 ± 0.05	88.52 ± 0.00	4.18 ± 0.00
TaskMeanShr	86.08 ± 0.00	4.29 ± 0.12	88.48 ± 0.00	4.17 ± 0.00
Top5-CosRR	86.59 ± 0.00	4.92 ± 0.12	87.05 ± 0.00	4.91 ± 0.10
OptFusion-MM	$\mathbf{87.04 \pm 0.00}$	4.82 ± 0.07	$\mathbf{89.23 \pm 0.00}$	4.17 ± 0.03
HardGate	73.60 ± 0.88	9.82 ± 0.17	86.56 ± 0.00	4.83 ± 0.48
Top5-TaskRR	75.02 ± 1.09	8.80 ± 0.53	86.71 ± 0.01	4.73 ± 0.46
TopK-TaskFilt	83.90 ± 0.15	6.30 ± 0.34	88.51 ± 0.01	4.25 ± 0.02
HardGate-LDA	73.83 ± 0.77	9.93 ± 0.14	86.97 ± 0.02	4.89 ± 0.53
SoftGate-LDA	81.07 ± 0.07	6.79 ± 0.14	87.19 ± 0.03	4.79 ± 0.62

4.5 Sample Efficiency and Incremental Class Inclusion

Figure 4.16 illustrates how the Diag-Cov and Full-Cov class-conditional models behave when samples of a new class arrive progressively. After learning an initial set of nine base classes, an additional class is introduced and its training samples are incorporated in steps of 10 % of the available data. Two representative target classes per benchmark are reported, chosen to span the dataset's class index range. On Split CIFAR-100 (**a**), the Diag-Cov classifier reaches a strong plateau already with 10 % of the samples, while Full-Cov improves more gradually and surpasses Diag-Cov once 40–70 % of the samples are available. On Split ImageNet-R (**b**), both estimators improve gradually and the gap between Diag-Cov and Full-Cov stays

Table 4.12: Generative–Discriminative Re-ranking Strategies on **Split ImageNet-R**.

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
SoftTaskFus	75.04 ± 0.02	$\mathbf{4.30 \pm 0.61}$	63.11 ± 0.00	6.72 ± 9.39
TaskMeanShr	75.21 ± 0.00	4.37 ± 0.68	62.80 ± 0.00	6.85 ± 9.57
Top5-CosRR	$\mathbf{76.20 \pm 0.00}$	5.01 ± 0.36	73.63 ± 0.01	6.31 ± 2.45
OptFusion-MM	76.65 ± 0.00	5.38 ± 0.42	$\mathbf{77.00 \pm 0.00}$	$\mathbf{4.94 \pm 1.23}$
HardGate	53.95 ± 0.59	11.65 ± 0.83	69.78 ± 0.09	4.54 ± 2.45
Top5-TaskRR	56.49 ± 0.52	9.82 ± 1.23	69.53 ± 0.11	4.64 ± 2.75
TopK-TaskFilt	68.65 ± 0.30	7.79 ± 0.35	65.46 ± 1.36	5.24 ± 8.95
HardGate-LDA	54.14 ± 0.54	11.88 ± 0.70	73.70 ± 0.09	6.08 ± 3.86
SoftGate-LDA	59.90 ± 0.25	9.78 ± 2.10	73.72 ± 0.12	6.15 ± 5.84

Table 4.13: Generative–Discriminative Re-ranking Strategies on **Split Caltech-256**.

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$
SoftTaskFus	90.87 ± 0.00	1.60 ± 0.35	$\mathbf{95.03 \pm 0.00}$	$\mathbf{1.68 \pm 0.03}$
TaskMeanShr	86.08 ± 0.00	4.29 ± 0.12	94.73 ± 0.00	1.84 ± 0.02
Top5-CosRR	91.99 ± 0.00	2.44 ± 0.13	94.03 ± 0.00	2.25 ± 0.01
OptFusion-MM	$\mathbf{93.96 \pm 0.00}$	2.28 ± 0.01	94.78 ± 0.00	1.94 ± 0.04
HardGate	80.36 ± 0.11	5.48 ± 0.16	92.77 ± 0.04	2.34 ± 0.09
Top5-TaskRR	90.53 ± 0.00	1.53 ± 0.04	92.94 ± 0.05	2.25 ± 0.10
TopK-TaskFilt	88.06 ± 0.06	3.20 ± 0.11	94.58 ± 0.01	1.99 ± 0.02
HardGate-LDA	82.10 ± 0.77	5.59 ± 0.36	92.57 ± 0.01	1.75 ± 0.10
SoftGate-LDA	87.64 ± 0.02	3.40 ± 1.41	92.26 ± 0.05	1.73 ± 0.06

Table 4.14: Generative–Discriminative Re-ranking Strategies on **Split Joint Datasets** (30 tasks).

Strategy	**Diag-Cov**		**Full-Cov**	
	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
SoftTaskFus	78.86 ± 0.00	$\mathbf{4.97 \pm 0.09}$	52.50 ± 0.00	8.44 ± 2.67
TaskMeanShr	78.13 ± 0.00	5.20 ± 0.13	52.14 ± 0.00	8.61 ± 2.87
Top5-CosRR	78.92 ± 0.00	5.41 ± 0.08	63.61 ± 0.00	7.08 ± 1.29
OptFusion-MM	$\mathbf{79.08 \pm 0.00}$	5.67 ± 0.08	$\mathbf{81.94 \pm 0.00}$	4.89 ± 0.02
HardGate	66.11 ± 0.00	9.70 ± 0.21	53.16 ± 0.00	5.53 ± 1.98
Top5-TaskRR	68.59 ± 0.00	7.80 ± 0.12	52.03 ± 0.00	5.83 ± 2.46
TopK-TaskFilt	75.41 ± 0.00	6.45 ± 0.03	47.61 ± 0.00	7.48 ± 3.87
HardGate-LDA	66.52 ± 0.00	8.66 ± 0.22	53.02 ± 0.00	4.75 ± 0.46
SoftGate-LDA	68.25 ± 0.00	8.94 ± 4.00	52.07 ± 0.00	$\mathbf{4.81 \pm 0.62}$

small. On Split Caltech-256 (**c**), Full-Cov already dominates from the very first 10 % slice, and Diag-Cov tracks the Full-Cov curve at a 1–2 point lower plateau.

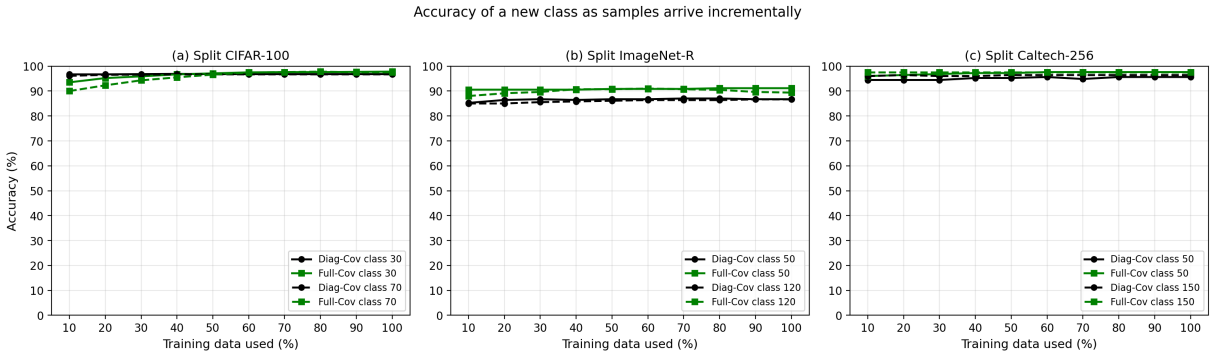


Figure 4.16: Accuracy of a new class as samples arrive incrementally: (**a**) Split CIFAR-100, (**b**) Split ImageNet-R, (**c**) Split Caltech-256. Two target classes per benchmark are shown.

4.6 Backbones Comparison

Table 4.15 reports the two best closed-form estimators (Diag-Cov and Full-Cov) when paired with three vision backbones: DINOv2 [12] (the default of this work), ViT-base patch-16/224 and ViT-medium patch-16/224, both supplied by the `timm` library [52].

Table 4.15: Backbone ablation. The two best closed-form estimators (Diag-Cov and Full-Cov) are evaluated with three vision backbones across all four benchmarks.

Method	**Backbone**	**Split CIFAR-100**		**Split ImageNet-R**		**Split Caltech-256**		**Split Joint**	
		$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{10}(\uparrow)$	$F_{10}(\downarrow)$	$A_{30}(\uparrow)$	$F_{30}(\downarrow)$
Diag-Cov	DINOv2	$\mathbf{86.07 \pm 0.00}$	4.30 ± 0.12	$\mathbf{77.12 \pm 0.00}$	7.44 ± 0.17	90.06 ± 0.00	1.69 ± 0.36	$\mathbf{78.32 \pm 0.21}$	6.05 ± 0.12
	ViT-m	81.41 ± 0.00	5.28 ± 0.13	66.11 ± 0.00	4.79 ± 0.14	$\mathbf{91.41 \pm 0.00}$	$\mathbf{1.63 \pm 0.12}$	50.63 ± 1.87	5.42 ± 0.07
	ViT-b	81.03 ± 0.00	$\mathbf{4.57 \pm 0.41}$	43.08 ± 0.00	6.41 ± 0.19	87.72 ± 0.00	1.74 ± 0.50	44.64 ± 1.43	$\mathbf{5.47 \pm 0.08}$
Full-Cov	DINOv2	$\mathbf{88.48 \pm 0.00}$	$\mathbf{4.18 \pm 0.00}$	$\mathbf{75.08 \pm 0.00}$	$\mathbf{5.07 \pm 0.03}$	90.01 ± 0.01	2.16 ± 0.05	$\mathbf{78.65 \pm 0.00}$	$\mathbf{4.69 \pm 0.02}$
	ViT-m	82.42 ± 0.00	5.25 ± 0.01	66.44 ± 0.00	5.18 ± 0.27	$\mathbf{90.11 \pm 0.00}$	$\mathbf{1.80 \pm 0.12}$	42.10 ± 0.00	9.39 ± 1.90
	ViT-b	80.40 ± 0.00	5.38 ± 0.36	45.72 ± 0.00	7.16 ± 0.17	87.13 ± 0.00	1.92 ± 0.80	66.51 ± 0.00	5.47 ± 0.08

4.7 Confidence and Calibration Metrics

Table 4.16 reports six metrics that characterise the probabilistic behaviour of the Gaussian classifier: the final-task accuracy (A_T in %), the Expected Calibration Error (ECE, %), the Confidence on Errors (CoE), Total Confidence (TC), Probability Margin (PM), and the Predictive Entropy. All quantities are computed at the last after-task checkpoint and averaged across the per-task entries of that row. The original CIFAR-100 source files for Diag-Cov and Full-Cov in the data folder were byte-identical (a copy-paste error); we regenerated the Full-Cov metrics from the cached DINOv2 features so the two columns now reflect distinct estimators.

For the Diag-Cov rows, TC and PM differ by 0.02–0.06 depending on the dataset (e.g. TC= 0.836, PM= 0.774 on CIFAR-100), with the gap being largest on ImageNet-R and smallest on Caltech-256.

Table 4.16: Probabilistic calibration metrics for the Bayesian classifier at the last after-task checkpoint. Accuracy and ECE are in percentage points; CoE, TC, PM and Entropy are in $[0, 1]$. PM (Probability Margin) is the average difference $p_{(1)} - p_{(2)}$ between the top-1 and top-2 softmax probabilities; TC is the average of $p_{(1)}$ alone. Lower is better for ECE and CoE; higher is better for Accuracy. CIFAR-100 Full was regenerated from the cached DINOv2 features because the original source file was a byte-identical copy of the Diag one.

Metric	**CIFAR-100**		**ImageNet-R**		**Caltech-256**		**Joint**	
	Diag	Full	Diag	Full	Diag	Full	Diag	Full
Accuracy (%)	86.07	88.48	77.12	75.08	90.06	90.01	78.32	78.65
ECE (%)	3.72	11.68	13.72	37.44	7.05	26.03	11.46	29.05
Confidence on Errors	0.467	1.000	0.656	1.000	0.805	1.000	0.742	1.000
Total Confidence	0.836	1.000	0.883	1.000	0.974	1.000	0.931	1.000
Probability Margin	0.774	1.000	0.825	1.000	0.955	1.000	0.890	1.000
Predictive Entropy	0.614	0.000	0.365	0.000	0.067	0.000	0.207	0.000

For the Full-Cov column, Confidence on Errors reaches 1.000, Total Confidence is identically 1.000, and predictive entropy collapses to 0.000 on every benchmark. For Diag-Cov, CoE increases with benchmark difficulty: CIFAR-100 ≈ 0.47, ImageNet-R ≈ 0.66, Caltech-256 ≈ 0.81.

Bibliography

[1] N. Henze and B. Zirkler, "A class of invariant consistent tests for multivariate normality," *Communications in Statistics – Theory and Methods*, vol. 19, no. 10, pp. 3595–3617, 1990.

[2] A. Thakur and A. Konde, "Fundamentals of neural networks," *International Journal for Research in Applied Science and Engineering Technology*, vol. 9, pp. 407–426, 08 2021.

[3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021. [Online]. Available: https://arxiv.org/abs/2010.11929

[4] J. M. Joyce, "Kurtosis," in *International Encyclopedia of Statistical Science*. Springer, 2011, pp. 722–723.

[5] K. Pearson, "Contributions to the mathematical theory of evolution.II. skew variation in homogeneous material," *Philosophical Transactions of the Royal Society of London. A*, vol. 186, pp. 343–414, 1895.

[6] M. Masana, X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. van de Weijer, "Class-incremental learning: Survey and performance evaluation on image classification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 5, pp. 5513–5533, 2023, doi: 10.1109/TPAMI.2022.3213473.

[7] R. French, "Catastrophic forgetting in connectionist networks," *Trends in cognitive sciences*, vol. 3, pp. 128–135, 05 1999, doi: 10.1016/S1364-6613(99)01294-2.

[8] G. M. van de Ven, Z. Li, and A. S. Tolias, "Class-incremental learning with generative classifiers," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2021, pp. 3606–3615, doi: 10.1109/CVPRW53098.2021.00400.

[9] R. Chitale, A. Vaidya, A. Kane, and A. Ghotkar, "Task arithmetic with lora for continual learning," 2023. [Online]. Available: https://arxiv.org/abs/2311.02428

[10] Y.-S. Liang and W.-J. Li, "Adaptive plasticity improvement for continual learning," in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 7816–7825, doi: 10.1109/CVPR52729.2023.00755.

[11] R. V. Sluyters, J. Atkinson, M. Banks, R. Held, K.-P. Hoffmann, and C. Shatz, *Visual Perception: The Neurophysiological Foundations*. San Diego, CA: Academic Press, 1990, ch. The Development of Vision and Visual Perception.

[12] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. HAZIZA, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jegou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024, featured Certification. Code available at: https://github.com/facebookresearch/dinov2. [Online]. Available: https://openreview.net/forum?id=a68SUt6zFt

[13] S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Upper Saddle River, NJ: Pearson, 2020, global Edition.

[14] C. M. Bishop, *Pattern Recognition and Machine Learning*, ser. Information Science and Statistics. New York, NY: Springer, 2006.

[15] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: The MIT Press, 2018.

[16] R. Szeliski, *Computer Vision: Algorithms and Applications*. Springer, 2010.

[17] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel, "Backpropagation applied to handwritten zip code recognition," *Neural Computation*, vol. 1, no. 4, pp. 541–551, 1989.

[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

[19] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*, F. Pereira, C. Burges, L. Bottou, and K. Weinberger, Eds., vol. 25. Curran Associates, Inc., 2012. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf

[20] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255, doi: 10.1109/CVPR.2009.5206848.

[21] T. Ridnik, E. B. Baruch, A. Noy, and L. Zelnik-Manor, "Imagenet-21k pretraining for the masses," *CoRR*, vol. abs/2104.10972, 2021. [Online]. Available: https://arxiv.org/abs/2104.10972

[22] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, and J. Gilmer, "The many faces of robustness: A critical analysis of out-of-distribution generalization," *arXiv preprint arXiv:2006.16241*, 2020.

[23] A. Krizhevsky, "The cifar-100 dataset," https://www.cs.toronto.edu/~kriz/cifar.html, 2009, accessed: 2025-10-02.

[24] G. Griffin, A. Holub, and P. Perona, "Caltech 256," 2022, doi: 10.22002/D1.20087.

[25] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.

[26] C. Eckart and G. Young, "The approximation of one matrix by another of lower rank," *Psychometrika*, vol. 1, no. 3, pp. 211–218, 1936.

[27] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.

[28] Y. Chen, A. Wiesel, Y. C. Eldar, and A. O. Hero, "Shrinkage algorithms for MMSE covariance estimation," *IEEE Transactions on Signal Processing*, vol. 58, no. 10, pp. 5016–5029, 2010.

[29] L. Wang, X. Zhang, H. Su, and J. Zhu, "A comprehensive survey of continual learning: Theory, method and application," *IEEE transactions on pattern analysis and machine intelligence*, vol. PP, 02 2024, doi: 10.1109/TPAMI.2024.3367329.

[30] M. Sazzad Hossain, P. Saha, T. Faisal Chowdhury, S. Rahman, F. Rahman, and N. Mohammed, "Rethinking task-incremental learning baselines," in *2022 26th International Conference on Pattern Recognition (ICPR)*, 2022, pp. 2771–2777.

[31] A. Chaudhry, M. Rohrbach, M. Elhoseiny, T. Ajanthan, P. K. Dokania, P. H. S. Torr, and M. Ranzato, "Continual learning with tiny episodic memories," *ArXiv*, vol. abs/1902.10486, 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:67855851

[32] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, "Efficient lifelong learning with A-GEM," *CoRR*, vol. abs/1812.00420, 2018. [Online]. Available: http://arxiv.org/abs/1812.00420

[33] P. Buzzega, M. Boschini, A. Porrello, D. Abati, and S. Calderara, "Dark experience for general continual learning: a strong, simple baseline," in *Proceedings of the 34th International Conference on Neural Information Processing Systems*, ser. NIPS '20. Red Hook, NY, USA: Curran Associates Inc., 2020.

[34] X. Liu, C. Wu, M. Menta, L. Herranz, B. Raducanu, A. D. Bagdanov, S. Jui, and J. van de Weijer, "Generative feature replay for class-incremental learning," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 915–924, doi: 10.1109/CVPRW50498.2020.00121.

[35] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, p. 3521–3526, Mar. 2017, doi: 10.1073/pnas.1611835114. [Online]. Available: http://dx.doi.org/10.1073/pnas.1611835114

[36] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ser. ICML'17. JMLR.org, 2017, p. 3987–3995.

[37] Z. Li and D. Hoiem, "Learning without forgetting," *CoRR*, vol. abs/1606.09282, 2016. [Online]. Available: http://arxiv.org/abs/1606.09282

[38] S. C. Y. Hung, C.-H. Tu, C.-E. Wu, C.-H. Chen, Y.-M. Chan, and C.-S. Chen, *Compacting, picking and growing for unforgetting continual learning.* Red Hook, NY, USA: Curran Associates Inc., 2019.

[39] S. Yan, J. Xie, and X. He, "Der: Dynamically expandable representation for class incremental learning," *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3013–3022, 2021. [Online]. Available: https://api.semanticscholar.org/CorpusID:232427923

[40] J. Yoon, E. Yang, J. Lee, and S. J. Hwang, "Lifelong learning with dynamically expandable networks," in *6th International Conference on Learning Representations, ICLR 2018*, 2018.

[41] Z. Wang, Z. Zhang, C.-Y. Lee, H. Zhang, R. Sun, X. Ren, G. Su, V. Perot, J. Dy, and T. Pfister, "Learning to prompt for continual learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 139–149.

[42] Z. Wang, Z. Zhang, S. Ebrahimi, R. Sun, H. Zhang, C.-Y. Lee, X. Ren, G. Su, V. Perot, J. Dy *et al.*, "Dualprompt: Complementary prompting for rehearsal-free continual learning," *European Conference on Computer Vision*, 2022.

[43] J. S. Smith, L. Karlinsky, V. Gutta, P. Cascante-Bonilla, D. Kim, A. Arbelle, R. Panda, R. Feris, and Z. Kira, "Coda-prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2023, pp. 11 909–11 919.

[44] Y.-S. Liang and W.-J. Li, "Inflora: Interference-free low-rank adaptation for continual learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 23 638–23 647.

[45] O. Moussadek, R. Salami, and S. Calderara, "Dolfin: Balancing stability and plasticity in federated continual learning," 2025. [Online]. Available: https://arxiv.org/abs/2510.13567

[46] H. Chen, P. Wang, Z. Zhou, X. Zhang, Z. Wu, and Y.-G. Jiang, "Achieving more with less: Additive prompt tuning for rehearsal-free class-incremental learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2025, pp. 340–349.

[47] G. Zhang, L. Wang, G. Kang, L. Chen, and Y. Wei, "Slca: Slow learner with classifier alignment for continual learning on a pre-trained model," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 19 148–19 158.

[48] H. Zhuang, Z. Weng, H. Wei, R. Xie, K.-A. Toh, and Z. Lin, "Acil: Analytic class-incremental learning with absolute memorization and privacy protection," *Advances in Neural Information Processing Systems*, vol. 35, pp. 11 602–11 614, 2022.

[49] D. Goswami, Y. Liu, B. Twardowski, and J. van de Weijer, "Fecam: Exploiting the heterogeneity of class distributions in exemplar-free continual learning," *Advances in Neural Information Processing Systems*, vol. 36, pp. 6582–6595, 2024.

[50] R. B. D'Agostino, "An omnibus test of normality for moderate and large size samples," *Biometrika*, vol. 58, no. 2, pp. 341–348, 1971.

[51] R. B. D'Agostino and E. S. Pearson, "Tests for departure from normality. empirical results for the distributions of b_2 and $\sqrt{b_1}$," *Biometrika*, vol. 60, no. 3, pp. 613–622, 1973.

[52] R. Wightman, "Pytorch image models," https://github.com/rwightman/pytorch-image-models, 2019.

[53] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," in *Advances in Neural Information Processing Systems*, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, Eds., vol. 28. Curran Associates, Inc., 2015. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/file/14bfa6bb14875e45bba028a21ed38046-Paper.pdf

Appendix A

Score Component Distributions

This appendix contains the empirical distributions of the score components—volume term $\ln A_c$, shape term d_{mah}, and final log-likelihood—for all three benchmarks (CIFAR-100, Caltech-256, ImageNet-R) and both covariance estimators (Diag-Cov and Full-Cov), computed separately on the training and validation splits. The two CIFAR-100 validation figures are shown in the main text (Figures 4.4 and 4.5); the remaining figures are collected here for completeness.

Diagonal Covariance — Training Splits

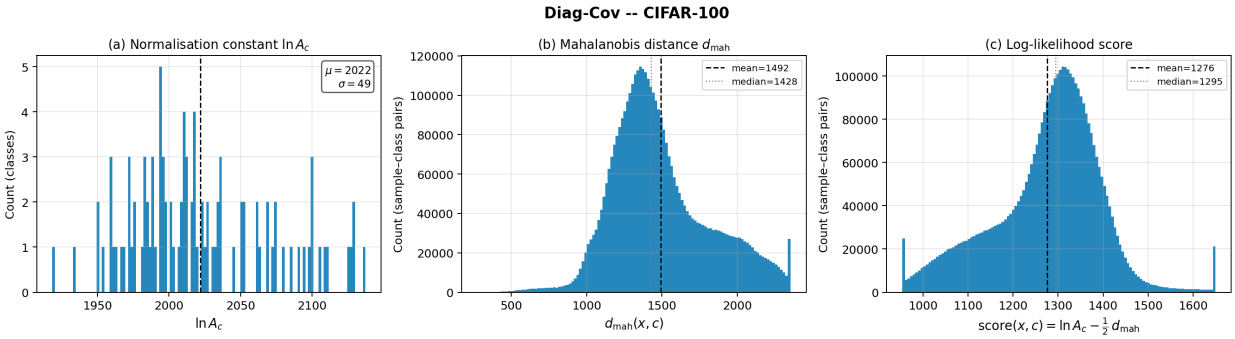


Figure A.1: **CIFAR-100 — Training set** (Diag-Cov). Empirical distributions of the volume term $\ln A_c$, the shape term d_{mah}, and the final log-likelihood score, computed on the data used to fit the Gaussian parameters.

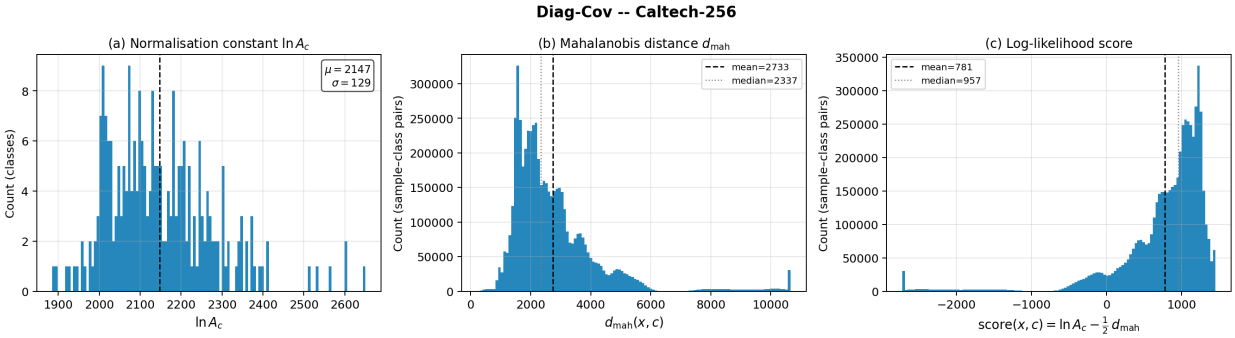


Figure A.2: **Caltech-256 — Training set** (Diag-Cov). Score component distributions on the training data.

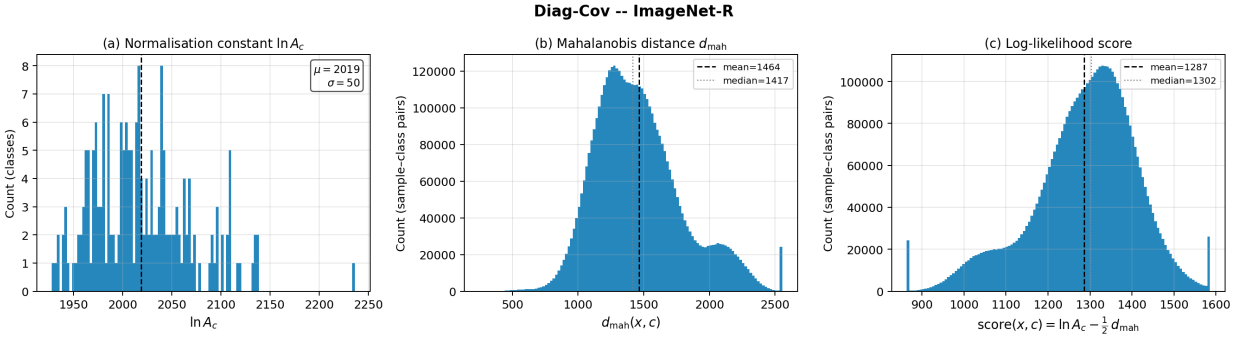


Figure A.3: **ImageNet-R — Training set** (Diag-Cov). Score component distributions on the training data.

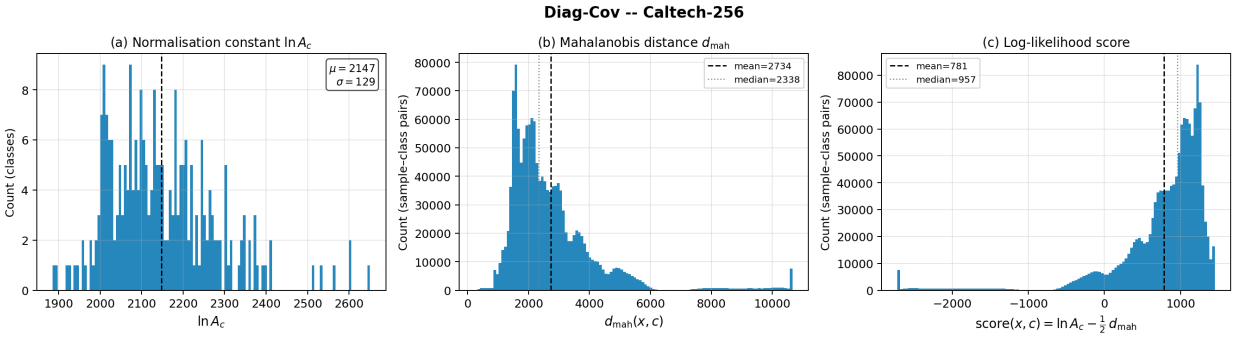


Figure A.4: **Caltech-256 — Validation set** (Diag-Cov). Score component distributions on held-out samples.

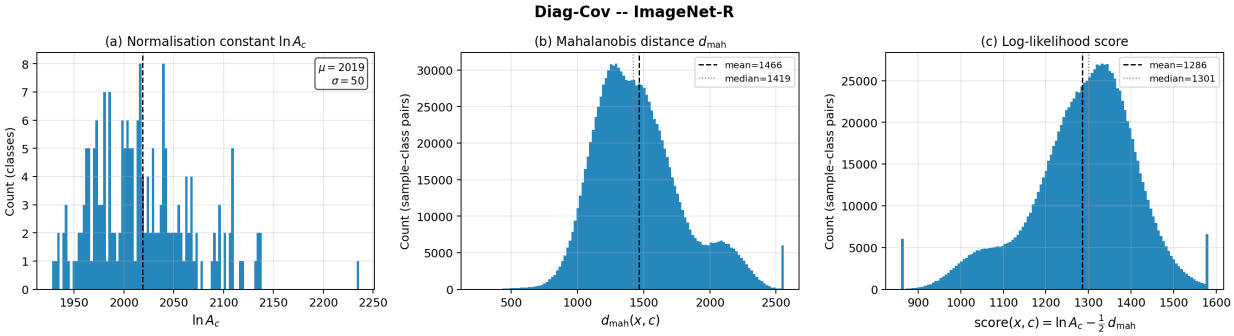


Figure A.5: **ImageNet-R** — **Validation set** (Diag-Cov). Score component distributions on held-out samples.

Diagonal Covariance — Validation Splits (Caltech-256 and ImageNet-R)

Full Covariance — Training Splits

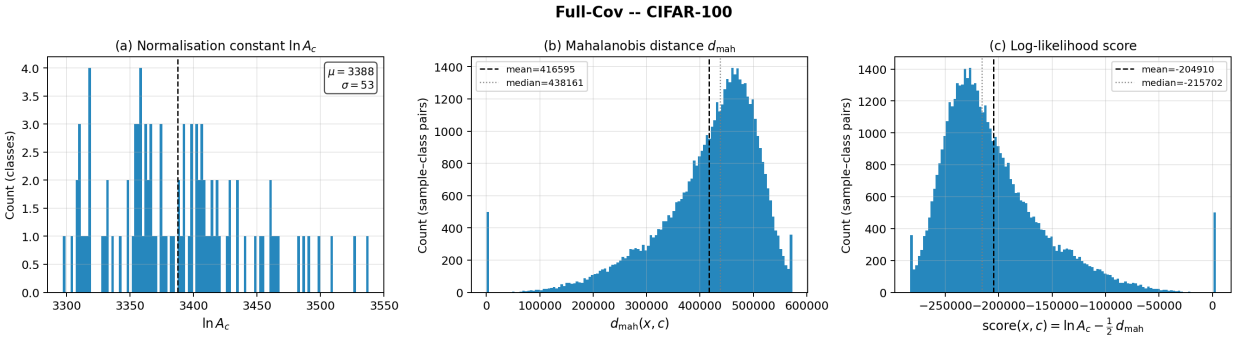


Figure A.6: **CIFAR-100** — **Training set** (Full-Cov). Score component distributions on the training data.

Full Covariance — Validation Splits (Caltech-256 and ImageNet-R)

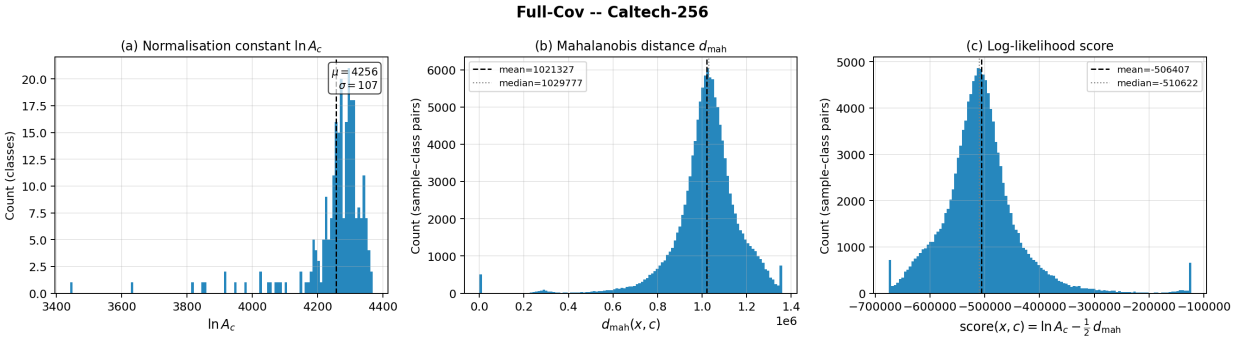


Figure A.7: **Caltech-256 — Training set** (Full-Cov). Score component distributions on the training data.

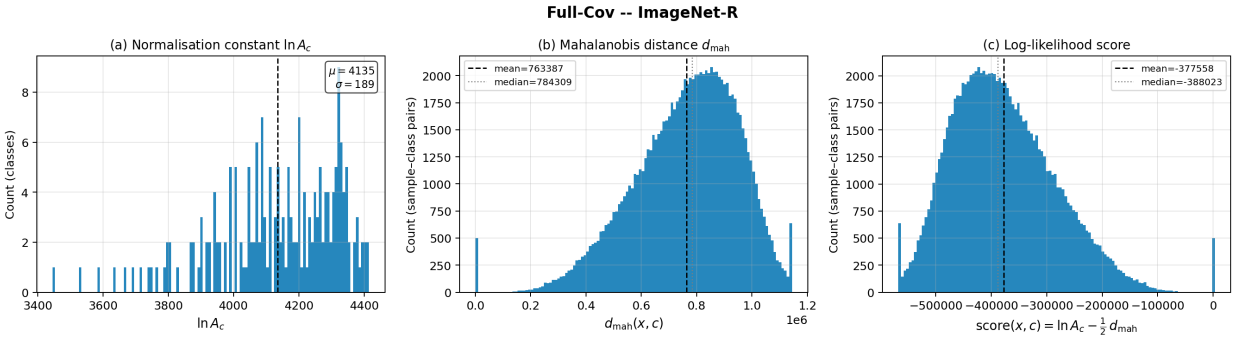


Figure A.8: **ImageNet-R — Training set** (Full-Cov). Score component distributions on the training data.

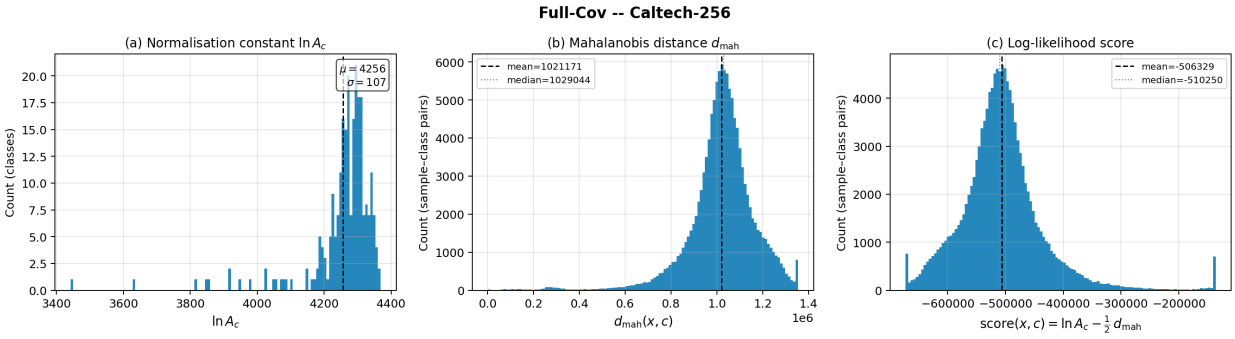


Figure A.9: **Caltech-256 — Validation set** (Full-Cov). Score component distributions on held-out samples.

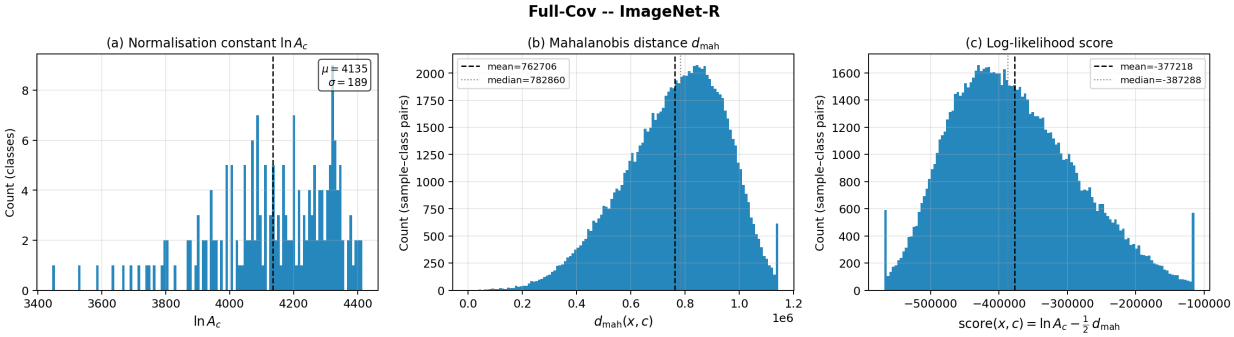


Figure A.10: **ImageNet-R — Validation set** (Full-Cov). Score component distributions on held-out samples.